Mathematical Tools for Econ PhD

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Chapter 0

Preface

This course is a handbook–precourse hybrid intended to walk (prospective) PhD students through the math concepts needed for (at least) the Micro sequence of the PhD. Each topic aims at introducing a key concept and fleshes out its relation to economic ideas and why it is important.

0.1 Style and Tone

The course is deliberately chatty in an attempt to stand apart from the mathematical appendices of economic textbooks. It also tries not to lose sight of the economic problem, in the hope of keeping on board those who care less about the beauty of the concept discussed and more about its application.

It is always a fine line between being comprehensive and complete, yet not losing track of the intuition. For the purposes I imagine with this course, I have chosen to sometimes not discuss things to the finest detail, but to leave that to a point in one's career when motivation is already built up sufficiently to have the stamina to survive (seemingly) unrelated lemmata. At the same time, I have tried to be precise and consistent and to flag any hand-wavy part. The hope is that this way the reader can at least see when we are using some magic to get us halfway there.

One thing that I aim at with these notes is to deliver stories behind concepts that capture at least the main intuition behind our topic. The goal is to have easy-to-remember items that deliver the gist of some buzz word. I am aware that there are slippery slopes on this road. Yet, I think it is sometimes better to have some idea before going back to the textbook.

0.2 Relation to Textbooks

One frustration I felt in my days as a PhD student was that basic math seemed either scattered across random mathematical appendices — all with different notation, many requiring you to read the textbook itself first — or buried in pure math books where it was unclear what the prerequisites were. This led to inefficient rabbit-holing at times (not to be confused with the efficient kind of rabbit hole which, I firmly believe, is key to surviving a PhD core sequence).

I believe that with AI at one's fingertips a new alternative arises: asking the machine every time to explain things — which, for many reasons, at least right now, leads to unpredictable outcomes because conveying for what purpose we are interested in these topics is hard for the machine to grasp, and difficult for the learner to explain to it.

So what I did here is curate a set of concepts in a language I like, at a precision level I can tolerate, and with a particular goal (the PhD core sequence) in mind. None of what happens here is new, but it comes together in one place and with limited noise surrounding it. Ideally, that makes it a useful tool.

Ideally for me, it also makes my life easier when teaching Microeconomics, because we do not get side tracked too much in discussions about who knows which math concepts and how well, but can focus on the **purpose** of the concept in the **economic** problem.

However, textbooks serve a purpose still, and Mas-Colell, Whinston, and Green (1995, MWG henceforth) remains incredibly powerful despite being older than most PhD students. Therefore, I aim to relate each topic to the relevant literature and in particular to MWG, hoping that readers can also use these topics as a guide through the many excellent textbooks out there.

Other books I refer to are Ok (2007), Aliprantis and Border (2006), and Mailath and Samuelson (2006). Kreps's writing style is always in the background, and all his books are worth getting into.

0.3 References

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Contents

Chapter 1

Topic 1: Sets, Mappings, and the Language of Economic Models

(Why economists spend so much time talking about functions)

1.1 Why this topic exists at all

When economists write down a model, they are almost never trying to compute a number.
They are trying to describe a **relationship**.

For example:

- how prices relate to demand,
- how preferences relate to choices,
- how information relates to outcomes,
- how today's actions relate to tomorrow's incentives.

Mathematically, a relationship is expressed by a **mapping**: something that takes inputs and assigns outputs.

Much of the difficulty students face with formal economic theory comes from not being fully comfortable with this basic ideas like:

- what are the objects?
- what is being mapped to what?
- what does it mean to "solve" the model?

This first topic is therefore about learning to **read and write the language in which economic relationships are expressed**.

1.2 Sets: describing what can possibly happen

We begin with the simplest idea: a **set**.

A set is simply a collection of objects. These could be **things you own**, **options you have**, **things that can be produced**, etc.

If, for example, a consumer chooses quantities of two goods, she then has a set of goods. There are many ways to write them down, one we often use is a pair of numbers, say (x_1, x_2).
The set of all possible bundles is written as a subset of the two-dimensional plane $\mathbb{R}^2$.

Already here we see an important idea: **economic objects are often vectors, not scalars**.

1.3 Cartesian products: building multidimensional spaces

Suppose we have two sets:

- X_1: possible choices of good 1,
- X_2: possible choices of good 2.

The **Cartesian product**

$$X_1 \times X_2$$

is the set of all ordered pairs (x_1, x_2) with $x_1 \in X_1$ and $x_2 \in X_2$.

Geometrically, you should think of a Cartesian product as a **multidimensional space**:

- one dimension per component,
- each point representing a complete specification of choices.

This construction appears everywhere in economics:

- a consumption bundle is a point in a product space,
- a strategy profile is a point in a product of players' strategy sets,
- a type profile is a point in a product of agents' type spaces.

Whenever you see notation like

$$x = (x_1, \ldots, x_n),$$

you are in a Cartesian product. Understanding this is essential for following later arguments about equilibrium and interaction.

1.4 Functions: when behavior is uniquely determined

A **function** is a rule that assigns **exactly one output** to each input.[1]

Formally, a function f from a set X to a set Y is written as

$$f : X \to Y.$$

This formulation means **every** element in set X is mapped to **some** element in set Y, or formally $\forall x \in X, \exists y \in Y$ such that $f(x) = y$.[2] It does *not* mean: For every element y in Y there is (at least one) x such that $f(x) = y$.

In economics, functions are used when the model predicts a **unique response**.

Examples:

- a utility function assigns a number to each consumption bundle,
- a production function assigns output to inputs,
- an allocation rule in mechanism design assigns an outcome to reported types.

What matters is not just the formula, but the **direction of the mapping**:

- What is taken as given?
- What is determined as a consequence?

Economic reasoning is often about fixing the domain and studying how the output changes when the input changes.

1.5 Injective, surjective, and bijective mappings

So far, we have said what a function is, but not all functions behave in the same way.
In economics, it is often important to understand *how much information is lost or preserved* by a mapping.

Let $f : X \to Y$ be a function.

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[1] Note that the opposite need not be true. So each output may have more than one input assigned to it. For more on this, see below when we talk about bijectivity and related concepts.

[2] You should read this subclause as: "… or formally *for each* ($\forall$) element x in set X *there exists* an element y in set Y such that plugging x into function f gives the result y. In terms of notation you may also sometimes see $x \mapsto f(x)$ when talking about specific elements rather than domain (X) and codomain (Y).

We say that f is **injective** (or one-to-one) if different inputs always lead to different outputs. Formally, this means:[3]

$$f(x) = f(x') \Rightarrow x = x'.$$

Intuitively, injectivity means that observing the output uniquely pins down the input. No two distinct elements of X are "collapsed" into the same element of Y.

In economic terms, injectivity is related to **identification**:

- if a mapping from types to outcomes is injective, observing the outcome reveals the type,
- if it is not injective, some information is necessarily lost (at least in some instances).

We say that f is **surjective** (or onto) if every element of Y is reached by *some* element of X. Formally:

$$\forall y \in Y, \exists x \in X \text{ such that } f(x) = y.$$

Surjectivity is about **coverage**:

- the codomain Y does not contain "irrelevant" elements that $f(x)$ never reaches, because
- everything in Y can actually arise as an outcome if we pick the right x.

In economics, surjectivity often fails by design. For example, not every allocation may be feasible given technological or incentive constraints.

Finally, a function is **bijective** if it is both injective and surjective.

A bijection establishes a perfect one-to-one correspondence between X and Y:

- no information is lost,
- no outcomes are redundant.

Bijective mappings allow us to "relabel" objects without changing structure. They play an important role when we talk about equivalent representations, such as different utility functions representing the same preferences.

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[3] Read the following sentence like this: the value of function f with input x is equal to the value of function f with input x' **only if** x is the same as x'. This sounds overly complicated, but it also teaches us something about negations. Here, we would say "if x and x' are not exactly the same, then $f(x)$ and $f(x')$ *cannot* be the same when f is injective."

It is worth emphasizing that **most economically interesting mappings are not bijective**. Demand is typically not injective, allocation rules are often not surjective, and equilibrium correspondences (see below) are almost never single-valued. Understanding this prevents a lot of confusion later on.

1.6 Correspondences: when economics refuses to choose uniquely

Very often, however, economic models do *not* deliver unique outcomes.

Consider a consumer who is indifferent between several bundles. Or a firm that has multiple profit-maximizing input choices.
Or a player who has several best responses.

In such cases, behavior cannot be described by a function. Instead, we use a **correspondence**.

A correspondence assigns **a set of outputs** to each input:

$$F : X \rightrightarrows Y.$$

This notation means:

> for each $x \in X$, $F(x)$ is a subset (not necessarily a single element) of Y. Other than that, correspondences behave *conceptually* very much like functions.[4]

Economically, looking at correspondences reflects an important modeling choice:

- non-uniqueness is not a bug,
- it is a realistic feature of optimization and strategic interaction.

1.7 Graphs: putting inputs and outputs together

To study functions and correspondences, it is often useful to look at their **graphs**.

Formally, the graph of a correspondence F is the set

$$\mathrm{Gr}(F) = \{(x, y) \in X \times Y \mid y \in F(x)\}.$$

Intuitively, the graph is the picture we would draw to represent the function. Geometrically, the graph lives in a higher-dimensional space that records both:

[4] Obviously, if f is a function it is also a correspondence. However if g is a correspondence, it may or may not be a function.

- the input,
- and all admissible outputs.

Later in the course, properties of graphs (such as being closed) will be crucial for proving:

- existence of equilibrium,
- stability of solutions,
- continuity of economic behavior.

For now, the key idea is simple with unidimensional functions:[5] **a graph makes the mapping visible**.

1.8 Where this appears in Mas-Collel, Whinston, and Green (MWG henceforth).

The basic language of sets, functions, relations, and correspondences is developed in:

- **MWG, Mathematical Appendix A**

The appendix is concise and formal. These notes are meant to provide the *verbal intuition* that helps you read it fluently.

1.9 An economic example: demand.

Consider a standard consumer problem:

$$\max_{x \in X} u(x) \quad \text{s.t. } p \cdot x \leq w.$$

What is the solution?

The correct answer is not necessarily "a bundle that solves the problem." The correct answer is: "**the set of bundles* that solve the problem."

This set is called the **demand correspondence**:

$$x(p, w) \subseteq X.$$

Why is that the case? Because if p is not the price for a single good but a **price vector** capturing the prices of several goods, the consumer may be indifferent between very different bundles for some p and w, but as prices change slightly,

[5]That is functions that map, e.g., from one number into another.

she may not anymore. Thus, keeping track of those indifferences (and changes therein) is going to drive some results.

If you mistakenly think demand must be a function, many later results will seem mysterious or false.

1.10 Why this matters for microeconomics

Once you see economic models as **mappings between sets**, many things fall into place:

- equilibrium is a fixed point of a correspondence,
- incentive compatibility is a constraint on a mapping,
- equilibrium payoffs in repeated games form a set.

Microeconomic theory is, in many ways, the study of **structured relationships**. Learning to read and write these relationships carefully is the first step toward understanding everything that follows.

1.11 What you should take away

After this topic, you should be comfortable with the following mental moves:

- asking what the domain and codomain of an object are,
- recognizing when outcomes are sets rather than points,
- visualizing multidimensional economic objects as product spaces.

You do **not** need to memorize formal definitions. You do need to develop the habit of *asking what lives where*.

Chapter 2

Topic 2: Preferences and Utility Representation

(How economists talk about what people want)

2.1 Why preferences come next

In Topic 1, we learned how economists describe **relationships** using sets and mappings. So far, however, we have not said *what drives choice*.

When we say that a consumer chooses one bundle rather than another, or that an agent prefers one outcome to another, we are making a statement about **preferences**. Preferences are the primitive objects economists use to encode tastes, goals, and objectives.

This topic introduces preferences as **formal objects**. Only later will we show how they can be represented by utility functions. For now, the goal is to understand what economists mean when they say that an agent "prefers" one alternative to another, and what structure we impose on such statements.

2.2 Alternatives and preference relations

Let X be a set of alternatives. Depending on the context, X could be:

- a set of consumption bundles,
- a set of allocations,
- a set of outcomes of a mechanism,

- or a set of lotteries.

A **preference relation** on X is a binary relation, usually denoted by $\succeq$.[1]

Formally, we write

$$x \succeq x'$$

to mean that x is weakly preferred to x'.[2]

From the weak preference relation $\succeq$, we define two *derived* shorthand notations:

- **Strict preference**:

$$x \succ y$$

  meaning "x is strictly preferred to y", or, formally, $x \succeq y$ but *not* $y \succeq x$.

- **Indifference**:

$$x \sim y$$

  meaning "the agent is indifferent between x and y".[3]

These are not new primitives. They are constructed from $\succeq$ and will inherit its properties.

2.3 Basic properties of preferences

Economists typically impose structure on preferences to make choice predictable and analyzable.

2.3.1 Completeness

Preferences are **complete** if for every pair $x, y \in X$, either

$$x \succeq y \quad \text{or} \quad y \succeq x \quad \text{(or both)}.$$

What this means:

> "Given any two alternatives, the agent can compare them."

Completeness rules out indecision. It does *not* rule out indifference.

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[1] The relation is "binary" because formally it compares elements from two sets. In our context the two sets are X and X again. If you want to be pedantic $\succeq$ in itself is a subset of $X \times X$. More on this after the definition.

[2] So in light of the set interpretation of $\succeq$, we see that $\succeq$ is indeed the set of all pairs (x, x') such that we weakly prefer x over x'. Think about the more familiar $\geq$ which of course is also a binary relation. Writing $x \succeq x'$ is then a simple and intuitive shorthand for this (perhaps) more complicated object.

[3] Think about how you would express this with $\succeq$!

2.3.2 Transitivity

Preferences are **transitive** if whenever

$$x \succeq y \quad \text{and} \quad y \succeq z,$$

it follows that

$$x \succeq z.$$

Read aloud:

> "If x is at least as good as y, and y is at least as good as z, then x is at least as good as z."

Transitivity rules out preference cycles. Without it, optimization problems would often be ill-defined.[4]

2.3.3 Rational preferences

A preference relation that is both complete and transitive is often called **rational**.

This terminology is not meant to be normative. It simply signals that preferences are structured enough to support meaningful choice and optimization (more on this later).

2.4 Choice and the meaning of rationality

Why do we impose completeness and transitivity?

Because together they ensure that the agent can make an (optimal) choice from a given set X

Formally, $b \in X$ is an optimal choice

$$b \succeq x \quad \text{for all } x \in X.$$

Notice that:

- there may be more than one such b,
- choices are therefore naturally a **correspondences**, not a functions.

This connects directly back to Topic 1.

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[4]An interesting fact in economics is that while individuals are often somewhat transitive in their preferences, it may be the case that a *group* of individuals with transitive preferences does *not* have transitive preferences on the group level.

2.5 Utility representation: turning preferences into numbers

Working directly with preference relations can be cumbersome. Economists therefore often ask whether preferences can be *represented* by a **utility function**.

A utility function represents a preference if we can find a mapping

$$u : X \to \mathbb{R}$$

such that

$$x \succeq x' \quad \Leftrightarrow \quad u(x) \geq u(x').$$

Read aloud:

> "The agent weakly prefers x to x' if and only if the utility assigned to x is at least as large as the utility assigned to x'."[5]

This equivalence is crucial. Utility is not an extra assumption – it is a **representation** of preferences.[6]

2.6 Ordinal versus cardinal utility

The definition above immediately implies an important point. If u represents preferences, then so does any function of the form

$$v(x) = f(u(x)),$$

as long as f is **strictly increasing**.[7]

Why is this the case? Well, if f is strictly increasing that means higher numbers in $\Rightarrow$ higher numbers out. But because $u(x) > u(x')$ if $x \succeq x'$, we know that this holds also for $f(u(x))$.

Why is this important? Because it implies that:

- only the **ordering** of utility values matters,
- differences in utility have no direct meaning.

[5]The term *if and only if* describes *equivalence* because it says the left hand side is true *if* the right hand side is true ($\Leftarrow$). But it also says, that the left hand side can be true *only if* the right hand side is true, which in turn means that the left hand side being true implies the right hand side being true ($\Rightarrow$).

[6]You may wonder why we need to "represent" something here. There reason is (as often) tractability. If we have the utility representation, we work on the real-line which makes it very easy for us to compare things. As we will see below, it also makes it easier for us to understand what properties mean, because we simply know very well how to operate and walk around in $\mathbb{R}$.

[7]Recall from school, that $f(u(x))$ implies we first plug x into u to obtain some $y = u(x)$ which we then plug into f to obtain $f(y) = f(u(x))$.

For this reason, utility in microeconomic theory is (without further assumptions) **ordinal**, not *cardinal*. That means, it gives us a ranking of alternative x vs x', but nothing on how much more we like x than x'. This observation will matter later when we discuss welfare comparisons, mechanism design, and incentives where sometimes we make additional assumptions to get a cardinal interpretation.

2.7 Existence of utility representations (informal)

Not every preference relation can be represented by a utility function. Some structure is needed.

A central result in microeconomic theory states that:

- if preferences are complete and transitive,
- and satisfy suitable continuity conditions, then a utility representation exists.[8]

We will not prove this result here nor make the conditions explicit. For now, the important message is that **utility functions do not come for free**. They summarize assumptions about preferences.

2.8 Where this appears in Mas-Colell, Whinston, and Green (MWG)

The formal treatment of preferences and utility representation appears in:

- **MWG, Chapter 1**
- **MWG, Mathematical Appendix A** (relations and functions)

The existence of utility representations is discussed in detail in Chapter 1, with formal proofs relying on the mathematical tools introduced in the appendix.

2.9 An economic example: indifference and choice

Consider a consumer who only cares about the total number of calories she consumes, not about the composition of goods.

[8] Note that this is an *only if* condition. So, that means other preferences may also have representations, but we would need to look at them carefully.

Many different bundles may deliver the same caloric content. Such bundles will be **indifferent** according to her preferences.

This indifference implies:

- multiple optimal choices,
- flat regions in utility representations,
- demand correspondences rather than demand functions.

Understanding this example prevents confusion later when demand fails to be single-valued.

2.10 Why this matters for microeconomics

Preferences are the starting point of almost all microeconomic models.

They:

- generate demand,
- shape incentives,
- determine equilibrium outcomes,
- constrain what mechanisms can achieve.

Much of microeconomic theory can be seen as studying how **different institutional environments transform preferences into outcomes**.

If you understand preferences as structured relations—rather than vague statements of taste—you are well prepared to follow everything that comes next.

2.11 What you should take away

After this topic, you should be comfortable with:

- reading preference notation,
- understanding what completeness and transitivity do and do not imply,
- interpreting utility functions as representations rather than psychological objects.

You should also see how this topic connects directly back to Topic 1: preferences live on sets, generate correspondences, and are often represented by functions.

Chapter 3

Topic 3: Convex Sets and the Geometry of Choice

(Why economists care so much about "nice" shapes)

3.1 Why geometry enters microeconomics

In Topic 2, we introduced preferences and choice. At that point, however, we were deliberately vague about *what the set of alternatives looks like.*

This was intentional.

To understand why many economic problems are tractable – and why others are not – we need to think about the **geometry** of choice. That is, we need to understand the *shape* of the sets over which agents choose.

Convexity is the key concept here. It will explain:

- why interior solutions exist,
- why averaging alternatives often makes sense,
- why optimization problems behave smoothly,
- and why equilibrium analysis is even possible in large economies.

3.2 Convex sets: the basic idea

Let $X \subseteq \mathbb{R}^n$ be a set.[1]

[1] This is not a requirement. But for the cleanest geometric interpretation, and for most of the applications we are interested in, assuming the underlying vector space is a Euclidean

We say that X is **convex** if for any two points $x, x' \in X$ and any $\lambda \in [0, 1]$,

$$\lambda x + (1 - \lambda)x' \in X.$$

Read aloud:

> "If x and x' are feasible, then any weighted average of x and x' is also feasible."

Geometrically, this means that the straight line segment connecting x and x' lies entirely inside the set. Or, in somewhat looser terms: If you want to walk from point x to point x' and you are not allowed to turn, you can do that by walking straight and without leaving the set X.

3.3 Economic interpretation of convexity

Convexity has a very natural economic interpretation.

Think of x and x' as two consumption bundles. The bundle

$$\lambda x + (1 - \lambda)x'$$

can be interpreted as consuming a fraction λ of x and a fraction $1 - \lambda$ of x'.

Convexity therefore formalizes the idea that:

> "Mixing feasible alternatives remains feasible."

This is a powerful assumption. It encodes divisibility of goods, absence of indivisibilities, and the ability to trade off quantities smoothly.

It, in fact, also says that you can do combinations of choices. It means, if you have the option to go left and right, you can also go anything in between like "straight" (50% left, 50% right) or somewhat left (80% left, 20% right) and so on.

Most standard consumption sets in microeconomics are convex for exactly this reason.[2]

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space isomorphic to $\mathbb{R}^n$ will be fine.

[2]There are of course important situations where this assumption fails. If you have to make the decision whether to move to the US or stay in Europe, moving half way into the middle of the Atlantic may not be feasible, for example.

3.4 Why convex problems are easier to solve

From a math point of view, convexity matters because it dramatically simplifies optimization.

If choice sets are non-convex we often

- cannot use standard optimization algorithms
- need to worry about discrete math
- have a hard time to figure out what small perturbations imply

Therefore much of economic theory is built on convexity. That sometimes makes things less realistic, but often just simplifies the amount of math needed to convey ideas accurately and intuitively. In practice, one needs to, of course, think about what convexity assumptions preclude, but often the convexity assumption in fact allows us to separate the wheat from the chaff among the set of possible solutions. More on this, once we are actually solving things.[3]

3.5 Convex combinations and notation

The expression

$$\lambda x + (1-\lambda)x'$$

is called a **convex combination** of x and x'.

More generally, if $x_1, \ldots, x_k \in X$ and $\lambda_1, \ldots, \lambda_k \geq 0$ with

$$\sum_{i=1}^{k} \lambda_i = 1,$$

then

$$\sum_{i=1}^{k} \lambda_i x_i$$

is a convex combination of the points $x_1, \ldots, x_k$.

Convex sets are exactly those sets that contain *all* convex combinations of their elements. Coming back to our ‘’walking’’ interpretation, here is how to think about this result.

Suppose you give me a set of points $x_1, \ldots, x_k$ (in a Euclidean space). And now I want construct a set that contains those points and everything ‘’inbetween’’ then what I am doing is constructing a convex set.[4]

[3] For the applied economist it is worthwhile to note that many of the estimators also assume explicitly or implicitly a somewhat convex choice set to be working.

[4] In fact, what I am constructing is what is called the **convex hull**—the smallest convex set containing all my points

In practice constructing this set is easy on a 2-dimensional plane (like $\mathbb{R}^2$). There, I would simply connect each point with every other point by a straight line. This gives me the shape of a fully connected network, with my initial points as *nodes* and the lines connecting those points as *edges*. Now if in a second step, I connect all points on the edges with all other points on all other edges through straight lines, I have constructed the convex hull. This type of reasoning, does not directly extend to more dimensions in the sense that it is not clear that I can always construct the convex hull using finitely many such steps. It is true, however, that if I were to follow that procedure *ad infinitum* I will eventually arrive at the convex hull.

3.6 A brief word on Concavity (will show up again)

If one talks about convexity, one has the urge to talk about concavity as well which is often seen as the antonym of convexity.

However, with the set interpretation, that is, in fact hard to do. So I will just clarify a few things that are not true (at least under our conventions).

- It is not the case that every non-convex set is concave. It is simply *non-convex*
- It is not the case that the complement of a convex set is concave. It is (sometimes) called *reverse convex*.

To sum up, there is no relevant notion of concave set in math.[5]

All that said, concavity will come back to these lecture notes, but only after we talk about other notions of *convexity,*

3.7 Convex preferences (informal)

One confusion that sometimes comes up in Microeconomics is that when we talk about convex preferences we do not talk about sets of preferences, but something else.

We call preferences **convex** if agents weakly prefer any combination of two bundles x and x' over a third bundle z when x and x' alone are (weakly) preferred to z.[6] This sounds, at first, totally unrelated to the geometric convexity notion apart from the fact that it talks about some form of mixing.

[5] I say relevant, because as often in math, there may be some esoteric subfield, at least, that has defined such a notion. But that is irrelevant to us.

[6] This holds especially if $z = x$ which has the interpretation that combining x with something weakly better than x cannot make it worse.

Formally, the statement is:

Preferences are convex if the following holds:

If for all x, x', z,

$$x \succeq z \text{ and } x' \succeq z$$

then

$$\lambda x + (1-\lambda)x' \succeq z \quad \text{for all } \lambda \in [0,1].$$

In words the formal part means

> "*If the preferences are convex and the agent likes x and x' better than z, then she prefers any combination $\lambda x + (1-\lambda)x'$ over z too."

How does that connect to our convexity notion? To understand this, we need to understand what a **better set** (MWG and mathematicians call it **upper contour set**—I will likely use both equivalently) is. The better set of some bundle x is the set of all things the agent weakly prefers to x.

So formally, the better set of x is

$$\{x' | x' \succeq x\}$$

which means[7]

> the set of all x' (often relative to a base set) such that x' is weakly preferred to x.

Now, you may have already understood the relationship here. Convex preferences are preferences where for any bundle x, the better set is a convex set.

3.8 Convex Functions

When we talk about functions, we also often talk about convexity. One may remember from school that $f : x \mapsto x^2$ is called a "convex function." Again, it may be hard to think about this given our notion of convexity on sets. It is not, however, if we think about it in the way we thought about it with our preferences.

Take a function that has domain X and codomain Y[8], We call a function's **epigraph** the set of all points in $X \times Y$ that lie *on or above* the graph of the

[7] Sometimes you will see a ":" instead of a "|" in the notation for conditioning (so in our case $\{x : x' \succeq x\}$). These are identical. I use the | because it is familiar from, e.g., *conditional expectations*.

[8] So, as we should know by now $f : X \to Y$.

function.[9]

For a real valued function $f : X \to \mathbb{R}$, we can write the epigraph as

$$\{(x, y) \in X \times \mathbb{R} | y \geq f(x)\}.$$

Then we can define convex functions as follows:

A function is convex if its epigraph is a convex set.

Perhaps you have heard a different definition before which says that a function is convex if the straight line connecting any two points on the graph lies weakly above the function, that is,

A function $f : X \to \mathbb{R}$ is convex if

$$\forall x, x' \in X, \text{ and } \lambda \in [0, 1] \text{ it holds that } \lambda f(x) + (1 - \lambda) f(x') \geq f(\lambda x + (1 - \lambda) x')$$

The inequality in this function is often known as "Jensen's inequality." This definition is *equivalent* to the epigraph definition, yet—perhaps—the connection to convex sets is less clear.

However, Jensen's inequality helps us to finally get to concavity which is well defined in this function interpretation, and very important in economics. The simplest definition is this

A function $f : X \to \mathbb{R}$ is concave if $-f$ is convex.

Alternatively, a function is concave if Jensen's inequality has $\leq$ instead of $\geq$.[10]

Clearly, of course, that means that a *linear* function $f(x) = mx$ or an *affine* function $f(x) = mx + t$ is both: convex *and* concave.

3.9 Utility functions and concavity

An important relationship between preferences and utility functions is that when it is possible to have a utility representation and this representation is a *concave utility function*, then the underlying preferences must be convex.

The intuition is simply that a concave utility function says, that if I take a combination $\lambda x + (1 - \lambda) x'$ then this gives me utility of

[9] You may, of course, wonder what *above* in this context now means, and you are correct in asking for a metric. This goes beyond what we cover here (where we focus on real-valued functions), but there are natural generalizations to codomains that are *partially ordered vector spaces*.

[10] You can also think of the counterpart of the epigraph, the *hypograph* to define concavity of a function.

$$u(\lambda x + (1-\lambda)x') \geq \lambda u(x) + (1-\lambda)u(x').$$

Now take a third bundle z that has $u(z) \leq \min\{u(x), u(x')\}$.[11]

Now we do know that this is true for any x, x' weakly better than z:

$$u(\lambda x + (1-\lambda)x') \geq \lambda u(x) + (1-\lambda)u(x') \geq u(z).$$

which means that u represents the preference relation[12]

$\lambda x + (1-\lambda)x' \succeq z$ for any x, x', z. That in turn is the very definition of convex preferences.

Concave utilities have many desirable properties that we are going to discuss over the course, but unfortunately the converse is not true. That is, assuming convexity does not get us a concave utility representation.

There are two reasons. One is the monotone transformations we discussed when talking about cardinal vs ordinal preferences.

The utility function $u(x,y) = x + y$ (which is concave) represents the same preferences as the utility $u'(x,y) = (x+y)^3$, if $x, y \geq 0$. But we see that

$$u'(2,1) = u'(1/2 * (4,0) + 1/2 * (0,2)) = 27 < 1/2u'(4,0) + 1/2u'(0,2) = 32 + 4.$$

This in itself would not cause too much trouble, because it would perhaps be enough if all convex preferences have *at least one* concave utility representation. Unfortunately, that is not true either.

This is not a result to be discussed here but Danish-German mathematician Werner Fenchel proved in 1953 that if indifference curves are lines that are not always parallel, then no concave utility representation exists. That means, restricting ourselves to concave utility functions is not enough.

3.10 Quasi-concavity and upper contour sets

Instead, economists use a weaker (and more annoying) concept called ''quasi-concavity.''

A function u is quasi-concave if, for every real number $\bar{u}$, the upper contour set** of the function

$$\{x \in X \mid u(x) \geq \bar{u}\}$$

[11] Read as: Take any bundle that fulfills this property.
[12] See Topic 2.

is convex.**[13]

That means, we are looking for the set of choices x such that $u(x)$ is larger than some given level $\bar{u}$. Now think back to our "better" sets. If u represents preferences and x is in the better set of some other x' which delivers utility $\bar{u}$ then it is weakly better than $\bar{u}$ and must thus be represented by a (weakly) higher u.

Thus, quasi-concavity is not really derived from convex preferences, it is the exact same thing!

However, there is another way to represent quasi concavity. It is (equivalently) true that

**a function u is quasi-concave if

$$\forall x, x' \quad u(\lambda x + (1-\lambda)x') \geq \min\{u(x), u(x')\}.$$

That means a function of a convex combination of the inputs is at least larger than the minimum of the two extremes. The main thing this rules out is that the function has an interior minimum (we will come back to what those are) like x^2 for example. Here if we take $x = -2$ and $x = 2$ then for any $\lambda \in (0,1)$, $\lambda(-2) + (1-\lambda)2 \in (-2,2)$ and for any $z \in (-2,2)$ $u(z) < u(2) = \min\{u(2), u(-2)\} = 4$.

3.11 Where this appears in Mas-Colell, Whinston, and Green (MWG)

Convexity and concavity are introduced and used throughout:

- **MWG, Chapter 1** (preferences and choice)
- **MWG, Chapter 3** (consumer theory)
- **MWG, Mathematical Appendix B** (convex sets and functions)

The appendix provides formal definitions and theorems. The chapters show how these ideas drive economic results.

3.12 An economic example: diversification

Consider an agent choosing between two risky assets.

If preferences are convex, then holding a diversified portfolio – a mixture of the two assets – is weakly preferred to holding either asset alone.

[13]The name is the same here like with the better sets, because mathematically it is the same thing (just with a different metric). But there is a connection to convex preferences!

This simple geometric fact underlies:

- portfolio choice,
- insurance demand,
- and risk-sharing in general equilibrium.

Without convexity, diversification need not be attractive.

3.13 Why this matters for microeconomics

Convexity is one of the main reasons microeconomic models "work".

It:

- makes optimization tractable,
- ensures equilibria exist,
- stabilizes comparative statics,
- and links geometry to economic intuition.

Later in the course, convexity will quietly sit in the background of:

- welfare theorems,
- mechanism design,
- monotone comparative statics,
- and repeated games with convex payoff sets.

Understanding convexity now will save you a lot of confusion later.

3.14 What you should take away

After this topic, you should be comfortable with:

- visualizing feasible sets geometrically,
- interpreting convexity economically,
- understanding why convex problems are easier to solve,
- and seeing how convexity shapes choice correspondences.

Most importantly, you should appreciate that geometry is not decoration in microeconomics – it is structure.

Chapter 4

Topic 4: Continuity, Closed Sets, and Existence

(Why solutions exist — and when they do not)

4.1 Why continuity and closedness matter

Up to now, we have talked about:

- sets of feasible alternatives,
- preferences over those alternatives,
- and geometric structure such as convexity.

But we have carefully avoided a fundamental question:

> Does an optimal choice actually exist?

This question is not rhetorical. Many economic models are internally consistent, well-motivated, and intuitive — and yet fail to have solutions.

The reason is usually not a lack of convexity, but a lack of **topological regularity**:[1]

- sets that are not closed,
- preferences that jump discontinuously,
- or objective functions that approach a maximum without ever reaching it.

[1] **Topology** is the study of properties of geometric objects. In our setting which ultimately talks about optimization, it often means which movements in the choice space we allow and how we restrict the effect on those moves on outcomes. For now this is very vague, but it helps us answer questions that economists like, such as: ''If we just add one more tiny thing? Would we be better or worse?''

This topic introduces the minimal tools needed to understand what later we will call **existence**.

4.2 Open and closed sets: intuition first

Again, let's think about Euclidean spaces and let $X \subseteq \mathbb{R}^n$.

Informally we say that X is

- an **open** set if wherever you start in the set, you can move a little in any direction and stay inside the set,
- a **closed** set if it contains all its boundary points.

At first glance, this sounds like two mutually exclusive notions. If the set contains the boundary, then walking away from the boundary will make me leave the set at least in one direction. It turns out, this intuition is often true, but not always.

First, let us be a bit more precise:

- X is **open** if for every $x \in X$ there exists $\varepsilon > 0$ such that

$$B_\varepsilon(x) \subseteq X,$$

  where $B_\varepsilon(x)$ denotes the open ball of radius ε around x.

- X is **closed** if its complement is an open set.

The notion of the ball $B_\varepsilon(x)$ may be unfamiliar at first glance, but it basically means that around x we can form some form of ''aura'' so that if we want to stand at x we miss slightly (in the slightest way of slightly) we will at least be in that ball. If the set is open and we miss in the slightest way of slightly the point x we wanted to be at, we will at least still be in the set.

Now, let us go back on the notions. It turns out they are really independent notions.

- a set can be open but not closed,
- closed but not open,
- both,
- or neither.

Intervals in $\mathbb{R}$ make that very clear. The interval $[1, 2]$ is a closed set, but not open, the interval $(1, 2)$ is open but not closed. The interval $(1, 2]$ is neither. The emptyset, $\emptyset$ is by definition open,[2] thus the entire underlying space—which is the complement to $\emptyset$ (here $\mathbb{R}$)—must be closed. However that space is also open, because if we slightly miss, we remain in $\mathbb{R}$. That means $\emptyset$ has to be closed too.

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[2] This sounds a bit fishy, but for reasons that are not yet clear is important to be able to work with closedness.

4.3 Why closedness is economically important

Closedness matters because **maxima** are typically on boundaries. They are the largest point there is (in some direction). So, they have to live on the boundary.

Consider a consumer choosing x from a feasible set X to maximize utility. Even if preferences are continuous and well-behaved, the optimal choice may lie:

- at a corner,
- on a constraint,
- or exactly at the boundary of feasibility.

If the boundary is not included, the optimum may be *approached* but never *attained*, so the problem would have no solution.

4.4 Example: A failure of existence

Consider the simple problem:

$$\max_{x\in(0,1)} x.$$

The objective function is continuous. The feasible set is convex. Preferences are trivial.

And yet: **there is no solution**.

The supremum is 1, but $1 \notin (0,1)$.

This example shows:

> Convexity alone does not guarantee existence.

Closedness is essential.

4.5 Compactness: putting everything together

In finite-dimensional spaces, existence results are often driven by **compactness**.

A set $X \subseteq \mathbb{R}^n$ is **compact** if it is:

- closed,
- and bounded.

This characterization is specific to $\mathbb{R}^n$ and is known as the **Heine–Borel property**.

Compactness is powerful because it ensures that sequences cannot ''escape to infinity'' and that limits stay inside the set.

4.6 Continuity of functions

Let $f : X \to \mathbb{R}$ be a function.

Informally, f is **continuous** if:

> small changes in the input lead to small changes in the output.

Formally, f is continuous at x if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\|x - x'\| < \delta \Rightarrow |f(x) - f(x')| < \varepsilon.$$

We will not work directly with ε–δ arguments in the course. Instead, we will use continuity as a structural assumption with powerful consequences.

4.6.1 A remark on metrics

You may have noticed that the definition of continuity relies on a notion of "distance": $\|x - x'\|$ measures how far apart two points are. In $\mathbb{R}^n$, this is the usual Euclidean distance — the one you know from school. It basically says, the distance between two points is the length of the line connecting the two points.

More generally, a **metric** on a set X is any function $d : X \times X \to \mathbb{R}$ that satisfies four properties:

1. $d(x, y) \geq 0$ for all x, y (distances are nonnegative),
2. $d(x, y) = 0$ if and only if $x = y$ (only identical points have zero distance),
3. $d(x, y) = d(y, x)$ for all x, y (distance is symmetric),
4. $d(x, z) \leq d(x, y) + d(y, z)$ for all x, y, z (the triangle inequality — going via a detour is never shorter).

A set equipped with a metric is called a **metric space**. Everything we have discussed in this topic — open sets, closed sets, continuity, compactness — can be defined in any metric space, not just in $\mathbb{R}^n$. The definitions look exactly the same, with $d(x, x')$ replacing $\|x - x'\|$.

For most of this course, we will work with the usual Euclidean metric on $\mathbb{R}^n$ and will not need to think about alternatives. But sometimes we need to work on different spaces in economics (the space of functions, environments, whatever). Then the natural metric is, perhaps, less clear. Therefore, we will later, in Topic 13, return to the question of why economists sometimes need more abstract metrics — for instance, when the objects being compared are not vectors of numbers but *functions*, *sets*, or *probability distributions*. Naming the concept now means it will not be unfamiliar when it reappears.

4.7 The Intermediate Value Theorem

Continuity has immediate consequences, and the simplest one is that continuous functions cannot "skip" values.

The following theorem says this:

> "If a continuous (one-dimensional) function starts below some value and ends above it, then it must pass through that value somewhere in between."

Theorem (Intermediate Value Theorem). Let $f : [a, b] \to \mathbb{R}$ be continuous, and suppose $f(a) \leq c \leq f(b)$ (or $f(b) \leq c \leq f(a)$). Then there exists some $\hat{x} \in [a, b]$ such that $f(\hat{x}) = c$.

Proof. Without loss assume $f(a) < f(b)$[3] Let $X = x \in [a, b] : f(x) \leq c$ be the set of points such that the function value is below c. X is clearly non-empty, because it contains a. It is also bounded above by b, so there has to be a supremum $\sup X = \hat{x} \in [a, b]$.

We now want to show that $f(\hat{x}) = c$.

First, $f(\hat{x}) < c$ cannot be true because either $\hat{x} = b$ in which case $f(\hat{x}){=}f(b) \geq c$ or $\hat{x} < b$ in which case we can, using continuity of f find some $x' \in (\hat{x}, b)$ such that $f(x') < c$ and thus $x' \in X$ contradicting the definition of $\hat{x}$ as the supremum of X.

Now $f(\hat{x}) > c$ cannot be true because either $\hat{x} = a$ in which case $f(\hat{x}){=}f(a) \leq c$ or $\hat{x} > a$ in which case we can, using continuity of f find some $x' \in (a, \hat{x})$ such that $f(x') > c$ and thus $x' \notin X$ contradicting the definition of $\hat{x}$ as the supremum of X.

Ths $f(\overline{x}) = c$ is the only remaining possibility, and thus the theorem is proven. □

The theorem says nothing about *where* that point is, or how many such points there are. It only guarantees existence. But that guarantee is surprisingly powerful.

Why does this work? Because continuity means no jumps. If the function starts at $f(a)$ and ends at $f(b)$ and never jumps, it has to pass through everything in between. A function that skips a value would need a discontinuity to do so.

[3] The equality case is trivial, the flipped inequality case is just a relabeling exercise.

This is our first taste of a pattern that will recur throughout the course: **continuity plus structure gives existence**. The Intermediate Value Theorem is, in a sense, the one-dimensional ancestor of the fixed-point theorems we will meet later. In fact, many fixed-point arguments in one dimension are just clever applications of this result.

4.8 The Weierstrass Theorem (existence of maxima)

Why do we do all this? Because of Weierstrass.

Theorem (Weierstrass).
If $X \subseteq \mathbb{R}^n$ is compact and $f : X \to \mathbb{R}$ is continuous, then:

$$\max_{x \in X} f(x)$$

exists.

The theorem in words: Suppose the choice set is a subset of a Euclidean space and compact. Assume the objective is continuous. Then a choice that maximizes the objective exists.

This theorem is the backbone of existence results in microeconomics.

4.9 Why economists care about continuity

Continuity is not a technical luxury. It encodes economically meaningful assumptions:

- small changes in prices do not cause jumps in demand,
- small changes in endowments do not cause discontinuous behavior,
- incentives vary smoothly with parameters.

Discontinuities often signal:

- indivisibilities,
- threshold effects,
- or institutional constraints.

These can be important—but they require different tools.

4.10 Continuity of preferences and utility representations

Preferences are **continuous** if small changes in alternatives do not flip rankings abruptly.

Formally, continuity of preferences means that for any x:

- the better set $\{y \mid y \succeq x\}$ is closed,
- and the *worse set* (or lower contour set) $\{y \mid x \succeq y\}$ is closed.

When preferences admit a utility representation, continuity of preferences corresponds to continuity of the representing utility function.

If we have that, our lives are simple. If not, they require other tools.[4]

4.11 Existence of optimal choices

Putting everything together:

If:

- preferences are complete and transitive,
- preferences are continuous,
- the feasible set is compact,

then:

- an optimal choice exists,
- and the choice correspondence is non-empty.

This result will be used repeatedly:

- in consumer theory,
- in mechanism design,
- in equilibrium analysis,
- and later in dynamic models.

4.12 Where this appears in Mas-Colell, Whinston, and Green (MWG)

The relevant material appears in:

[4] Note, that continuity rarely comes for free, and sometimes is explicitly not desired. But if we have it, we can use a lot of machinery to understand what is going on even if we can't describe a solution—the knowledge of its existence is enough.

- **MWG, Chapter 1** (preferences and continuity)
- **MWG, Chapter 3** (consumer optimization)
- **MWG, Mathematical Appendix A** (topological notions)
- **MWG, Mathematical Appendix C** (existence theorems)

The appendices give formal statements. The chapters show how these ideas power economic analysis.

4.13 Another economic example: non-existence of demand

Consider a consumer with utility $u(x) = x$ and feasible set $X = (0, 1)$.

Even though preferences are continuous and convex, demand is empty. Which of course does not make much economic sense. Often, if a model fails to produce existence, it is a bad model. It is not useful to think about situation where we don't know what is happening, and often they are not applicable to the real world in which things typically tend to happen. But, it is a useful example, because it explains why economists insist on (in their models):

- closed budget sets,
- local nonsatiation,
- and sometimes artificial bounds.

Existence is not automatic — it is engineered.

4.14 Why this matters for microeconomics

Much of microeconomic theory is concerned with **existence before uniqueness**.

Before we ask:

- how many equilibria exist,
- how they compare,
- how they change with parameters,

we must first ensure that *something exists at all*.

Continuity and closedness are the quiet assumptions that make the rest of the theory possible.

4.15 What you should take away

After this topic, you should be comfortable with:

- distinguishing open and closed sets,
- understanding why compactness matters,
- seeing continuity as an economic assumption,
- and recognizing when existence arguments apply.

These tools will soon be used implicitly. Knowing where they come from will help you spot when they fail.

Chapter 5

Topic 5: Optimization, Correspondences, and Existence of Solutions

(Why optimal behavior is rarely a point and often a set)

5.1 Why this topic sits where it does

By now we have introduced:

- sets and mappings,
- preferences and utility,
- convexity and geometry,
- continuity, closedness, compactness, and existence.

What we have *not* yet done is put all of this together in a single place.

This topic does exactly that. It explains how optimization problems in economics actually behave once we take all previous structure seriously.

The key message is simple but important:

> Even when optimization problems are well behaved, their solutions are often **sets**, not points.

Understanding this is essential before moving on to equilibrium, mechanisms, or games.

5.2 Optimization problems revisited

A generic economic optimization problem takes the form

$$\max_{x \in X} f(x),$$

where:

- $X \subseteq \mathbb{R}^n$ is a feasible set,
- $f : X \to \mathbb{R}$ is an objective function.

From Topics 3 and 4 we know that:

- convexity of X simplifies geometry,
- continuity of f and compactness of X guarantee existence.

What these assumptions do *not* guarantee is **uniqueness**.

5.3 Linear versus convex problems

An important special case is when:

- X is convex,
- f is **linear**.

In this case, if a maximum exists, it is always attained at an **extreme point** of X.

So the message is

> "Linear objectives on convex sets push solutions to the boundary."

Why is that the case? If a function is linear, that means if we walk a little bit in any given direction in X, then f changes the same, no matter where we start. In the 1-dimensional case this is obvious because linear functions have a constant slope. But then, whatever direction we take, this function either goes up or down everywhere (or is flat throughout). In each of these cases, picking a point at the boundary of X in that dimension is optimal. Because X is compact, that boundary exists.

This observation explains why:

- prices support extreme allocations,
- corner solutions are common,
- incentive constraints in mechanism design are linear.

However, in a more general world linearity is a knife-edge case.

5.4 Convex (concave) objectives and non-uniqueness

Now suppose instead:

- X is convex,
- f is strictly[1] **concave**.

Now what do we know then? As we walk along the slope diminishes. So if we are at a point where in all directions the slope (in the intuitive sense) goes down, then this is the unique hill top on our landscape, because concavity tells us we will never recover.

If we give up on strict concavity, there may be flat regions, so although every top is a global max, there may be plateau so the set of optima may not be a single point.

5.5 The argmax leads to a correspondence

Given an optimization problem, define the **argmax**

$$\text{argmax}_X\, f = \{x \in X \mid f(x) \geq f(x') \text{ for all } x' \in X\}.$$

This equation means: The $\arg\max$ of a function f is the set of all x in the choice set that maximize f.

The argmax is an important piece of notation, as often we are not interested in what the *value* of the best choice is, but what the actual *choice* is.[2] Often, however, f is not strictly concave (or even concave at all), so the $\arg\max$ may be more than a singleton for a given f. Thus the mapping $g : \mathcal{F} \to X$ which maps choice functions to their maximizers is in general a correspondence.

5.6 Why are correspondences unavoidable?

There are three distinct reasons why optimal behavior is typically set-valued:

1. **Indifference**
   Preferences may be flat along some directions.
2. **Linear constraints**
   Linear objectives on convex sets produce faces of maximizers.

[1] Strict concavity means Jensen's inequality is strict.

[2] And often we can determine it without the need to be specific about f, think of the linear case we had looked at before.

3. **Symmetry**
   Different choices may be payoff-equivalent.

Recognizing this early prevents confusion later when:

- demand is a correspondence,
- best responses are sets,
- equilibria are not unique.

5.7 Regularity of solution correspondences

Existence alone is not enough. We also want solution sets to behave *stably* when the problem changes.

Theory's purpose is to deliver guidance and predictions on how to think about a changing world. So we want to know what happens when institutions change, prices change, players' types change. All these things are embedded in f.

At least for small changes we want to know whether solutions change drastically, which is why we ask:

> "Does the **set of solutions** vary in a controlled way?"

In functions, continuity helps with that. We now need to understand what continuity means when facing correspondences.

5.8 Upper hemicontinuity (definition)

Let $F : X \rightrightarrows Y$ be a correspondence.

F is **upper hemicontinuous** at $x \in X$ if:

For every open set $V \supseteq F(x)$ there is an open set $U \subset X$ containing x

such that for every $u \in U$, $F(u) \subset V$.

That means: We take a "fuzzy approach" at the image $F(x)$—that is we consider some (open) set V in the space of the *codomain* that contains the whole image $F(x)$. Then it is possible for us to find *one* (open) set U in the space of the *domain* that also contains our starting point x and for all points in that set U the image lies in V.

The important conditions in this definition are: (i) this needs to hold for *any* open set V that contains the *whole* image, but we only need to find *one single* set U in the domain.

Practically, what upper hemicontinuity means is that there are no floating stairs in the graph. Suppose we look at the graph from afar with our binoculars and sharpen them in a way that we just cover the graph at a given point. Now, as we move a little bit to the left or right in the domain (as in, turn our head a little), we shouldn't suddenly lose the image out of sight because it, for example, explodes. As long as small head movements are possible, we're good on hemicontinuity.

If F is upper hemicontinuous at *any* x, then F in itself is upper hemicontinuous.

5.9 Lower hemicontinuity (definition)

The "no floating staircases" property is not the only thing that continuity in the classical function space implies. So, for correspondences we need another continuity notion.

Let $F : X \rightrightarrows Y$ be a correspondence.

F is **lower hemicontinuous** at $x \in X$ if:

For every open set $V \subseteq Y$ that contains some element of $F(x)$ there is an open set U containing x such that some element of $F(x')$ is in V for all $x' \in U$.

Again, we take our "fuzzy approach," but now we provide more focus. Instead of looking at a subset of the codomain that covers the whole image $F(x)$, we look at a subset that covers some part of the image. Now for *every* such subset (that means also those that contain very little of the image) we can find one (open) set U around our starting point x that makes sure that again *some* elements of the image $F(x')$ is also in V for any x'.

The important qualifier here is now that we look only at a set that contains *some element* of $F(x)$. So instead of making the focus of our binoculars close to the whole graph, we focus on one point in the graph. Now, if we turn our head a little, we may lose our graph because suddenly we look at nothing. So for lower hemicontinuity, we need that there is no sharp "cliff" in the set. For upper hemicontinuity, we need to exclude the floating stair.

5.10 Hemicontinuity taken together

Here are two graphs hopefully showing the difference

Upper hemicontinuous but not lower hemicontinuous

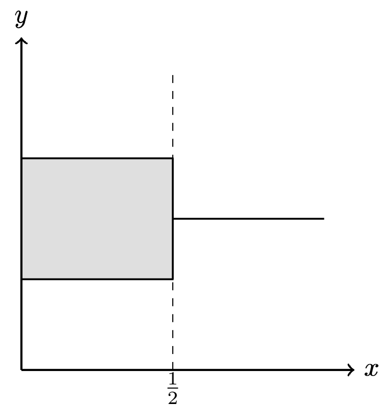

Figure 5.1: Upper hemicontinuous but not lower hemicontinuous

In this example, the correspondence is **upper hemicontinuous** at $x = 1/2$ because the image does not suddenly expand outward: every neighborhood of the image at $x = 1/2$ contains the images of nearby points. However, it is **not lower hemicontinuous**, since points in the vertical segment disappear immediately to the right of $1/2$.

Lower hemicontinuous but not upper hemicontinuous

In this example, the correspondence is **lower hemicontinuous** at $x = 1/2$ because points in the image can be approximated by points in the images of nearby x. However, it is **not upper hemicontinuous**, since the image suddenly expands at $x = 1/2$.

5.10.1 Closed Graphs

The *closed-graph property* is another concept that sometimes shows up, and there is some relation to upper hemi-continuity.

A correspondence $f : X \to Y$ is said to have a closed graph if the set $\{(x, y)|y \in f(x)\}$ is a closed subset of $X \times Y$.

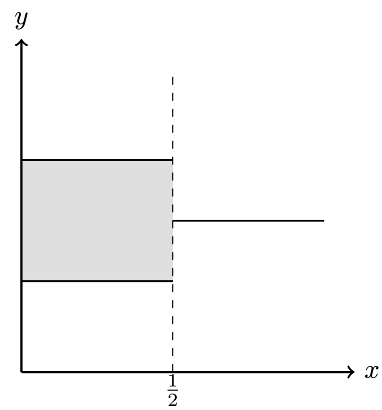

Figure 5.2: Lower hemicontinuous but not upper hemicontinuous

While upper-hemicontinuity and closed-graphs are not the same in general, the following is true

*If f is **compact-valued**[3], then f has a closed graph if and only if f is upper hemicontinuous.*

5.11 Berge's Maximum Theorem (economic reading)

Berge's Maximum Theorem connects all previous ideas.

Informally:

If:

- the objective function is continuous,
- the feasible set correspondence is compact-valued and upper hemicontinuous,

then:

- the argmax correspondence is non-empty (there is a solution),
- compact-valued,
- and upper hemicontinuous.

Or in short:

> "Well-behaved problems have well-behaved solution sets."

[3] f is compact valued if the image $f(x)$ is a compact set for all x

This theorem explains why demand, best responses, and equilibrium correspondences exist and vary regularly.

5.12 Economic example: demand as a correspondence

Consider the consumer problem

$$\max_{x \in B(p,w)} u(x),$$

where:

- u is continuous,
- $B(p, w)$ is the budget correspondence.

Under standard assumptions:

- $B(p, w)$ is compact-valued and upper hemicontinuous,
- the demand correspondence is non-empty,
- and it is upper hemicontinuous in (p, w).

Comparative statics rely on exactly this structure.

5.13 Why this prepares equilibrium analysis

Equilibrium concepts are defined as:

- fixed points of correspondences,
- intersections of best-response sets,
- solutions to systems of optimization problems.

Without the machinery developed here:

- existence arguments collapse,
- comparative statics break down,
- equilibrium becomes fragile.

This topic is the last step before equilibrium theory proper.

5.14 Where this appears in Mas-Colell, Whinston, and Green (MWG)

The core references are:

- **MWG, Chapter 3** (consumer optimization)
- **MWG, Chapter 5** (duality and linear structure)
- **MWG, Mathematical Appendix A** (correspondences)
- **MWG, Mathematical Appendix C** (Berge's theorem)

The technical tools introduced here are used repeatedly and often implicitly.

5.15 What you should take away

After this topic, you should be comfortable with:

- viewing optimization solutions as correspondences,
- understanding why non-uniqueness is normal,
- reading upper hemicontinuity statements,
- and seeing how existence and stability interact.

From here on, equilibrium and strategic interaction can be treated without apology.

Chapter 6

Topic 6: Differentiability and Marginal Reasoning

(Why economists care about slopes, not just optima)

6.1 Why differentiability enters now

Up to now, we have focused on two fundamental questions:

1. **Does an optimal choice exist?**
2. **What does the set of optimal choices look like?**

We now turn to a different question:

> **How do optimal choices change when the environment changes?**

This is the core question of **comparative statics**.

To answer it, economists often rely on *local approximations*:

- small changes in prices,
- small changes in income,
- small changes in parameters.

Differentiability is the mathematical structure that makes such reasoning precise.

6.2 From global behavior to local behavior

Recall the generic optimization problem

$$\max_{x \in X} f(x).$$

So far, our analysis has been **global**:

- existence relies on compactness,
- uniqueness relies on strict concavity,
- stability relies on hemicontinuity.

Differentiability allows us to zoom in and ask:

> What happens to $f(x)$ when we move x *a little bit*?

This shift — from global structure to local behavior — is what makes marginal analysis possible.

6.3 Differentiability: the basic idea

Let $f : \mathbb{R}^n \to \mathbb{R}$.

We say that f is **differentiable at** x if there exists a linear map[1]

$$\mathrm{D}f(x) \colon \mathbb{R}^n \to \mathbb{R}$$

such that

$$f(x+h) \approx f(x) + Df(x)[h]$$

for small vectors h.

What does it mean:

> $Df(x)[h]$ is the predicted change in f if I take a small step h away from x.

[1] A linear map is a function that has the following property: $L(ax + by) = aL(x) + bL(y)$ for any vectors x and y and scalars a and b

This definition, however, uses a vector h which is important. So it says, basically: If you give me a direction in which to walk, and (what we will later call) the *gradient* (the linear extrapolation of the change in the function value in our desired direction) at your starting point, then I can tell you where you end up in ''height'' **relative to your starting point** even without knowing the precise function or the starting point.

Important: Differentiability does *not* mean that the function is linear. It means that **locally**, it can be well approximated by something linear.

Sidenote: The relation $\approx$ is one that is used in various ways in mathematics always trying to mean something like the left hand side is close to the right hand side. You may correctly ask, ''what is the metric in which we should measure that closeness?' And in general I have no answer. In this particular case, the notion of closeness we have in mind behind this is that as we let $h \to 0$ the error we are making vanishes somewhat continuously. The more classical (and also more precise) definition of differentiability would be that $f : U \to \mathbb{R}$ is differentiable at $x \in U$ if

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

exists. However, the one we have works for multidimensions[2] and carries more economic meaning, and once we talk about it, the notion of $\approx$ we mean here, will be clear.

6.4 Partial derivatives and gradients

When f is differentiable, the linear map $Df(x)$ can be represented by a vector of **partial derivatives**

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right).$$

The gradient vector encodes:

- the direction of steepest ascent,
- the marginal effect of each component.

What is a partial derivative?

> The partial derivative with respect to x_i measures how the objective changes when we increase x_i slightly, holding all other components fixed.

[2] Look up Frechet derivatives for the more precise definition of the unidimensional one I have given in this sidenote

> The gradient collects all partial derivatives, that is the changes in each direction.

From the gradient we know how we would walk on the landscape given by the function in each direction by making *linear combinations* of the partial derivatives.

The ‘’holding everything else fixed’ ’ language the gradient encapsulates is fundamental in economic reasoning (we often call it ceteris paribus or c.p. reasoning).

6.5 Marginal interpretation in economics

Marginal reasoning appears everywhere in microeconomics:

- **Marginal utility**: how utility changes with one more unit of a good,
- **Marginal cost**: how cost changes with one more unit of output,
- **Marginal willingness to pay**: how much an agent is willing to give up for a small increase.

All of these are formalized using partial derivatives. In fact, often that is a shortcut we take because differentiability implies continuity which means that ‘small changes’ are possible in the first place. As economists we have decided to often assume continuity, because it makes the statements clearer and the error in translating back to economics is not too big. However, one has to, of course, always be vigilant whether such an assumption that only serves the economist, is costly in terms of the interpretation.

Why is it great to have differentiability? It gives us a disciplined way to talk about ‘’small changes’ ’ without ambiguity.

6.6 First-order conditions (interior solutions)

Suppose:

- $X \subseteq \mathbb{R}^n$ is open,
- f is differentiable,
- x^* is an interior maximizer.

Then a necessary condition for optimality is

$$\nabla f(x^*) = 0.$$

What this means:

> If the function has our differentiability properties, then at an optimum, we must be in a (locally) flat area.

The reason is (hopefully) clear: If we are at the hill top, anywhere we go, we must go down. However, any path that *crosses* the hilltop from somewhere lower to somewhere lower must first go up and then down again. Differentiability gives us that everywhere we can 'think linearly' if our next step is small enough. But then, it must be that when switching from going uphill to downhill, we eventually pass a part where the linear approximation is flat (even if only for very short). And this passing must happen exactly at the hill top.

The condition above is called a **first-order condition**.

It does *not* guarantee optimality by itself.
It identifies **candidates** for optimality (so called critical points).

So if we are looking for an optimum in a smooth problem like the one described above, it is *first order* for us to find points that satisfy that condition. Once we have those, we can worry about which of those to eliminate.

6.7 Why first-order conditions are not enough

A point where $\nabla f(x) = 0$ could be:

- a maximum,
- a minimum,
- a saddle point.

A Maximum is what we want, a hill top. A minimum is the lowest point in the valley and the opposite of what we want. A saddle point is something tricky. It is the lowest point of a valley in some direction, but the hill top in another. So standing on the saddle point, means it goes down in some direction, and up in others (like at that one point at the back of a saddle), in some it may be even flat. It is, however, clearly not a hill top.

Moreover, we need to distinguish between local and global optimality. If we are on a hill top in a mountain range, that means we are at a maximum (it goes down in all directions, locally), but nothing says that this is the highest hill top.

The first-order condition tells us nothing about these things.

This is why first-order conditions must be combined with **global structure**:

- concavity of the objective ensures that any critical point is a global maximum,
- differentiability ensures that no jumps or kinks occur that bring about points that can trump the critical points without being critical points themselves

- convexity of the choice set ensures that no local traps exist where we are not allowed to go to.

This connects directly back to Topics 3 and 4.

6.8 Second-order intuition (informal)

Second-order information captures **curvature**.

If f is twice differentiable, curvature is summarized by the **Hessian matrix**

$$D^2 f(x).$$

We will not work with Hessians explicitly here, but with help of the Hessian one can verify optimality of a critical point (yet often this is not very elegant).

The key intuition is:

> Concavity means the function bends downward everywhere.

Under global concavity of the objective:

- first-order conditions become sufficient,
- local optima are global.

6.9 Differentiability and constraints (preview)

Most economic problems involve constraints:

$$\max_{x \in X} f(x).$$

When constraints bind, optimality conditions involve:

- trade-offs,
- shadow prices,
- Lagrange multipliers.

We will introduce this machinery later.

For now, the key message is:

> Differentiability allows us to express trade-offs precisely.

6.10 An economic example: marginal utility

Consider a utility function $u(x_1, x_2)$.

The **partial derivative**

$$\frac{\partial u}{\partial x_1}(x)$$

measures the marginal utility of good 1 at bundle x.[3]

Comparing marginal utilities across goods explains:

- substitution,
- demand responses,
- interior versus corner solutions.

Without differentiability, these ideas lose precision.

There is another form of differentiation, which is the **total derivative**. The way to think about this is to think about the partial derivative as exactly what it is, the change in one direction of the choice vector. So in the above example, $\frac{\partial u}{\partial x_1}(x)$ means we keep x_2 unchanged and only change x_1 (marginally). But of course, there are many ways to change things "slightly." We could walk in direction 1, in direction 2 or any weighted combination of it. The *total derivative* tells us what that means for our function. And it is (because of the linear map we defined in the beginning) a remarkably simple object. So here it is[4]

$$\mathrm{d}u(x) = \frac{\partial u}{\partial x_1}(x)\mathrm{d}x_1 + \frac{\partial u}{\partial x_2}(x)\mathrm{d}x_2$$

The d here is a way to think about small changes which (however) have a relative size. So if we go in the x_1 direction only, then $\mathrm{d}x_2 = 0$ is 0 and $\mathrm{d}x_1 = 1$.[5]

6.11 Why this matters for comparative statics

Differentiability underlies a bunch of concepts we will see in Microeconomics:

- Slutsky decompositions,
- envelope theorems,
- local comparative statics.

It allows us to replace

[3] These are the standard derivatives we know from undergraduates and school.

[4] This is just spelling out the definition of derivative!!

[5] Important: While $\mathrm{d}x_1$ is an object we can move around (multiply by $\mathrm{d}x_1$) for example, ∂x_1 is **not,** instead, here the object is the entire $\frac{\partial u}{\partial x_1}(x)$.

''What happens when something changes?''

with

''What is the derivative with respect to that change?''

This is a powerful simplification — but one that rests on strong assumptions.

6.12 Where this appears in Mas-Colell, Whinston, and Green (MWG)

The main references are:

- **MWG, Chapter 3** (consumer theory)
- **MWG, Mathematical Appendix B** (differentiation)
- **MWG, Chapter 5** (duality and marginal conditions)

Differentiability is often used implicitly; these notes make the logic explicit.

6.13 What you should take away

After this topic, you should be comfortable with:

- interpreting derivatives as local approximations,
- reading gradients as vectors of marginal effects,
- understanding when first-order conditions apply,
- seeing how differentiability complements convexity.

Differentiability does not replace global reasoning.
It refines it.

Chapter 7

Topic 7: Order, Lattices, and Monotonicity

(How to do comparative statics without calculus)

7.1 Why order shows up in economic theory

In economics, we care about how changes in the environment change outcomes. Because we aim to work on understanding human behavior, we care about how (equilibrium) choices change when the environment changes. The simplest way to say something here is what we call a comparative static, that is an answer to this question:

> "If an exogenous parameter goes up, does the optimal choice also go up?"

Examples:

- If income rises, does the consumer buy more of a good?
- If a bonus becomes more powerful, does effort increase?
- If one firm becomes more aggressive, do other firms respond by becoming more aggressive too?

Now as long as things are smooth enough, there is a straightforward way to deal with these questions (it still may not be an easy one, of course). We will briefly discuss that and then move on to the more general case which often is, unfortunately, also the more relevant one.

7.2 Excursion: The Implicit Function Theorem

Suppose we live in a perfectly smooth world. Everything is as differentiable and concave as we want it to be. Moreover, for the sake of this argument, let's assume we are working on a single dimension. So, what we are looking at is a function $f(x;\theta)$ where x is our variable of choice, and θ is a parameter. Now, we know the necessary and sufficient condition for an optimum is that

$$f'(x;\theta) := \frac{\partial f(x;\theta)}{\partial x} = 0.$$

Solving this for x gives us our optimum x^*. Now if we change θ, that optimum is likely to change, but[1] we know that even under another θ' it must be true, that x'^* solves

$$f'(x'^*;\theta') = 0.$$

Knowing this, means, we know also how the first order condition needs to change if we change θ slightly, simply by using the fact that although x is our choice and θ is a parameter (if you like the *state* of the problem), mathematically f is a function of x and θ. Moreover, we can (in principle at least) describe $x(\theta)$ as the choice function that describes how the optimal choice changes in θ.

So in our example problem $x(\theta) = x^*$ and $x(\theta') = x'^*$. Hopefully, you now see where this is going. The object we want to understand fundamentally is $\partial x(\theta)/\partial\theta$ or in words, how does our choice x change if the environment changes slightly.[2]

One complication could be that if f is complicated, explicitly describing $x(\theta)$ in *closed form*[3] may be difficult.[4]

But, our knowledge, that x (whatever it is) has to solve the first-order condition tells us *implicitly* what x is, and more relevantly, it tells us *how it changes in* θ. Why the latter?

Well, think about the function $f(x(\theta);\theta)$. For that function, we know that

$$f'(x(\theta);\theta) = 0$$

So if $\hat{x} := x(\hat{\theta})$ for some $\hat{\theta}$, then we have $f'(\hat{x},\hat{\theta}) = 0$ and as long as we keep it this way, this is always true. Then, for small changes $\mathrm{d}\theta$, that means[5]

[1] still assuming the first-order conditions are necessary and sufficient in our problem

[2] For example, if unemployment benefit (θ) goes up, how does the labor supply respond (does it go up? How much does it go up?).

[3] That means as an actual function.

[4] For example, think of a first order condition $xe^{\theta x} - 3 = 0$. There is no closed-form in x.

[5] Important: Note we are taking derivatives of the *derivative* here! Not of the original function!

$$\mathrm{d}f'(x;\theta) = \left(\frac{\partial f'(x;\theta)}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f'(x;\theta)}{\partial \theta}\right)\mathrm{d}\theta = 0$$

if we want to ensure that we stay on the "solution" part in x.[6]

But this can only be true if the part in the large brackets is 0, which in turn means that

$$\frac{\partial x}{\partial \theta} = -\frac{\frac{\partial f'(x;\theta)}{\partial \theta}}{\frac{\partial f'(x;\theta)}{\partial x}}.$$

This last relation is called the *implicit function theorem* as it tells us that just knowing the first order equation, is enough to understand how the optimal choice, x, changes in the parameter θ.

In the example where the first order condition is

$$xe^{\theta x} - 3 = 0$$

we know there is no closed form solution $x(\theta)$,[7] yet we can solve

$$\frac{\partial x(\theta)}{\partial \theta} = -\frac{(x(\theta))^2}{1 + \theta x(\theta)}$$

via the implicit function theorem.

7.3 Back to general problems

The above works nicely under great simplifying assumptions, but they can fail for reasons we have already seen:

- the choice may be multidimensional,
- the solution may be a set, not a point,
- the objective may not be smooth (kinks are common),
- the action space may be discrete ("choose a technology" is not a calculus problem).

[6] We implicitly assume here that $\partial f'(x;\theta)/\partial x \neq 0$ (otherwise we divide by zero). But recall, that for a (strictly) concave function that is no problem.

[7] You can solve it with the quasi-closed-form usage of the Lambert-W. We will not get into those details.

The order-theoretic approach we will now discuss is the alternative. It is designed for exactly the situations where you do not want (or cannot) to lean on calculus.

The punchline of this topic is:

> If you can express your model using an order ("higher" versus "lower") and you can show a certain kind of complementarity, then monotonicity results fall out cleanly.

7.4 Partially ordered sets

Let X be a set. A binary relation $\preceq$ on X is a subset of $X \times X$ (Topic 2 already used this viewpoint for preferences).

We say that $(X, \preceq)$ is a **partially ordered set** (a "poset") if $\preceq$ satisfies:

1. **Reflexivity:** for every $x \in X$,

$$x \preceq x.$$

   Every element is weakly below itself.

2. **Antisymmetry:** for all $x, y \in X$,

$$x \preceq y \text{ and } y \preceq x \;\Rightarrow\; x = y.$$

   If each is weakly below the other, then they are the same element.

3. **Transitivity:** for all $x, y, z \in X$,

$$x \preceq y \text{ and } y \preceq z \;\Rightarrow\; x \preceq z.$$

   If x is below y and y is below z, then x is below z.

A partial order is "partial" because not every pair can be compared under $\preceq$.

7.4.1 Total orders

A poset is a **total order** (sometimes called "linear order") if any two elements can be compared: for all $x, y \in X$,

$$x \preceq y \quad \text{or} \quad y \preceq x.$$

The real line $(\mathbb{R}, \leq)$ is a total order.

7.5 Two orders economists use all the time

7.5.1 The usual order on $\mathbb{R}$

This is the one you already know: $x \leq y$.

7.5.2 The product order on $\mathbb{R}^n$

In consumer problems, choices are usually vectors. So we need an order for vectors.

For $x, y \in \mathbb{R}^n$, define

$$x \preceq y \quad \iff \quad x_i \leq y_i \text{ for all } i = 1, \ldots, n.$$

"y is higher than x if it is at least as high in every coordinate.

This is a partial order, not a total order. For instance, in $\mathbb{R}^2$, $(1, 0)$ and $(0, 1)$ are not comparable under $\preceq$.

Economically, the product order is the natural order when "more of every component" is interpreted as "bigger".

7.6 Monotone maps

Let $(X, \preceq_X)$ and $(Y, \preceq_Y)$ be posets.

A function $g : X \to Y$ is **monotone increasing** if

$$x \preceq_X x' \;\Rightarrow\; g(x) \preceq_Y g(x').$$

"If you increase the input, the output does not go down."

A function g is **monotone decreasing** if

$$x \preceq_X x' \ \Rightarrow \ g(x') \preceq_Y g(x).$$

These definitions are deliberately minimal. They do not require differentiability, continuity, or convexity. They only require that the words "higher" and "lower" make sense.

7.7 Lattices: when any two points have a meet and a join

A poset $(X, \preceq)$ is a **lattice** if for any two elements $x, y \in X$ there exist:

- a **join** (least upper bound), written $x \vee y$, and
- a **meet** (greatest lower bound), written $x \wedge y$,

such that:

1. $x \preceq x \vee y$ and $y \preceq x \vee y$ (it is an upper bound), and if z is any upper bound of both x and y, then $x \vee y \preceq z$ (it is the least upper bound);
2. $x \wedge y \preceq x$ and $x \wedge y \preceq y$ (it is a lower bound), and if z is any lower bound of both x and y, then $z \preceq x \wedge y$ (it is the greatest lower bound).

In English:

- $x \vee y$ (the least upper bound) is "the smallest element that is above both x and y,"
- $x \wedge y$ (the greatest lower bound) is "the largest element that is below both x and y."

What does it mean, practically? It means in particular, that any element z in X that is (under our order) larger than both x and y, then it has to be ranked with (and larger than) a single, defined *join*. And analogously for the meet. Important, this does *not* imply a total order, as we will see in this (often relevant) example.

7.7.1 Example: $\mathbb{R}^n$ with the product order

Under the product order on $\mathbb{R}^n$,

$$(x \vee y)_i = \max\{x_i, y_i\}, \qquad (x \wedge y)_i = \min\{x_i, y_i\}.$$

So the n–dimensional Euclidean space under the product order is indeed a lattice.

7.8 Complete lattices (and why you might care later)

A lattice $(X, \preceq)$ is a **complete lattice** if every subset $A \subseteq X$ has:

- a least upper bound, and
- a greatest lower bound.

"A lattice is complete if we not only can take joins and meets of two points; but we can take them for any collection of points."

Why this matters (preview): complete lattices are the right environment for fixed-point results like **Tarski's fixed-point theorem**, which is a workhorse in monotone equilibrium arguments.

In terms of notation, we call the least upper bound of a set its *supremum* and the greatest lower bound its *infimum.* **Important:** As in the two-point case, there is nothing that says these bounds need to lie *in the subset*, it just says that any point that we can compare to the elements of the set, we can also compare to those, and the order is in the right way.

7.9 Increasing differences: the core monotonicity condition

Now we move from ordering choices to ordering how incentives change.

Let X be a poset (often a lattice) and let T be a poset of parameters (often $T = \mathbb{R}$ with $\leq$).

Consider an objective function

$$f : X \times T \to \mathbb{R}.$$

We say that f has **increasing differences** in (x, t) if for all $x' \succeq x$ and $t' \succeq t$,

$$f(x', t') - f(x, t') \geq f(x', t) - f(x, t).$$

In English:

> "f has increasing differences if, when the parameter t is higher, increasing x becomes (weakly) more attractive."

There is a very concrete way to think about it:

- fix two actions $x \preceq x'$,
- compare the advantage of choosing x' rather than x,
- increasing differences says that this advantage does not fall when t rises.

7.9.1 A quick differentiable sanity check (only as intuition)

If X and T are intervals in $\mathbb{R}$ and f is twice differentiable, increasing differences corresponds to a nonnegative cross-partial:

$$\frac{\partial^2 f}{\partial x \, \partial t}(x, t) \geq 0.$$

This is not the definition, but it is a useful mental picture: cross-partials capture complementarity.

7.10 Supermodularity: complementarity inside the choice vector

Increasing differences is about complementarity between a choice x and a parameter t.

Supermodularity is about complementarity between components of the choice vector itself.

Let X be a lattice and $f : X \to \mathbb{R}$.

We say that f is **supermodular** if for all $x, y \in X$,

$$f(x \vee y) + f(x \wedge y) \geq f(x) + f(y).$$

In English:

"If I take the coordinatewise higher bundle and the coordinatewise lower bundle, their total value is at least as large as the total value of the original two bundles."

This inequality encodes "components like to move together." In $\mathbb{R}^n$, it is closely related to nonnegative cross-effects (or cross-partial derivatives).

(Again: cross-partials are the differentiable picture, not the definition.)

7.11 Topkis' theorem: monotone comparative statics without derivatives

We now connect this back to Topic 5, where we treated optimal choices as correspondences.

Let X be a lattice, T a poset, and let $f : X \times T \to \mathbb{R}$.

For each $t \in T$, define the argmax correspondence

$$X^*(t) = \mathrm{argmax}_{x \in X}\, f(x, t).$$

Topkis' monotonicity theorem (one standard version) says:

If (i) $f(\cdot, t)$ is supermodular on X for each t, and (ii) f has increasing differences in (x, t), then the set of maximizers moves upward with t in the following sense:

- If $t' \succeq t$, and $x \in X^*(t)$, $x' \in X^*(t')$, then there exist maximizers $\underline{x} \in X^*(t)$ and $\overline{x} \in X^*(t')$ such that

$$\underline{x} \preceq \overline{x}.$$

In English:

> "Higher parameters do not force optimal choices to move down. Even if the optimizer is not unique, you can choose maximizers consistently so that they move upward."

This is why the lattice approach is so useful: you can get monotone comparative statics even when:

- the solution is set-valued,
- the objective is not differentiable,
- you do not want to solve the model explicitly.

7.12 A single-agent economic example: effort choice with a bonus parameter

An agent chooses effort e from an interval $X = [0, 1]$ (ordered by $\leq$). A principal chooses a bonus level $b \in T = \mathbb{R}_+$.

Suppose the agent's payoff is

$$f(e, b) = b \cdot e - c(e),$$

where c is increasing and convex.

Interpretation:

- $b \cdot e$ is the marginal reward from effort,
- $c(e)$ is the cost.

Now check increasing differences.

Take $e' > e$ and $b' > b$. Then

$$f(e', b') - f(e, b') = b'(e' - e), \qquad f(e', b) - f(e, b) = b(e' - e).$$

Since $b' > b$ and $e' - e > 0$, we have

$$b'(e' - e) \geq b(e' - e),$$

so f has increasing differences in (e, b).

What it means:

> "A higher bonus makes higher effort relatively more attractive."

So the optimal effort correspondence $E^*(b) = \operatorname{argmax}_{e \in [0,1]} f(e, b)$ is (weakly) increasing in b.

Notice what we did not use:

- no first-order conditions,
- no differentiation of a solution function.

We only used the order structure and a simple algebraic inequality.

7.13 Outlook: why this matters for micro (and for games later)

Order and lattice methods matter because they let you prove robust qualitative statements that survive model variations.

In microeconomics, they show up in at least three places:

1. **Monotone comparative statics in optimization**
   This is what we did: higher parameters lead to higher optimal choices under complementarity.
2. **Strategic complements in games**
   When players' best responses are increasing, equilibrium sets often have lattice structure. This is one route to existence and to sharp predictions about how equilibrium moves with parameters.
3. **Mechanism design and auctions**
   "Single-crossing" and monotonicity of optimal reports are close cousins of increasing differences. They explain why allocation rules tend to be monotone in types.

So this topic is not just a mathematical curiosity. It is the vocabulary behind a large family of economically meaningful monotonicity results.

7.14 Where this appears in MWG (and where it does not)

- The language of relations and orders is closest in spirit to **MWG, Mathematical Appendix A** (relations and basic set-theoretic language).
- The lattice-theoretic approach to monotone comparative statics (increasing differences, supermodularity, Topkis' theorem) is not developed systematically in MWG's mathematical appendices.

For this topic, MWG is best treated as background micro motivation, while lattice methods are typically learned from dedicated comparative statics or game-theory notes.

7.15 What you should take away

After this topic, you should be comfortable with:

- reading statements like $x \preceq x'$ and understanding what order is being used,

- recognizing $\mathbb{R}^n$ with the product order as the natural "more in every dimension" comparison,
- understanding increasing differences as "higher parameters strengthen incentives to choose higher actions,"
- seeing why monotone comparative statics is compatible with set-valued argmax correspondences.

Once this is in place, supermodular games and monotone equilibrium arguments will feel like a continuation rather than a new language.

Chapter 8

Topic 8: Probability and Expectations

(How economists talk about uncertainty, types, and randomization)

8.1 Why probability enters now

Up to now, the pre-course has built a toolkit for:

- sets and mappings,
- preferences and utility,
- convexity and geometry,
- continuity, closedness, and existence,
- optimization and correspondences,
- differentiability and marginal reasoning,
- order, lattices, and monotone comparative statics.

All of these tools were developed in a world of certainty. The agent knows her choice set, she knows her utility function, and she knows the parameters of the problem. The only question was: given all this information, what does she choose, and how does that choice change when the environment changes?

But a large part of microeconomics does not live in such a world.

Agents often face:

- uncertainty about payoffs (will this investment pay off?),
- uncertainty about other agents' characteristics (what is the other bidder's valuation?),
- uncertainty about signals (what does this test result mean?),

- or deliberate randomization (a player who mixes between strategies).

So before we can talk cleanly about auctions, Bayesian games, or mixed strategies, we need a minimal probability language.

The point of this topic is not to turn you into a probability theorist. It is to make sure that when economists write "expected payoff" or "distribution over types," those words are no longer mysterious.

8.2 The state space: what might happen

Start with the most primitive object: the set of things that could happen.

We call this set Ω and refer to it as the **state space**. Each element $\omega \in \Omega$ is a **state of the world** – a complete description of everything that is uncertain.

In economics, the state space might describe:

- which of several possible cost realizations occurs,
- which type each player in a game has,
- which signal each agent observes.

> "Ω is the set of all possible states of the world. An element ω is one particular resolution of uncertainty."

Mathematically, this is just a set like any other. It becomes interesting only once we start measuring how likely different parts of it are.

8.3 Events and probabilities

An **event** is a subset of the state space: $A \subseteq \Omega$.

> "An event is a collection of states. The event A occurs if the realized state ω happens to lie in A."

A **probability** is a rule that assigns a number to events in a coherent way. Formally, a probability (often written Pr or P) satisfies three properties:

1. **Nonnegativity**: for every event A,

$$\Pr(A) \geq 0.$$

2. **Normalization**: the probability that *something* happens is one,

$$\Pr(\Omega) = 1.$$

3. **Additivity**: if A and B are disjoint events (they share no states), then

$$\Pr(A \cup B) = \Pr(A) + \Pr(B).$$

> "Probability is a coherent way of assigning weights to events. The weights are nonnegative, the total is one, and non-overlapping events add up."

These three properties are sometimes called the **Kolmogorov axioms**. They are the foundation on which everything else rests.

Note that we said nothing about *where* probabilities come from. In some models they represent objective frequencies (how often something happens if we repeat the experiment). In others they represent subjective beliefs (how confident the agent is). The mathematics is the same either way. Microeconomics uses both interpretations depending on the context.

8.4 Random variables

A **random variable** is a mapping from states of the world into numbers:

$$X : \Omega \to \mathbb{R}.$$

> "A random variable turns uncertainty about states into uncertainty about a number."

This is just a function in the sense of Topic 1. The domain is the state space Ω, the codomain is $\mathbb{R}$. What makes it "random" is that we do not know which ω will occur, and therefore we do not know what value $X(\omega)$ will take.

In economics, random variables represent objects like:

- a consumer's income (uncertain because it depends on the state of the economy),
- a bidder's valuation in an auction (uncertain because it is private information),
- a firm's cost (uncertain because of technological shocks),
- a player's type in a game of incomplete information.

Why does this matter? Because once uncertainty is encoded as a random variable, we can use all the tools of analysis – integration, optimization, comparison – to study it systematically.

8.5 Distributions

Knowing that X is a random variable is not enough. We also need to know *how likely* different values of X are. This is what a **distribution** tells us.

8.5.1 Discrete distributions

If X takes finitely many values $x_1, \ldots, x_n$, the distribution is simply a list of probabilities:

$$p_i = \Pr(X = x_i), \quad i = 1, \ldots, n.$$

The probabilities are nonnegative and sum to one:

$$\sum_{i=1}^{n} p_i = 1.$$

This is the simplest case. It covers situations like: a firm faces two possible cost levels, or a buyer's type is drawn from a finite set.

8.5.2 Continuous distributions and the CDF

In many economic models, the random variable takes a continuum of values. We do that because it makes our lives often much much easier. In that case, the probability that X equals any single value is typically zero, so we need a different way to describe the distribution.

The **cumulative distribution function** (CDF) is defined as

$$F(x) = \Pr(X \leq x).$$

> "The CDF at x is the probability that the random variable takes a value no larger than x."

Properties of the CDF:

- F is nondecreasing (higher x means weakly more probability accumulated),
- $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to+\infty} F(x) = 1$,
- F is right-continuous.

If the CDF is differentiable, we can define the **probability density function** (PDF) as

$$f(x) = F'(x).$$

The density tells us how "concentrated" probability is near x. A higher density means more probability mass near that value.

Economists often write things like "let v be drawn from F" or "types are distributed according to F." This simply means: v is a random variable with cumulative distribution function F.

8.6 Expectation: the probability-weighted average

The **expectation** (or **expected value**) of a random variable is its probability-weighted average.

For a discrete random variable with values $x_1, \ldots, x_n$ and probabilities $p_1, \ldots, p_n$,

$$\mathbb{E}[X] = \sum_{i=1}^{n} p_i \, x_i.$$

For a continuous random variable with density f,[1]

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \, f(x) \, \mathrm{d}x.$$

> "Expectation is the weighted average value you would get if the uncertainty were resolved many times."

8.6.1 Why economists care about expectation

Expectation is the central bridge between probability and optimization. Whenever an agent faces uncertainty and must choose *before* the uncertainty resolves, the standard approach in economics is to evaluate choices by their **expected payoff**.

This appears as:

- **expected utility** in consumer theory under risk,
- **expected profit** in a firm's production decision,
- **expected revenue** in auction design,
- **expected continuation payoff** in dynamic games.

8.6.2 Linearity of expectation

One of the most useful properties of expectation is **linearity**:

$$\mathbb{E}[aX + bY] = a \, \mathbb{E}[X] + b \, \mathbb{E}[Y]$$

for any constants a, b and random variables X, Y.

> "The expected value of a sum is the sum of the expected values, regardless of whether the random variables are related."

[1] I usually claim that integrals $\int$ are nothing but a fast way of writing a sum $\sum$. Although mathematically there is a difference and the laws are a bit different, this intuition is often good for *economic* purposes. In reality people have a hard time thinking in terms of continuous state spaces, so even if they were they behave as if things were discrete. If you result relies on one of the tricks where the integral behaves different from the sum, there may be an issue in what you are modeling.

This property holds even when X and Y are not independent. It is one of the reasons expectation is so tractable.

8.7 Conditional probability and Bayes' rule

Sometimes new information arrives. A bidder observes a signal. A firm learns a partial cost estimate. An agent sees another agent's action.

When information arrives, beliefs must be updated. This is where **conditional probability** enters.

8.7.1 Conditional probability

The probability of an event A conditional on observing that event B has occurred is

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)},$$

provided $\Pr(B) > 0$.

In English the above equation says:

> "The conditional probability of A given B is the probability that both A and B occur, divided by the probability that B occurs."

Think of it as zooming in on the world where B happened and asking: within that world, how likely is A?

8.7.2 Bayes' rule

Bayes' rule is a way to represent the previous equation as a relation between conditional probabilities. It thus follow directly from the above rule

$$\Pr(A \mid B) = \frac{\Pr(B \mid A)\ \Pr(A)}{\Pr(B)}.$$

> "Bayes' rule says that when we observe B, the probability of A is the probability that B has occurred *because* A is true, times the relative liklihod of A occurring in the first place, all divided by the overall probability of B."

This formula is the formal engine behind belief updating in microeconomics. The components have names:

- $\Pr(A)$ is the **prior** – what we believed before seeing B,
- $\Pr(B \mid A)$ is the **likelihood** – how probable the evidence B is if A were true,
- $\Pr(A \mid B)$ is the **posterior** – what we believe after seeing B.

Bayes' rule is used everywhere in information economics. Whenever you read "the agent updates beliefs," the formal content behind that phrase is Bayes' rule.

8.8 Independence

Two events A and B are **independent** if

$$\Pr(A \cap B) = \Pr(A) \cdot \Pr(B).$$

So the probability that both A and B occur is just the product of their individual probabilities.

Two random variables X and Y are **independent** if learning the value of one tells you nothing about the other. Formally, this means that for all values x and y,

$$\Pr(X \leq x \text{ and } Y \leq y) = \Pr(X \leq x) \cdot \Pr(Y \leq y).$$

> "Independence means that observing one thing does not change your beliefs about the other."

Independence is a simplifying assumption that appears constantly in economic models. For instance:

- in auctions, bidders' valuations are often assumed to be independently drawn,
- in mechanism design, agents' types are frequently independent,
- in repeated games, stage-game shocks may be independent across periods.

Important: Independence is substantive. It is not a default. In reality, everything is connected with everything and many economically interesting situations involve correlation (e.g., common-value auctions, correlated types). When independence is assumed, it should be recognized as a modeling choice, not a law of nature.

8.9 Conditional expectation

When we combine expectation with conditional probability, we get **conditional expectation**:

$$\mathbb{E}[X \mid B] = \sum_i x_i \Pr(X = x_i \mid B)$$

in the discrete case.

> "The conditional expectation of X given B is the expected value of X knowing that B is true."

In economics, conditional expectations appear naturally whenever agents form beliefs:

- a buyer's expected valuation given a signal,
- a principal's expected output given the agent's reported type,
- a player's expected payoff given the other players' strategies.

A useful property that connects conditional and unconditional expectation is the **law of iterated expectations** (sometimes called the law of total expectation):

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big].$$

> "If I first compute the expected value of X for each possible value of Y, and then average those conditional expectations over the distribution of Y, I recover the unconditional expectation of X."

This property is extremely useful in sequential models where information is revealed gradually.

It is also super important for consistency and tells us one way to think about distributions (that will show up in other settings, too). We could think of a larger sate space Ω that has two dimensional elements (x, y) and a probability distribution over those (x, y). Now, we are in the end only interested in x, but y carries some information about x.

So, if we learn y, we update about x. But: it has to be the case that if we don't know y yet, we cannot learn anything about x, so our expectations on x are just $E[X]$. But, we do know how likely all thedifferent y will show up and we know what we would do then in our beliefs about x, so we do know all $E[X|Y]$. But if we are consistent, then this information does not tell us anything about x yet (because we haven't seen y yet), so we have to have that $E[X] = E[E[X|Y]]$.

8.10 An economic example: first-price auction with private values

Consider two bidders in a sealed-bid first-price auction. Each bidder has a private valuation v_i drawn independently from a distribution F on $[0, 1]$. Each bidder submits a bid b_i, the highest bid wins, and the winner pays her bid.

Bidder 1's expected payoff from bidding b when her valuation is v and the other bidder uses a bidding strategy $\beta(\cdot)$ is

$$\mathbb{E}[\text{payoff}] = (v - b) \cdot \Pr(\text{win}).$$

The probability of winning depends on the other bidder's bid, which depends on her valuation, which is random. So

$$\Pr(\text{win}) = \Pr(\beta(v_2) < b) = \Pr(v_2 < \beta^{-1}(b)) = F(\beta^{-1}(b)),$$

where the second equality uses the fact that β is (typically) increasing.

This example uses almost everything from this topic:

- a random variable (v_2, the other bidder's type),
- a distribution (F),
- expectation (the expected payoff),
- independence (valuations are independently drawn).

Without the probability language, writing down and solving this problem would be impossible.

8.11 Where this appears in Mas-Colell, Whinston, and Green (MWG)

Probability is used throughout MWG but is not developed as a standalone mathematical appendix topic in the same way as convexity or correspondences.

The most relevant places where probability language is used heavily include:

- **MWG, Chapter 6** (choice under uncertainty, expected utility),
- **MWG, Chapter 8** (competitive markets with uncertainty),
- **MWG, Chapter 13** (adverse selection),
- **MWG, Chapter 23** (mechanism design and auctions).

8.12 Why this matters for microeconomics

Probability gives us the language to treat uncertainty as part of the model rather than as narrative.

Without it:

- expected utility cannot be defined,
- Bayesian equilibrium cannot be formulated,
- mixed strategies appear as magic rather than precise objects,
- information economics loses its formal language,
- mechanism design under incomplete information becomes inaccessible.

The tools in this topic are modest – we have not touched measure theory, stochastic processes, or advanced convergence results. But they are exactly what is needed to read the core microeconomics curriculum without stumbling over probabilistic language. And they already get us a long way into the heart of modern microeconomic theory.

8.13 What you should take away

After this topic, you should be comfortable with:

- reading statements about random variables and distributions,
- computing and interpreting expectations as weighted averages,
- using conditional probability and Bayes' rule to update beliefs,
- recognizing independence as a substantive modeling assumption,
- and seeing why uncertainty is unavoidable in modern microeconomics.

You should also see how this topic connects back to earlier material: random variables are functions (Topic 1), distributions live on sets (Topic 1), expected utility combines preferences (Topic 2) with probability, and optimization under uncertainty (Topics 4 and 5) evaluates choices by expected payoffs.

Chapter 9

Topic 9: Fixed Points and Equilibrium Existence

(Why equilibrium is a mathematical object, not a metaphor)

9.1 Why fixed points enter now

The pre-course has so far given us the tools to talk about:

- how agents choose (optimization, Topics 2–6),
- how those choices depend on the environment (comparative statics, Topics 6–7),
- how to deal with uncertainty (probability, Topic 8).

But microeconomics is not just about a single agent choosing in isolation. Most of the interesting questions involve **interaction**: consumers react to prices, firms react to competitors, bidders react to other bidders' strategies.

In such settings, the fundamental question is no longer "what does the agent choose?" but rather:

> "Is there a situation where everyone's behavior is mutually consistent?"

That situation is what economists call an **equilibrium**. And mathematically, an equilibrium is almost always a **fixed point**.

This topic introduces fixed points as the formal backbone of equilibrium existence results. It connects directly back to Topics 4 and 5 (existence, compactness, correspondences) and Topic 7 (lattices and order), and it sets the stage for everything in the PhD course that involves strategic interaction.

9.2 What is a fixed point?

Let $f : X \to X$ be a function that maps a set into itself.

A point $x^* \in X$ is a **fixed point** of f if

$$f(x^*) = x^*.$$

> "A fixed point is a point that the function leaves unchanged. You plug it in, you get it back."

This sounds simple, but it is the key idea behind equilibrium. Here is why.

Think about a game. Each player has a **best response**: given what the others do, what is optimal for her? A best response is a mapping from the other players' strategies to one's own. Now, a **Nash equilibrium** is a strategy profile where every player is already playing a best response to everyone else. Nobody wants to deviate.

But that means: if we collect all best responses into a single mapping from strategy profiles to strategy profiles, an equilibrium is exactly a point where this mapping sends the profile back to itself. That is a fixed point.

So the question "does an equilibrium exist?" becomes: "does this mapping have a fixed point?"

9.3 Brouwer's fixed-point theorem

The most classical fixed-point result is due to Brouwer.

Theorem (Brouwer). Let $X \subseteq \mathbb{R}^n$ be nonempty, compact, and convex, and let $f : X \to X$ be continuous. Then f has a fixed point.

> "If you continuously map a compact convex set into itself, at least one point stays put."

The conditions matter:

- **Compact**: the set is closed and bounded, so sequences cannot escape (Topic 4).
- **Convex**: the set has no holes or missing pieces (Topic 3).
- **Continuous**: the mapping does not jump (Topic 4).
- **Into itself**: the function sends every point in X back into X.

Proof sketch A one-dimensional version is easy to see. Take $X = [0, 1]$ and $f : [0, 1] \to [0, 1]$ continuous. Consider the function $g(x) = f(x) - x$. We know $g(0) = f(0) \geq 0$ (because f maps into $[0, 1]$) and $g(1) = f(1) - 1 \leq 0$. By the intermediate value theorem, there exists some x^* where $g(x^*) = 0$, which means $f(x^*) = x^*$. Done.

In one dimension, the argument is clean because we can use the Intermediate Value Theorem (Topic 4). In higher dimensions, the IVT is no longer available, and the proof requires a genuinely different idea. The classical geometric route goes through a result by Borsuk (1931).[1]

9.3.1 The idea of the proof

In higher dimensions, we prove Brouwer by contradiction. Suppose $f : B^n \to B^n$ is continuous and has *no* fixed point, where B^n is the closed unit ball in $\mathbb{R}^n$.[2] We will show that this leads to an impossible geometric construction.

Before we start, some *notation.* The **boundary** of B^n is the set of points that are exactly at distance 1 from the origin. This is the sphere S^{n-1}.[3] In our usual 1-D world of functions mapping scalars into scalars, the unit ball is the interval $[-1, 1]$, and the sphere is just the two endpoints $\{-1, 1\}$. In two dimensions, B^2 is a disk and S^1 is the unit circle around it. In three dimensions, B^3 is the solid ball and S^2 is its surface. We will also use *rays.* A **ray** from a point a through a point b is the (half-)line that starts at a, passes through b, and continues beyond b indefinitely.

Now to the argument. Because we are looking at a self-map both the domain and the image are in the same space. That means, in particular, if $f(x) \neq x$ for every x, then for each $x \in B^n$ we can draw a ray starting at $f(x)$ and passing through x. Because $f(x) \neq x$, this ray has a well-defined direction.[4] Since x is inside the ball (or on its boundary), and the ray continues beyond x, it must eventually hit the boundary sphere S^{n-1} at some unique point.[5] Call that point $r(x)$.

[1] The proof is in fact pretty involved. For a bit more discussion, see Ok, "Real Analysis with Economic Applications" (2007, chapters D.8.2 and D.8.3). Franklin (1980), "Methods of Mathematical Economics" provides 3 proofs of Brouwer. One is combinatorial, something we haven't looked at here, so the tools are a little unfamiliar. One is somewhat indirect (yet very elegant) and one is brute force. That one implicitly constructs the Borsuk observation.

[2] That is, a ball of radius 1 around the origin.

[3] That means, it is one dimension lower than the ball.

[4] If $f(x) = x$, the "ray from $f(x)$ through x" has no direction because every line starting at x goes, trivially, also through $f(x)$. But if there is no fixed point, then the direction of the line is pinned down.

[5] The ball is convex, so if we connect two points in the ball with each other by a line, that line segment does not leave the ball. Moreover, the sphere is strictly curved in the sense that no straight line segment lies entirely on it, so a ray through an interior point can intersect S^{n-1} at most twice – and since the ray continues beyond x, it must exit the ball and hit S^{n-1}.

The map $r : B^n \to S^{n-1}$ is continuous (because f is continuous and the ray construction depends continuously on x). Moreover, if x is already on the boundary S^{n-1}, the ray from $f(x)$ through x hits the sphere at x itself, so $r(x) = x$ for all $x \in S^{n-1}$.

A continuous map from B^n onto S^{n-1} that fixes every point of S^{n-1} is called a **retraction**. So our no-fixed-point assumption has produced a retraction of the ball onto its boundary sphere.

But Borsuk (1931) proved that no such retraction can exist: *S^{n-1} is not a retract of B^n*.[6]

This is a contradiction. So our assumption was wrong, and f must have a fixed point.

> "If a continuous function on the ball had no fixed point, we could use it to continuously retract the ball onto its boundary sphere. But that is geometrically impossible – you cannot flatten a ball onto its skin without tearing. So a fixed point must exist."

The intuition behind Borsuk's result is worth pausing on. Think about the two-dimensional case: try to continuously push every point of a disk onto its boundary circle, while keeping the boundary points fixed. You will find that no matter what you do, some interior point gets "stuck." The ball is simply too full to be compressed onto its boundary without a discontinuity. This is a topological obstruction, and it is what ultimately powers Brouwer's theorem.[7]

What matters for economics is the punchline: if your model gives you a continuous self-map on a compact convex set, a fixed point is guaranteed. The geometry takes care of the rest.

9.3.2 Why Brouwer is not enough for economics

Brouwer requires the mapping to be a **function** – single-valued. But as we saw in Topic 5, optimal behavior is often set-valued. Best responses are typically correspondences, not functions. So we need a fixed-point theorem for correspondences.

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[6] For $n = 1$, this is easy to see directly. The ball $B^1 = [-1, 1]$ is connected, but a retraction onto $S^0 = \{-1, 1\}$ would be a continuous surjection from a connected set onto a disconnected one, which is impossible. For $n \geq 2$, the argument is harder and typically uses tools from algebraic topology, but the geometric content is the same: you cannot continuously flatten a ball onto its boundary without tearing something.

[7] For the empiricists among you: a retraction is a continuous map from a set onto a subset that leaves the subset's points fixed – the same idea as a projection, but with a continuity requirement. So Borsuk's result says that there is no continuous projection from a ball onto its boundary sphere.

9.4 Kakutani's fixed-point theorem

Kakutani generalizes Brouwer to correspondences.

Theorem (Kakutani). Let $X \subseteq \mathbb{R}^n$ be nonempty, compact, and convex, and let $F : X \rightrightarrows X$ be an upper hemicontinuous correspondence with nonempty, convex values.[8] Then F has a fixed point: there exists $x^* \in X$ such that $x^* \in F(x^*)$.

> "If you map a compact convex set into itself using a well-behaved correspondence with convex values, at least one point is contained in its own image."

Compare this with Brouwer:

- the mapping is now a correspondence (set-valued),
- instead of continuity, we need upper hemicontinuity (Topic 5),
- instead of the mapping being single-valued, we need the image sets to be convex.

The convexity-of-values condition is the new ingredient. It says: for any given x, the set of "responses" $F(x)$ has no gaps. This is often ensured by concavity of the objective function (which makes the set of optimizers convex, as we discussed in Topic 5).

9.4.1 Why this is the workhorse theorem for equilibrium

Kakutani's theorem is behind the existence of:

- **Nash equilibrium** (in mixed strategies),
- **Walrasian equilibrium** (in general equilibrium theory),
- **Bayesian Nash equilibrium** (in games of incomplete information).

The logic is always the same:

1. define a correspondence that collects all agents' best responses,
2. verify the conditions (compact convex domain, upper hemicontinuity, convex values),
3. apply Kakutani to conclude that a fixed point – an equilibrium – exists.

This is why Topics 3, 4, and 5 spent so much time on convexity, compactness, and upper hemicontinuity. They were preparing exactly these conditions.

[8] Convex values means: for each $x \in X$, the set $F(x)$ is convex. In our setting – a correspondence on a compact subset of $\mathbb{R}^n$ – upper hemicontinuity with compact values implies that F has a closed graph (recall Topic 5, Section 10). The proof sketch below uses the closed graph property, which is thus a consequence of the assumptions stated here, not an additional requirement.

9.4.2 How Kakutani follows from Brouwer (sketch)

The idea behind Kakutani's theorem is to reduce it to Brouwer. The correspondence F is set-valued, but we can try to "approximate" it by functions and then take a limit. The formal tool behind this construction is known as Cellina's approximation theorem.[9]

Step 1: Approximate the correspondence by functions. Cover our set X with a grid of mesh size $1/n$ (think of dividing X into small cells). At each grid point x_k, pick some element $y_k \in F(x_k)$.[10] Now, we know that $f_n(x_k) \in F(x_k)$ for any grid point x_k. For x that is not a grid point, we define $f_n(x)$ by taking a weighted average (convex combination) of the selected y_k's at nearby grid points, with weights depending on how close x is to each grid point.[11] The result is a continuous (in fact piecewise linear) function $f_n : X \to X$.

Step 2: Apply Brouwer to each approximation. Each f_n is a continuous function from the compact convex set X into itself. So by Brouwer, each f_n has a fixed point x_n:

$$f_n(x_n) = x_n.$$

Step 3: Take a limit. We now have a sequence (x_n) in the compact set X. By compactness, this sequence has a convergent subsequence with some limit x^* (this is the Bolzano-Weierstrass property from Topic 4).

Now, here is the key subtlety. As the grid becomes finer (n grows), the fixed point x_n comes closer and closer to satisfying $x_n \in F(x_n)$, but it may never quite get there, because f_n is a function, not a correspondence. So each x_n is an *approximate* fixed point of F, not an exact one.

What saves us is the closed graph property. If a sequence of points $(x_n, f_n(x_n))$ converges and each $f_n(x_n)$ is close to $F(x_n)$, then the limit point (x^*, x^*) lies in the graph of F. The graph of F is closed (which follows from upper hemicontinuity and compact values – recall Topic 5), so it contains its limit points. "Almost in" becomes "actually in."

That gives us

$$x^* \in F(x^*).$$

That is our fixed point.

> "Kakutani is Brouwer in the limit: approximate the correspondence by functions, use Brouwer on each approximation, and let the approximation quality go to infinity. Compactness keeps the sequence

[9]For the details in an economics-oriented presentation, see Ok, "Real Analysis with Economic Applications" (2007), Chapter E.5.

[10]Note that y_k need not be a grid point, but is *any one point* in $F(x_k)$.

[11]This is where we use convexity of $F(x)$. Because each $F(x)$ is convex, and the values y_k come from nearby images of F, these weighted averages remain close to $F(x)$. Without convex values, we could not average and stay "near" the correspondence.

> from escaping, and the closed graph turns approximate fixed points into an exact one."

This sketch shows why *every* assumption in Kakutani earns its place:

- compactness of X gives us the convergent subsequence,
- convexity of X keeps us inside X throughout,
- convexity of $F(x)$ allows the continuous approximations to exist (we need to take weighted averages),
- upper hemicontinuity (closed graph) ensures that the limit of approximate fixed points is an actual fixed point.

9.5 Tarski's fixed-point theorem

There is a completely different route to fixed points that does not use convexity or continuity at all. Instead, it uses **order**.

Theorem (Tarski). Let $(X, \preceq)$ be a nonempty complete lattice and let $f : X \to X$ be monotone increasing. Then the set of fixed points of f is nonempty and itself forms a complete lattice.

> "If the choice space is a complete lattice and the mapping is monotone, then fixed points exist – and the set of fixed points has the same lattice structure."

This connects directly to Topic 7. Recall:

- a complete lattice is a partially ordered set where every subset has a least upper bound and a greatest lower bound,
- monotone increasing means: higher input, higher output ($x \preceq x'$ implies $f(x) \preceq f(x')$).

No convexity, no continuity, no topological assumptions at all. Just order and monotonicity.

9.5.1 When is Tarski useful?

Tarski is the right tool when:

- the action space is discrete or has natural lattice structure,
- best responses are monotone (strategic complements, as in Topic 7),
- you want existence without mixed strategies.

For example, in a game where each firm chooses a price and higher rival prices make higher own prices optimal (strategic complements), Tarski gives existence of a pure-strategy equilibrium directly. No need to convexify or randomize.

The fact that the fixed-point set is itself a lattice is a bonus: it means there is a largest and a smallest equilibrium, which is often useful for comparative statics.

9.5.2 Why Tarski works (constructive sketch)

Unlike Brouwer and Kakutani, Tarski's theorem has a remarkably direct proof. It is constructive: we can *find* the fixed points, not just prove they exist.

Here is the argument for the greatest fixed point. Start at the top of the lattice, $\bar{x} = \sup X$ (which exists because X is a complete lattice). Since f maps X into X, we know $f(\bar{x}) \in X$, so $f(\bar{x}) \preceq \bar{x}$.

Now apply f to both sides. Because f is monotone increasing, $f(\bar{x}) \preceq \bar{x}$ implies $f(f(\bar{x})) \preceq f(\bar{x})$. So the sequence

$$\bar{x} \succeq f(\bar{x}) \succeq f(f(\bar{x})) \succeq \cdots$$

is decreasing. In a complete lattice, every decreasing sequence (in fact, every subset) has a greatest lower bound. Call it x^*.

One can then show that $f(x^*) = x^*$.[12]

The argument for the least fixed point is symmetric: start at $\inf X$ and iterate upward.

> "Start at the top, apply the monotone function, and walk downward. In a complete lattice, you cannot fall forever – you must land on a fixed point. The same works from the bottom, walking up."

This is why Tarski requires no topological assumptions. There is no limit-taking in the topological sense, no continuity, no convexity. The lattice structure and monotonicity do all the work. The "convergence" here is order-theoretic, not metric.

The constructive nature of Tarski is practically useful, too. In games with strategic complements, you can often *compute* the largest and smallest equilibria by starting at the extremes of the strategy space and iterating best responses. If best responses are monotone, this procedure converges to equilibrium.

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[12]The details require checking that $f(x^*)$ is both $\preceq x^*$ and $\succeq x^*$. For $\preceq$: since $x^* \preceq f^k(\bar{x})$ for all k (it is the greatest lower bound), monotonicity gives $f(x^*) \preceq f^{k+1}(\bar{x})$ for all k, so $f(x^*)$ is also a lower bound of the sequence, hence $f(x^*) \preceq x^*$. For $\succeq$: applying f to $f(x^*) \preceq x^*$ gives $f(f(x^*)) \preceq f(x^*)$ by monotonicity, so $f(x^*)$ is itself a lower bound that x^* must be at least as large as – but x^* is the *greatest* lower bound, so $x^* \preceq f(x^*)$.

9.6 Comparing the three theorems

The three fixed-point theorems serve different purposes:

Brouwer requires a continuous function on a compact convex set. It is the simplest result and the one closest to geometric intuition. It works when the mapping is single-valued.

Kakutani extends Brouwer to correspondences. It is the standard tool for proving equilibrium existence in games and markets when best responses are set-valued. It requires convexity of the domain and of the image sets, plus upper hemicontinuity.

Tarski replaces all topological assumptions with order-theoretic ones. It is the right tool when the environment has lattice structure and monotonicity, especially in games with strategic complements.

Each theorem has assumptions that the others do not need. None dominates the others. The choice depends on which structure the economic model provides.

9.7 An economic example: Nash equilibrium existence

Consider a finite game: n players, each with a finite set of pure strategies S_i.

A **mixed strategy** for player i is a probability distribution σ_i over S_i (Topic 8). The set of mixed strategies is

$$\Delta(S_i) = \left\{ \sigma_i \in \mathbb{R}_+^{|S_i|} \;\middle|\; \sum_{s_i \in S_i} \sigma_i(s_i) = 1 \right\}.$$

This set is compact and convex (it is a simplex).

The product of all players' mixed-strategy sets,

$$\Sigma = \Delta(S_1) \times \cdots \times \Delta(S_n),$$

is also compact and convex (Cartesian product of compact convex sets).

Now define the **best-response correspondence** $BR : \Sigma \rightrightarrows \Sigma$ that maps each strategy profile to the set of profiles in which every player is playing a best response.

Under standard assumptions:

- BR is upper hemicontinuous (because payoffs are continuous in mixed strategies and the constraint set is well-behaved – this is Berge's Maximum Theorem from Topic 5),

- BR has convex values (because if two mixed strategies are both best responses, any mixture of them is also a best response – expected payoff is linear in one's own mixing probabilities).

Kakutani applies. A fixed point exists. That fixed point is a Nash equilibrium.

This is Nash's theorem (1950), and the proof is exactly the logic above.[13] Notice how it pulls together compactness, convexity, upper hemicontinuity, Berge's theorem, and mixed strategies – essentially everything from Topics 3 through 8.

9.8 Why existence matters (even if we cannot compute)

A common reaction is: "So we know an equilibrium exists. But we still don't know what it is. What's the point?"

The point is that existence is a minimal consistency check on the model. If an equilibrium does not exist, the model is making contradictory assumptions. Agents are supposed to optimize, but no configuration of choices is mutually consistent. That means the model is broken, not the world.

Beyond that, existence results often come with structure:

- Kakutani tells us fixed points exist in a compact convex set,
- Tarski tells us the fixed-point set is a lattice with largest and smallest elements,
- Berge's theorem tells us solutions vary continuously with parameters.

This structure is what makes comparative statics, welfare analysis, and mechanism design possible. We do not always need to *find* the equilibrium. We often just need to know it *exists* and *behaves well*.

9.9 Where this appears in Mas-Colell, Whinston, and Green (MWG)

Fixed-point theorems are used at several points in MWG:

- **MWG, Chapter 8** (Nash equilibrium existence, via Kakutani),
- **MWG, Chapter 17** (existence of Walrasian equilibrium),
- **MWG, Mathematical Appendix C** (fixed-point theorems stated).

[13]Nash's original proof used Brouwer rather than Kakutani, constructing a particular continuous function whose fixed points are Nash equilibria. The Kakutani route is the more standard modern presentation.

Tarski's theorem is not developed in MWG but is standard in the monotone comparative statics literature that builds on the lattice methods introduced in Topic 7.

9.10 Why this matters for microeconomics

Fixed-point theorems are the mathematical engine behind equilibrium.

They explain:

- why Nash equilibria exist in finite games,
- why competitive equilibria exist under standard assumptions,
- why mechanism design problems have solutions,
- why games with strategic complements have well-structured equilibrium sets.

Without fixed-point results, "equilibrium" would be a word without guaranteed content. These theorems are what turn it into a precise and operational concept.

9.11 What you should take away

After this topic, you should be comfortable with:

- reading "equilibrium as fixed point" without it feeling like a metaphor,
- knowing the conditions of Brouwer, Kakutani, and Tarski and which model features they require,
- seeing how the pre-course machinery (convexity, compactness, upper hemi-continuity, lattices) feeds directly into equilibrium existence,
- and understanding why existence is the first question economists ask, even before uniqueness or characterization.

From here on, equilibrium is not an assumption. It is a consequence of structure.

Chapter 10

Topic 10: Discounting and Infinite Horizons

(Why the future is cheaper, and why that makes infinite problems tractable)

10.1 Why discounting enters now

The pre-course has so far been mostly static. Agents choose, they optimize, they interact — but everything happens at a single point in time.

In much of microeconomics, however, decisions have consequences that unfold over time:

- a firm invests today and earns returns tomorrow,
- a player cooperates now in exchange for future cooperation,
- a consumer saves part of her income for later.

Once time enters, we need to compare payoffs that arrive at different dates. A euro today is not the same as a euro next year. This is not a psychological claim — it is an assumption that economists encode with **discounting**.

This topic introduces discounting and the infinite horizon as mathematical objects. The goal is to make two things precise: how economists aggregate payoffs over time, and why infinite-horizon problems are often *easier* than finite-horizon ones.

10.2 The discount factor

Let $\delta \in (0, 1)$ be a number called the **discount factor**.

> "The discount factor δ captures how much the agent values tomorrow relative to today. A payoff of x received one period from now is worth δx today."

If δ is close to 1, the agent is **patient**: she treats the future almost like the present. If δ is close to 0, the agent is **impatient**: the future is heavily discounted.

Where does δ come from? In different models it has different interpretations:

- a pure time preference (the agent simply prefers earlier consumption),
- an interest rate ($\delta = 1/(1+r)$, where r is the rate of return on savings),
- a probability of continuation (in each period, the game continues with probability δ and ends with probability $1-\delta$).

The third interpretation is worth flagging because it shows up in repeated games: discounting and random termination are mathematically the same thing. A patient player is one whose game is likely to continue.

The key restriction is $\delta < 1$. If $\delta \geq 1$, the agent values the future at least as much as the present, and infinite sums of payoffs will (generically) diverge. Everything in this topic depends on $\delta < 1$.

10.3 Geometric series: the basic calculation

The mathematical object behind discounting is the **geometric series**.

10.3.1 Finite geometric series

The sum of a finite geometric series is

$$\sum_{t=0}^{T} \delta^t = 1 + \delta + \delta^2 + \cdots + \delta^T = \frac{1-\delta^{T+1}}{1-\delta}.$$

This formula is worth knowing by heart. It comes from a simple trick: if $S = 1 + \delta + \cdots + \delta^T$, then $\delta S = \delta + \delta^2 + \cdots + \delta^{T+1}$, so $S - \delta S = 1 - \delta^{T+1}$, and dividing both sides by $(1-\delta)$ gives the result.

10.3.2 Infinite geometric series

Now let $T \to \infty$. Since $\delta < 1$, the term $\delta^{T+1} \to 0$, and we get

$$\sum_{t=0}^{\infty} \delta^t = \frac{1}{1-\delta}.$$

> "The infinite sum of a geometric series with ratio $\delta < 1$ is $1/(1-\delta)$. The smaller δ, the smaller the sum — because the future matters less."

This formula is the backbone of every present-value calculation in economics. It works because $\delta < 1$ ensures the terms shrink fast enough that the infinite sum converges. Convergence here is in the sense of Topic 4: the partial sums form a bounded, increasing sequence in a complete space, so a limit exists.

10.3.3 Shifted series

A useful variant: if the series starts at period s instead of period 0,

$$\sum_{t=s}^{\infty} \delta^t = \delta^s \cdot \frac{1}{1-\delta}.$$

This says: the "value of the future as seen from period s" is just a scaled-down version of the value as seen from period 0. The future always looks the same, just discounted by how far away you are from it. This observation will be important shortly.

10.4 Present value of a stream

Suppose an agent receives a stream of payoffs $(x_0, x_1, x_2, \dots)$. The **present value** (or **discounted sum**) of this stream is

$$V = \sum_{t=0}^{\infty} \delta^t x_t.$$

> "The present value weights each future payoff by the appropriate power of the discount factor. Earlier payoffs count more; later ones are discounted away."

For the present value to be well-defined, we need the sum to converge. If the payoffs are bounded — say $|x_t| \leq M$ for all t — then

$$|V| \leq \sum_{t=0}^{\infty} \delta^t M = \frac{M}{1-\delta},$$

so the sum converges absolutely. Bounded payoffs and $\delta < 1$ are enough.

A special case that comes up constantly: if the stream is constant, $x_t = c$ for all t, then

$$V = c \cdot \sum_{t=0}^{\infty} \delta^t = \frac{c}{1-\delta}.$$

10.5 The normalized (average) payoff

The present value $V = \sum_{t=0}^{\infty} \delta^t x_t$ has an awkward feature: it scales with $1/(1-\delta)$. A constant stream of c per period gives present value $c/(1-\delta)$, which blows up as $\delta \to 1$. This makes it difficult to compare payoffs across different levels of patience, and — more practically — difficult to compare the value of a stream with the value of a single period's payoff.

The standard fix is to **normalize** by multiplying with $(1-\delta)$:

$$\bar{V} = (1-\delta) \sum_{t=0}^{\infty} \delta^t x_t.$$

> "The normalized payoff rescales the present value so that a constant stream of c per period has normalized value exactly c."

Why is this a natural thing to do? Think of $(1-\delta)$ as the weight on "this period" in the discounting scheme. The full weights $(1-\delta), (1-\delta)\delta, (1-\delta)\delta^2, \ldots$ sum to one:

$$(1-\delta) \sum_{t=0}^{\infty} \delta^t = 1.$$

So the normalized payoff is a genuine weighted average of the per-period payoffs, with weights that sum to one and decline geometrically. It tells us: "what constant per-period payoff would give the same present value as this stream?"

This normalization is not just cosmetic. It has a very practical consequence: the **continuation value** from any future period, when normalized, is directly comparable to a single period's payoff. If the agent is considering a one-time deviation — taking a different action today and then returning to the original plan — the gain from deviating is a single-period payoff, and the loss is a change in the continuation value. Both are now measured in the same units. This is exactly the structure behind the **one-shot deviation principle** in dynamic games, which says that checking for profitable one-period deviations is enough to verify optimality of an entire strategy.[1]

10.6 Finite versus infinite horizons

Why do economists so often work with infinite horizons when the real world is clearly finite?

The honest answer has two parts.

[1] We do not develop the one-shot deviation principle here, but the normalization is what makes it work cleanly. Without it, the comparison between a one-period gain and a multi-period loss requires carrying around factors of $(1-\delta)$ everywhere.

First, infinite horizons avoid the "last period" problem. In a finite-horizon repeated game, the last period has no future. So there is no incentive to cooperate in the last period. But then the second-to-last period is effectively the last period with future incentives... and by backward induction, cooperation unravels all the way to the beginning. An infinite horizon eliminates this unraveling because there is no last period.[2]

Second, infinite horizons have a stationarity property that makes them mathematically tractable. The problem "starting from period t" looks exactly like the problem "starting from period 0," just with different initial conditions. This self-similarity is what allows recursive methods to work: instead of solving a problem with infinitely many variables, we solve a single-period problem that references its own solution.

This is the key insight that connects this topic to Topic 11 (dynamic programming). The infinite horizon is not a complication — it is a simplification. Finite horizons, paradoxically, often require more work because the structure changes as you approach the end.

10.7 Stationarity and the recursive structure

Let us make the stationarity observation precise, using the normalized formulation from Section 5.

The normalized value of a stream starting at period 0 is

$$\bar{V}_0 = (1-\delta)\sum_{t=0}^{\infty} \delta^t x_t.$$

Now look at the normalized value of the same stream starting from period 1:

$$\bar{V}_1 = (1-\delta)\sum_{t=1}^{\infty} \delta^{t-1} x_t.$$

The relationship between the two is

$$\bar{V}_0 = (1-\delta)x_0 + \delta \bar{V}_1.$$

> "Today's normalized value is a weighted average of today's payoff and the continuation value, with weights $(1-\delta)$ and δ."

[2] This is the core of why the folk theorem requires an infinite (or indefinitely repeated) horizon. With a known finite horizon, backward induction often forces very different — and less cooperative — behavior.

This is a **recursive decomposition**. It says: the infinite-horizon problem can be split into "what happens today" and "the value of the future," and the weights on the two parts sum to one. If the environment is stationary — the same structure repeats every period — then $\bar{V}_1$ solves the same problem as $\bar{V}_0$, possibly with different state variables.

In such cases, we can write

$$\bar{V} = (1 - \delta)x + \delta\bar{V},$$

where $\bar{V}$ is the normalized value and x is the per-period payoff (assuming stationarity). Solving:

$$\bar{V} = x.$$

This is exactly the point of the normalization: a constant stream of x per period has normalized value x. But now we have derived this from the recursive structure rather than from summing a series. The two approaches give the same answer — and they must, by the uniqueness of limits.

The recursive approach generalizes: when the per-period payoff depends on a state variable that evolves over time, we get a **functional equation** (the Bellman equation). That is the subject of Topic 11.

10.8 A brief look at continuous time

Everything so far has been in discrete time: periods $t = 0, 1, 2, \ldots$, with a discount factor δ applied once per period. But economists sometimes work in continuous time, where decisions happen not at integer dates but at every instant $t \in [0, \infty)$.

In continuous time, the discount factor δ^t is replaced by e^{-rt}, where $r > 0$ is the **discount rate** (or interest rate). The present value of a payoff stream $x(t)$ becomes

$$V = \int_0^\infty e^{-rt} x(t)\, \mathrm{d}t$$

instead of a sum. The integral plays the same role as the geometric series: it aggregates future payoffs, weighting earlier ones more heavily.

The two formulations are closely related. If we set $\delta = e^{-r}$ (or equivalently $r = -\ln\delta$), then evaluating the continuous formula at integer dates recovers the discrete one. So the mathematics is not fundamentally different — it is the same idea expressed in a different language.

10.8.1 What gets simpler

Continuous time has genuine advantages. Derivatives replace difference equations: instead of writing $V_{t+1} - V_t$ and worrying about discrete jumps, we write $\dot{V}(t)$ and use calculus. For models where timing is flexible — when does the firm invest? when does the worker quit? — continuous time provides a clean framework because there is no need to artificially discretize the decision.[3] Many equilibrium conditions also take simpler forms: the Hamilton-Jacobi-Bellman equation (the continuous-time counterpart of the Bellman equation) is often a differential equation, and differential equations have a very well-developed solution theory.

10.8.2 What gets harder

The cost is technical. Sums become integrals, difference equations become differential equations, and the probability theory needed for continuous-time stochastic models (Brownian motion, Itô calculus) is substantially more demanding than the discrete-time counterpart. Convergence arguments that are straightforward in discrete time — like the bounded payoffs argument we used above — require measure-theoretic care in continuous time.

For strategic interaction, continuous time also introduces subtleties about what "simultaneous" means when time is a continuum. In discrete time, "both players move in period t" is unambiguous. In continuous time, the order of actions within an instant can matter, and formalizing strategies requires more care.

10.8.3 Why economists use both

The choice between discrete and continuous time is usually a modeling convenience, not a deep commitment. Roughly:

- **Discrete time** is the default for game theory, repeated interaction, contracting, and mechanism design. The strategic structure is clearer when periods are distinct, and the recursive machinery (Topic 11) is easier to set up.
- **Continuous time** is common in finance, macroeconomics, search theory, and dynamic optimization with flexible timing. The calculus tools are often more powerful, and closed-form solutions are sometimes available where the discrete case only gives numerical answers.

Many results have exact counterparts in both settings. The folk theorem, dynamic programming, and optimal stopping all have discrete-time and continuous-time versions. For this pre-course and the PhD microeconomics course that follows, we work in discrete time. But knowing that continuous time exists — and that the economic ideas translate — prevents confusion when you encounter it in other courses.

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[3] Optimal stopping problems, real options, and search models are natural examples where continuous time simplifies the analysis.

10.9 An economic example: sustaining cooperation

Consider two firms that interact every period. In each period, each firm can either **cooperate** (keep prices high) or **defect** (undercut).

Per-period payoffs:

- if both cooperate, each earns π_C,
- if both defect, each earns $\pi_D < \pi_C$,
- if one defects while the other cooperates, the defector earns $\pi_H > \pi_C$ and the cooperator earns $\pi_L < \pi_D$.

Consider the **trigger strategy**: cooperate as long as the other firm cooperates; if the other firm ever defects, defect forever.

Under this strategy, the normalized payoff from cooperating forever is π_C.

The temptation to deviate is: earn π_H today (the one-period gain from undercutting), but suffer π_D forever after (because the other firm switches to permanent defection). The normalized payoff from deviating is

$$(1-\delta)\pi_H + \delta\pi_D.$$

Cooperation is sustainable if

$$\pi_C \geq (1-\delta)\pi_H + \delta\pi_D,$$

which rearranges to

$$\delta \geq \frac{\pi_H - \pi_C}{\pi_H - \pi_D}.$$

> "Cooperation is sustainable if the firms are patient enough. The more tempting the short-run gain from defection (higher $\pi_H - \pi_C$), the more patience is required."

Notice how the normalization made the comparison clean: a per-period cooperation payoff on the left, a weighted average of a one-period deviation gain and a permanent punishment on the right. Without normalization, both sides would carry factors of $1/(1-\delta)$ that obscure the economics.

10.10 Where this appears in Mas-Colell, Whinston, and Green (MWG)

Discounting and infinite horizons are not developed as a standalone mathematical topic in MWG. They appear implicitly in:

- **MWG, Chapter 20** (equilibrium and time, intertemporal choice),

- **MWG, Chapters 12 and 13** (repeated games, where discounting is a primitive).

The geometric series itself is assumed as background knowledge. These notes make the economic logic behind the calculation explicit.

10.11 Why this matters for microeconomics

Discounting is the language of time in economic theory.

It appears in:

- repeated games (where δ determines whether cooperation can be sustained),
- dynamic contracting (where continuation values serve as incentive instruments),
- asset pricing (where prices are present values of dividend streams),
- dynamic mechanism design (where future transfers and allocations are discounted).

The recursive structure — "today plus the discounted future" — is what makes infinite-dimensional problems manageable. Without it, every dynamic model would require solving for infinitely many variables simultaneously.

Understanding discounting now means that when the course introduces trigger strategies, continuation values, or Bellman equations, the mathematical objects will already be familiar. The economic ideas can then take center stage.

10.12 What you should take away

After this topic, you should be comfortable with:

- computing present values using geometric series,
- interpreting the discount factor as patience, interest, or continuation probability,
- understanding why the normalized payoff is a weighted average and why it makes continuation values comparable to per-period payoffs,
- seeing why infinite horizons create stationarity rather than complexity,
- recognizing the recursive decomposition $\bar{V} = (1-\delta)x + \delta\bar{V}$ as the seed of dynamic programming,
- and knowing that continuous time is an alternative formulation with the same economic ideas but different technical tools.

Chapter 11

Topic 11: Dynamic Programming and Recursive Thinking

(How to solve infinite-dimensional problems one step at a time)

11.1 Why dynamic programming enters now

Topic 10 introduced discounting, infinite horizons, and the recursive decomposition

$$\bar{V} = (1 - \delta)x + \delta\bar{V}.$$

That decomposition was for a constant payoff stream. But in most economic problems, payoffs are not constant — they depend on the **state** of the world, and on the **actions** of the agents. Often, those actions affect the future state as well.

For example, a firm's profit depends on its capital stock, which depends on past investment. A consumer's utility depends on her wealth, which depends on past savings decisions. A player's continuation payoff in a repeated game depends on the history of play, which determines the current "state" of the relationship.

In all these cases, the agent faces the same question every period: given the current state, what should I do today, knowing that my choice affects not only today's payoff but also tomorrow's state?

Dynamic programming is the method that turns this infinite-dimensional problem —choosing a full program of state-dependent choices—into a manageable one-step problem. The key idea is to find a **value function** that summarizes the future, so that the agent can optimize one period at a time.

11.2 The ingredients

A dynamic optimization problem has three components:

1. **State variable** $\theta \in \Theta$: a summary of everything from the past that matters for the future. The state might be a capital stock, a wealth level, a vector of past actions, or a belief.

2. **Action** $a \in A(\theta)$: what the agent chooses in the current period, given the state. The feasible set may depend on the state (you cannot invest more than your current wealth).

3. **Transition**: a rule that determines the next state θ' given the current state θ and action a. In the deterministic case this is a function $\theta' = g(\theta, a)$. In the stochastic case, θ' is drawn from a distribution that depends on (θ, a).

The agent's per-period payoff is $u(\theta, a)$, and she discounts the future with discount factor $\delta \in (0, 1)$.

The agent's problem is to choose a sequence of actions $(a_0, a_1, a_2, \dots)$ to maximize the normalized present value

$$(1-\delta)\sum_{t=0}^{\infty} \delta^t u(\theta_t, a_t),$$

where $\theta_{t+1} = g(\theta_t, a_t)$ and θ_0 is given.

This is, in principle, a problem with infinitely many choice variables. Dynamic programming reduces it to one.

11.3 The principle of optimality

The key insight is due to Bellman:

> "If a plan is optimal from period 0 onward, then its continuation from period 1 onward must be optimal given the state at period 1."

This sounds obvious, but it is powerful. It says that optimal behavior has a recursive structure: you cannot be doing the best thing overall without also doing the best thing going forward from every intermediate point.

The contrapositive makes the logic clear: if the continuation were not optimal from period 1, you could improve the overall plan by swapping in a better continuation — contradicting optimality of the original plan.

The principle of optimality is what justifies reducing the infinite-horizon problem to a one-period problem. Instead of choosing an entire sequence, the agent asks: "given the current state θ, what action a maximizes today's payoff plus the discounted value of being in state θ' tomorrow?" If we know the value of being in each state, the problem is solved.

11.4 The Bellman equation

The **value function** $V : \Theta \to \mathbb{R}$ assigns to each state θ the normalized value of behaving optimally from θ onward.

By the principle of optimality, V must satisfy the **Bellman equation**:

$$V(\theta) = \max_{a \in A(\theta)} \left\{ (1-\delta)\, u(\theta, a) + \delta\, V(g(\theta, a)) \right\}.$$

> "The value of being in state θ is the best you can do today — choosing an action that maximizes the weighted sum of today's payoff and the discounted value of tomorrow's state."

Although the Bellman equation does simplify the problem enormously, it, depending on the eye of the beholder, it is (un)fortunately not yet trivial. Why? Because the Bellman equation is a **functional equation**: the unknown is not a number but a *function* V. The value function appears on both sides of the equation, because the value of today's state depends on the value of tomorrow's state, which is itself determined by V.

Compare this to the simple recursive decomposition in Topic 10. There, the "state" was trivial (the payoff was the same every period), so V was just a number. Now V is a function of the state, and the Bellman equation tells us what this function must look like.

11.5 The Bellman equation as a fixed-point problem

The next step is now to solve this equation. In Topic 9, we studied fixed points: a point x^* such that $f(x^*) = x^*$. The Bellman equation has exactly this structure, but in a different space.

Define an *operator* T that takes a function $W : \Theta \to \mathbb{R}$ and produces a new function $TW : \Theta \to \mathbb{R}$ by

$$(TW)(\theta) = \max_{a \in A(\theta)} \left\{ (1-\delta)\, u(\theta, a) + \delta\, W(g(\theta, a)) \right\}.$$

Then the Bellman equation says: $V = TV$.

> "The value function is a fixed point of the Bellman operator. It is a function that, when you plug it into the right-hand side, you get back the same function on the left."

Let's reiterate how we get there.[1]

1. First, we notice that the agent's problem is described by the principle of optimality. She does whatever is best for her which means assuming that wherever she ends up tomorrow she will do what is best from tomorrow onward. Knowing that, today's choice boils down to choosing what happens today and where to end up tomorrow.
2. Second, using this "truth" we can determine what describes the value of starting at a particular state θ. It is the best that can come out from wherever we are. If we want to describe that **for all** possible states, we get the value function.
3. But then, this is *one single function*, completely pinned down by our inputs (state space, action space, transition function).
4. Finally, this means we can write the right hand side as a *function over functions* that for each value function W we pick on the right hand side (optimal or not) gives us the best action for today, that is,

$$T(W, \theta) = \max_{a \in A(\theta)} \left\{ (1-\delta)\, u(\theta, a) + \delta\, W(g(\theta, a)) \right\}$$

Notice that on the right hand side a is already pinned down by the max. So we are left with θ which enters through $W(g())$ and $u()$. So the only "choice" we have is that of W. If we can find a $W : \Theta \to \mathbb{R}$ such that $W(\theta) = T(W, \theta)$ for all θ, then we know that for this W the agent always takes the correct decision. But because θ is still an element here, we have to find our fixed point in the space of functions, not just numbers. This is what makes it a functional equation.

Our function space is the space of all (bounded) functions from Θ to $\mathbb{R}$.[2] The question is: does such a fixed point exist? Is it unique? And can we find it? The answer to all three is yes, thanks to a new fixed-point theorem.

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[1] My macro professor in the PhD classified people in two intelligence levels: Those that can solve Bellman equations and those that cannot. I do not subscribe to his view, but maybe it is still worth it to invest into understanding them, if only to impress a Macro Economist.

[2] If u is bounded, and $\delta < 1$, then any solution sequence is going to yield a bounded payoff, thus any function that could work must be a function that is bounded too.

11.6 The contraction mapping theorem

The fixed-point theorems in Topic 9 — Brouwer, Kakutani, Tarski — guarantee existence but not uniqueness (except Tarski for extremal fixed points). For the Bellman equation, we construct something stronger: a result that gives both existence and uniqueness, and tells us how to compute the solution.

The right tool is the **contraction mapping theorem** (also called the Banach fixed-point theorem).

11.6.1 Complete metric spaces

To state it, we need one concept. A **complete metric space** is a set X with a metric d (recall the remark in Topic 4) where every Cauchy sequence converges.[3] The real line $\mathbb{R}$ is complete (this is a fundamental property of the reals). The space of bounded functions from Θ to $\mathbb{R}$, equipped with the "sup metric" $d(f,g) = \sup_{\theta\in\Theta} |f(\theta) - g(\theta)|$, is also complete.[4]

11.6.2 The theorem

Theorem (Contraction Mapping / Banach). Let (X, d) be a complete metric space and let $T : X \to X$ be a **contraction**: there exists $\beta \in (0, 1)$ such that

$$d(T(x), T(y)) \le \beta\, d(x, y) \quad \text{for all } x, y \in X.$$

Then T has exactly one fixed point x^*, and for any starting point $x_0 \in X$, the sequence $x_0, T(x_0), T(T(x_0)), \ldots$ converges to x^*.

> "A contraction squeezes points closer together. If you keep applying it, everything collapses to a single fixed point."

11.6.3 Proof sketch

The argument is constructive. Start anywhere: pick any $x_0 \in X$. Define $x_1 = T(x_0)$, $x_2 = T(x_1)$, and so on.

Because T is a contraction,

$$d(x_1, x_2) = d(T(x_0), T(x_1)) \le \beta\, d(x_0, x_1).$$

[3] A Cauchy sequence is a sequence where the terms get arbitrarily close to each other as the sequence progresses: for every $\varepsilon > 0$, there exists N such that $d(x_m, x_n) < \varepsilon$ for all $m, n \ge N$. Completeness means such sequences actually have a limit in X — they don't "converge to a hole."

[4] The sup metric measures the largest pointwise difference between two functions. Two functions are close under this metric if they are close *everywhere*. An exercise to work with metrics is to prove that the sup metric is indeed a metric.

Repeating,

$$d(x_n, x_{n+1}) \leq \beta^n \, d(x_0, x_1).$$

Since $\beta < 1$, these distances shrink geometrically — exactly like the geometric series from Topic 10. For any $m > n$, the triangle inequality gives

$$d(x_n, x_m) \leq d(x_n, x_{n+1}) + d(x_{n+1}, x_{n+2}) + \cdots \leq \beta^n \, d(x_0, x_1) \cdot \frac{1}{1-\beta}.$$

As $n \to \infty$, the right hand side goes to 0, because $d(x_0, x_1)$ is bounded, because the functions are bounded, $1 - \beta > 0$ and $\beta^n \to 0$ because $\beta < 1$, so the sequence is Cauchy. By completeness, it converges to some x^*.

To see that x^* is a fixed point: T is continuous (any contraction is), so $T(x^*) = T(\lim x_n) = \lim T(x_n) = \lim x_{n+1} = x^*$.

Uniqueness: if there were two fixed points x^* and y^*, then $d(x^*, y^*) = d(T(x^*), T(y^*)) \leq \beta \, d(x^*, y^*)$. Since $\beta < 1$, this implies $d(x^*, y^*) = 0$, so $x^* = y^*$ because in every metric $d(x, y) = 0 \Leftrightarrow x = y$.

> "Start anywhere, keep applying the contraction, and you converge to the unique fixed point. The proof is the algorithm."

11.6.4 Comparison with Topic 9

The contraction mapping theorem has a very different character from Brouwer, Kakutani, and Tarski:

- it requires no convexity and no lattice structure,
- it requires a metric and completeness (topological structure that the others did not need),
- it gives **uniqueness** (the others generally do not),
- it is **constructive** (the iteration $x, T(x), T(T(x)), \ldots$ converges to the solution).

The price is the contraction property: the mapping must strictly squeeze distances. This is a strong requirement, but one that the Bellman operator naturally satisfies.

Another practical limitation is that value function iteration is often not very elegant. You can solve the problem, but the solution is often not closed-form, and — I am out of my depth here, but at least in my days — it is not very computationally efficient. That's why in Macro you often use other methods than value function iteration to determine the optimum. See more on this below.

11.7 Why the Bellman operator is a contraction

This is the result that makes everything work. Under standard assumptions — bounded payoffs, $\delta < 1$ — the Bellman operator T is a contraction on the space of bounded functions, with contraction factor $\beta = \delta$.

The intuition is direct. Suppose two candidate value functions W and W' differ by at most ε everywhere: $\sup_\theta |W(\theta) - W'(\theta)| \le \varepsilon$. Now compute TW and TW'. The only way W and W' enter the Bellman equation is through the continuation value, which is multiplied by δ. So the difference between TW and TW' is at most $\delta\varepsilon$.[5]

This means:

$$d(TW, TW') \le \delta\, d(W, W').$$

Since $\delta < 1$, the operator T is a contraction. The contraction mapping theorem then guarantees:

- the Bellman equation has a **unique** solution V,
- starting from any initial guess W_0 and iterating $W_{n+1} = TW_n$ converges to V.

> "Discounting is what makes the Bellman operator a contraction. Because the future is worth less than the present, disagreements about continuation values get squeezed by δ every time you iterate."

11.8 Value iteration and policy functions

The constructive nature of the contraction mapping theorem gives us an algorithm: **value iteration**.

1. Start with any bounded function W_0 (a common choice is $W_0 = 0$).
2. Compute $W_1 = TW_0$, then $W_2 = TW_1$, and so on.
3. The sequence $W_0, W_1, W_2, \ldots$ converges to the true value function V.

At each step, computing TW_n requires solving a static optimization problem — maximizing over today's action — for each state. This is where the single-agent tools from Topics 2–6 come in: for each state, we solve a one-period problem.

Once we have the value function V, the **policy function** (or policy correspondence) is the argmax of the Bellman equation:

$$a^*(\theta) = \text{argmax}_{a \in A(\theta)} \left\{ (1-\delta)\, u(\theta, a) + \delta\, V(g(\theta, a)) \right\}.$$

[5]More precisely: for any state θ and any action a, $|(1-\delta)u(\theta,a) + \delta W(g(\theta,a)) - (1-\delta)u(\theta,a) - \delta W'(g(\theta,a))| = \delta |W(g(\theta,a)) - W'(g(\theta,a))| \le \delta\varepsilon$. Taking the max over a on each side and then the sup over θ gives $d(TW, TW') \le \delta\, d(W, W')$.

> "The policy function tells the agent what to do in each state. It is the decision rule that, together with the value function, solves the entire infinite-horizon problem."

This connects directly to Topic 5: the argmax is in general a correspondence (set-valued), and its properties — upper hemicontinuity, convexity of values — depend on the same assumptions that Berge's Maximum Theorem requires.

11.9 An economic example: optimal savings

Consider a consumer with wealth θ who must split it between consumption c and savings $\theta - c$ each period. She earns a gross return $R > 0$ on savings, so next period's wealth is $(\theta - c) \cdot R$. Her per-period utility from consumption is $u(c)$, increasing and concave.

The state is wealth θ. The action is how much to consume, $c \in [0, \theta]$. The transition is $\theta' = (\theta - c)R$.

The Bellman equation spells this out:

$$V(\theta) = \max_{0 \le c \le \theta} \left\{ (1-\delta)\, u(c) + \delta\, V((\theta - c)R) \right\}.$$

The consumer's trade-off is transparent: consuming more today (higher $u(c)$) means saving less and arriving at a lower wealth state tomorrow (lower $V((\theta - c)R)$). The weights $(1-\delta)$ and δ govern how she balances present enjoyment against future wealth.

Under standard assumptions (u bounded and continuous, $\delta R < 1$ to ensure the problem is well-behaved), the contraction mapping theorem guarantees a unique value function V, and the policy function $c^*(\theta)$ tells us how much to consume at each wealth level.

This example shows dynamic programming in its most natural habitat: the consumer does not need to plan her entire life's consumption at once. She just needs to know V and solve a one-period problem. The value function encodes everything about the future that matters for today's decision.

11.10 Where this appears in Mas-Colell, Whinston, and Green (MWG)

Dynamic programming is not developed as a standalone topic in MWG's mathematical appendices. It appears in the economic chapters:

- **MWG, Chapter 20** (intertemporal choice and dynamic optimization),

- **MWG, Chapters 12 and 13** (repeated games, where continuation values play the role of the value function).

The contraction mapping theorem is stated in **MWG, Mathematical Appendix C**. The connection between the Bellman equation and the contraction mapping theorem is typically developed in more specialized references.[6]

11.11 Why this matters for microeconomics

Dynamic programming is how economists tame infinite-horizon problems.

It matters because:

- it converts infinite-dimensional optimization into a sequence of one-period problems,
- it provides both existence and uniqueness of optimal behavior (via the contraction mapping theorem),
- it gives a computational method (value iteration),
- and it underlies the recursive formulation of repeated games, dynamic contracts, and mechanism design.

In the PhD course, you will see continuation values used as incentive instruments — a principal can promise future rewards or threaten future punishments to influence behavior today. The Bellman equation is what makes "the value of the future" a precise, well-defined object that can be manipulated and optimized over.

11.12 What you should take away

After this topic, you should be comfortable with:

- reading Bellman equations and understanding what the value function represents,
- seeing the Bellman equation as a fixed-point problem in function space,
- knowing what the contraction mapping theorem says and why it gives both existence and uniqueness,
- understanding that $\delta < 1$ is what makes the Bellman operator a contraction,
- and recognizing that dynamic programming reduces infinite-horizon problems to sequences of static problems.

[6] Stokey, Lucas, and Prescott, "Recursive Methods in Economic Dynamics" (1989), is the standard reference for economists. It develops the contraction mapping approach to dynamic programming in detail. Kreps (1995), "A Course in Microeconomic Theory," has a more micro oriented treatment.

This is the last major mathematical tool. From here, the remaining topics refine and extend what we have already built.

Chapter 12

Topic 12: Linear Constraints and Incentives as Geometry

(Why prices are perpendicular and constraints have shape)

12.1 Why geometry enters now

Much of what we have done so far has been algebraic or analytic: writing down objective functions, taking derivatives, applying fixed-point theorems. But many of the most important ideas in microeconomics have a clean geometric interpretation that makes them easier to understand and harder to forget.

This topic develops the geometry of linear constraints. The central objects are **hyperplanes** and **half-spaces** — the building blocks of budget sets, feasibility constraints, and incentive conditions. The central results are the **separating** and **supporting hyperplane theorems**, which are the geometric engine behind the welfare theorems, the existence of prices, and the structure of mechanism design.

The connection to earlier material is direct: convexity (Topic 3) is what makes separation possible, and optimization (Topics 5–6) is what supporting hyperplanes formalize.

12.2 Hyperplanes and half-spaces

Hyperplanes and half-spaces are very simple and intuitive concepts. It turns out however that they are also very powerful. Notation wise they look much more scary than they actually are, the picture in 2-D essentially tells it all and the great thing about is, is that it generalizes to arbitrary dimensions (the machinery behind is the Hahn-Banach extension theorem—a theorem we won't discuss here in detail).

12.2.1 Hyperplanes

A **hyperplane** in $\mathbb{R}^n$ is a set of the form

$$H = \{x \in \mathbb{R}^n : p \cdot x = c\},$$

where $p \in \mathbb{R}^n$ is a nonzero vector and $c \in \mathbb{R}$ is a scalar.

> "A hyperplane is a flat surface of dimension $n-1$ that divides the space into two sides."

In $\mathbb{R}^2$, a hyperplane is a line. In $\mathbb{R}^3$, it is a plane. In general, it is the set of points whose inner product with a fixed direction p equals a constant, so in $\mathbb{R}^2$ these are all points (x, y) such that $ax + by = c$ for some given $p = (a, b)$ and c, e.g., if $a = 1, b = 2, c = 3$ we need to find $x + 2y = 3 \Leftrightarrow x = 3 - 2y \Rightarrow$ a line.

An interesting property is that the vector p is the **normal** to the hyperplane: it is perpendicular to H and points in the direction of increasing $p \cdot x$.

In our notation choice you see the tight connection to a budget set. If c is the budget, and (x, y) is the bundle of goods, than $p = (a, b)$ is the price vector. The hyperplane is then the set of points the consumer may buy if she usses *all* her budget. If there are more than 2 goods (like, e.g., in reality), your choice set has more dimensions, and so has the hyperplane. Because the hyperplane is a linear equation with n unknowns, we can solve for 1 of those unknowns, leading to a sphere in $n-1$ dimensions. However, if we allow points to be *negative*,[1] then there is a whole space of $n-1$ dimensions floating around in the n dimensions.

Another way to think of this is to say. Let's start with the an n dimensional space and consider all points $(x_i)_{i=1}^n$ such that $x_n = 0$. Here it is (I hope) clear to see that we get an $n-1$ dimensional space because there are no restrictions on $(x_i)_{i=1}^n$. But then that space is the hypeplane that we get from selecting $p = (0, 0, ..., 1)$ and $c = 0$. Now, let's imagine the x_n dimension is the one going straight up. Then changing $c \neq 0$ while keeping p implies we shift our hyperplane up and down. Changing p on the other hand means we rotate it.[2]

12.2.2 Half-spaces

A hyperplane cuts $\mathbb{R}^n$ into two **half-spaces**:

$$H^- = \{x : p \cdot x \leq c\} \quad \text{and} \quad H^+ = \{x : p \cdot x \geq c\}.$$

[1] For example because you could sell them on the market at market price, or, as in financial markets go short on an assest without even owning it.

[2] One should note that p is not unique for a given direction (a fact you may remember from highschool algebra), we can scale it, by any scalar $m \neq 0$ without changing the direction.

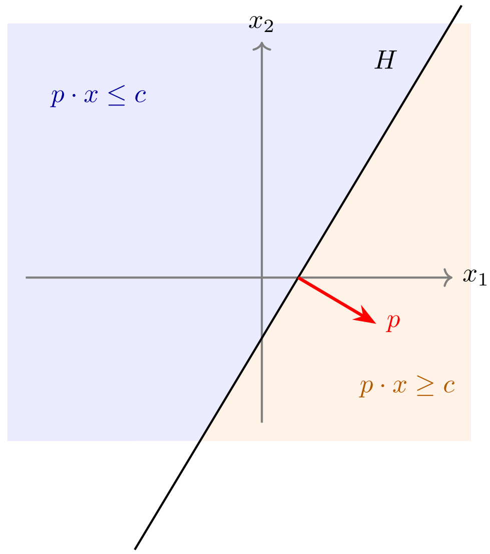

Figure 12.1: Figure 1

> "A half-space is everything on one side of a hyperplane, including the hyperplane itself."

Again, think about the budget interpretation. The hyperplane (the budget line) divides the set of possibilities into those that you can afford given budget c (H^-) and those in which leaves you with no money leftover (H^+).

Half-spaces are convex sets (Topic 3). This is immediate: if $p \cdot x \leq c$ and $p \cdot y \leq c$, then for any $\lambda \in [0, 1]$,

$$p \cdot (\lambda x + (1 - \lambda)y) = \lambda(p \cdot x) + (1 - \lambda)(p \cdot y) \leq c.$$

Intersections of half-spaces are also convex (the intersection of convex sets is convex). This means that any set defined by a system of linear inequalities is convex. Such sets are called **polyhedra**, and they are everywhere in economics.

12.3 Budget sets as half-spaces

Let's now discuss the budget set a bit more structured. The budget set is probably the most familiar linear constraint in economics is the **budget set**:

$$B(p, w) = \{x \in \mathbb{R}^n_+ : p \cdot x \leq w\}.$$

This is the intersection of a half-space ($p \cdot x \leq w$) with the nonnegative orthant ($x \geq 0$). The **budget line** is the hyperplane $p \cdot x = w$ intersected with the half space defined by $x \geq 0$; the boundary of the budget set.

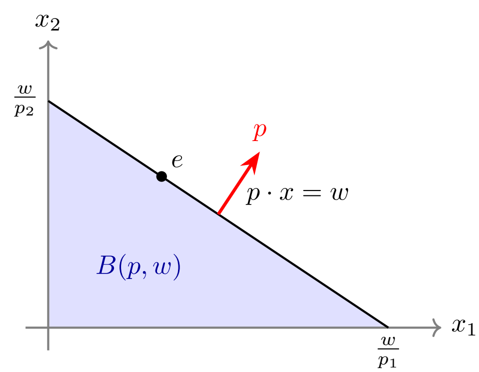

Figure 12.2: Budget Set

The geometry makes several things visible that algebra obscures:

- The price vector p is **perpendicular** to the budget line. Moving along the budget line (spending the same amount) means moving perpendicular to p. Moving in the direction of p means spending more without change the relative proportions.
- The **slope** of the budget line is $-p_1/p_2$: the rate at which the market allows you to trade good 1 for good 2.
- A change in prices **rotates** the budget line. A change in wealth **shifts** it.

This geometric perspective is not just a visualization aid. It is the basis for understanding why Walrasian equilibrium prices exist and why they have the structure they do.

12.4 The separating hyperplane theorem

The most important geometric result for economics is the **separating hyperplane theorem**. It says: two disjoint convex sets[3] can always be separated by a hyperplane. This sounds trivial and to some extend it is. So don't try to read more into it than there is. Often the difficulty is defining the hyperplanes.

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[3] Sets are disjoint if their intersection is empty.

Theorem (Separating Hyperplane). Let $A, B \subseteq \mathbb{R}^n$ be nonempty, disjoint, convex sets. Then there exists a nonzero vector $p \in \mathbb{R}^n$ and a scalar $c \in \mathbb{R}$ such that

$$p \cdot a \leq c \quad \text{for all } a \in A, \qquad p \cdot b \geq c \quad \text{for all } b \in B.$$

> "If two convex sets don't touch, there is a flat surface that passes between them, putting each set entirely on one side."

A corollary to the separating hyperplane theorem is that we can always find a halfspaces H such that A is entirely in H and B is entirely in the complement to it H.

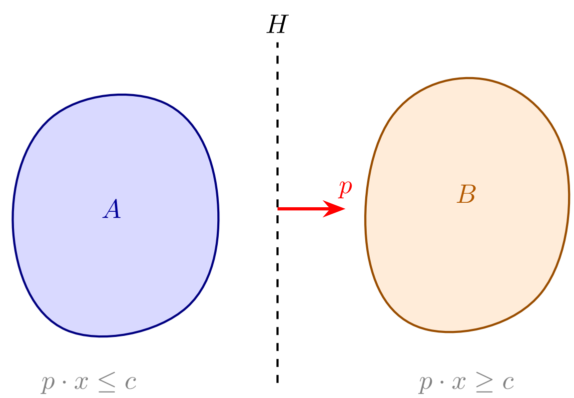


Figure 12.3: Separating Hyperplane

The vector p defines the separating hyperplane $\{x : p \cdot x = c\}$. The set A lies in the half-space $p \cdot x \leq c$ and B lies in $p \cdot x \geq c$.

We do not prove this result here. But the geometric content is intuitive: if two convex blobs don't overlap, you can always slide a flat surface between them. If one wants to prove separating hyperplanes there are two roads. One is to restrict oneself to finite dimensions as we do up here. In that case the separating hyperplanes can be proven using the *supporting hyperplane* theorem below—we will come back to that road.

Alternatively, if one wants to go for infinite dimension, one would need to restrict to *compact* sets, too. Otherwise, the theorem does not hold. Then there is proof that uses the Hahn-Banach theorem. We will not push in that dimension further here.

12.4.1 Why separation matters for economics

The separating hyperplane theorem is the mathematical engine behind:

The Second Welfare Theorem. Any Pareto efficient allocation can be supported as a competitive equilibrium with appropriate transfers. The proof works by separating the "better-than" set of one consumer from the aggregate production set. The separating hyperplane gives us the **prices**.

The existence of supporting prices. When we say "an efficient allocation can be decentralized by prices," we are saying: there exists a hyperplane (defined by prices) that separates the sets of preferred allocations from what is feasible. The prices are literally the normal vector of the separating hyperplane.

Duality in optimization. The connection between primal and dual problems in constrained optimization is fundamentally about separating hyperplanes. The dual variables (Lagrange multipliers, shadow prices) are the normals of hyperplanes that separate the feasible set from the "better" set.

12.5 The supporting hyperplane theorem

A closely related result is the **supporting hyperplane theorem**. Instead of separating two disjoint sets, it finds a hyperplane that "touches" a convex set at a boundary point without cutting into it.

Theorem (Supporting Hyperplane). Let $C \subseteq \mathbb{R}^n$ be a nonempty, convex set, and let x^* be a point on the boundary of C. Then there exists a nonzero vector $p \neq 0$ such that

$$p \cdot x \leq p \cdot x^* \quad \text{for all } x \in C.$$

> "At any boundary point of a convex set, there is a flat surface that touches the set without entering its interior. The entire set lies on one side."

The supporting hyperplane $\{x : p \cdot x = p \cdot x^*\}$ is tangent to C at x^*. The normal vector p points "outward" — away from the interior of C.

The proof uses one geometric fact that is both intuitive and powerful: if you stand outside a closed convex set, there is a unique closest point in the set, and the line connecting you to that closest point defines a separating direction.

12.5.1 Proving supporting hyperplanes and with it separating ones

We prove things here for a closed convex set. It doesn't matter though really, because the closure of C is also convex and the smallest closed set containing C.

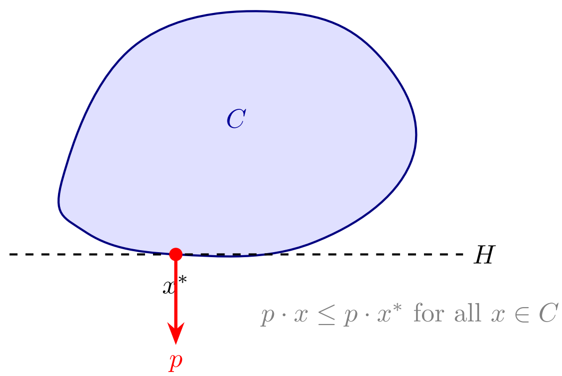


Figure 12.4: Supporting Hyperplane

So if the result holds for a closed C, it holds for any convex set whose closure is C.

Step 0: The closest-point property. Let C be a closed convex set and $y \notin C$. Then there exists a unique point $z \in C$ that is closest to y.[4]

Now set $p = y - z$. We claim that $p \cdot x \leq p \cdot z$ for all $x \in C$. In words: the hyperplane through z with normal p puts all of C on one side and y on the other.

Why? Suppose some $x \in C$ had $p \cdot x > p \cdot z$. Then look at the point $z_t = z + t(x - z)$ for small $t > 0$. By convexity of C, $z_t \in C$. A direct calculation gives

$$\|y - z_t\|^2 = \|y - z\|^2 - 2t\, p \cdot (x - z) + t^2 \|x - z\|^2.$$

The second term is $-2t(p \cdot x - p \cdot z) < 0$ for small enough t (because we assumed $p \cdot x > p \cdot z$), and the third term is bounded by t^2, or $O(t^2)$, so for small t the whole expression is less than $\|y - z\|^2$. But that means z_t is closer to y than z is, contradicting the fact that z is the closest point. So $p \cdot x \leq p \cdot z$ for all $x \in C$.

Here is the geometric intuition for step 0:

> "Drop a perpendicular from y to the convex set. The foot of the perpendicular is the closest point, and the hyperplane at the foot separates y from C."

Step 1: From separation to support. Now we use the closest-point property to prove the supporting hyperplane theorem. Take x^* on the boundary of C. Because x^* is a boundary point, there exist points outside C that are arbitrarily close to it. Of those we now take a sequence $y_k \notin C$ with $y_k \to x^*$.

[4]Existence follows from closedness and boundedness of the relevant sublevel set (the same logic as the Weierstrass theorem in Topic 4). Uniqueness follows from strict convexity of the distance function: if two points $z_1, z_2 \in C$ were both closest, the midpoint $(z_1 + z_2)/2 \in C$ (by convexity) would be strictly closer — a contradiction.

For each y_k, let $z_k = \mathrm{proj}_C(y_k)$ be the closest point in C, and define the unit normal

$$p_k = \frac{y_k - z_k}{\|y_k - z_k\|}.$$

By Step 0, we know that $p_k \cdot x \le p_k \cdot z_k$ for all $x \in C$.

Step 2: Extract a limit. By construction, each p_k is a unit vector (it lives on the unit sphere in $\mathbb{R}^n$). In finite dimensions, the unit sphere is compact (Topic 4), so the sequence (p_k) has a convergent subsequence with limit p satisfying $\|p\| = 1$ (in particular, therefore, $p \neq 0$).

Along this subsequence, $z_k \to x^*$ (because z_k is between y_k and C, and $y_k \to x^*$, which lies on the boundary of C, the closest points must approach x^* as well).

Taking the limit in the inequality $p_k \cdot x \le p_k \cdot z_k$ gives

$$p \cdot x \le p \cdot x^* \quad \text{for all } x \in C.$$

That is the supporting hyperplane.

> "Approach the boundary from outside, project onto the set at each step, and take a limit. The limit of the projection directions is the normal to the supporting hyperplane."

Every ingredient in this proof comes from the course: convexity gives us the closest point and keeps the set well-behaved (Topic 3), compactness of the unit sphere gives us the convergent subsequence (Topic 4), and the limit argument uses continuity of the inner product.

12.5.2 From supporting to separating

The separating hyperplane theorem then follows from the supporting one. The construction is worth seeing because it is a beautiful example of reducing one problem to another.

If A and B are disjoint convex sets, form the so-called **Minkowski difference**

$$A - B = \{a - b : a \in A,\ b \in B\}.$$

This set is convex (the "sum" of two convex sets is convex, and $A - B = A + (-B)$). Because A and B are disjoint, $0 \notin A - B$ (if $0 = a - b$ for some $a \in A, b \in B$, then $a = b \in A \cap B$, contradicting disjointness).

So 0 is a point outside the convex set $\overline{A - B}$. Find the closest point $z^* \in \overline{A - B}$ to the origin, and set $p = -z^*$ (the direction from z^* toward the origin). By the closest-point property (Step 0), $p \cdot z \le p \cdot z^*$ for all $z \in \overline{A - B}$. Moreover, $p \cdot z^* = -\|z^*\|^2 < 0 = p \cdot 0$, so the origin is strictly on the other side.

In particular, for every $a \in A$ and $b \in B$, the element $a - b \in A - B$ satisfies $p \cdot (a - b) \leq p \cdot z^* < 0$, which gives

$$p \cdot a < p \cdot b \quad \text{for all } a \in A,\ b \in B.$$

Setting $c = \sup_{a \in A} p \cdot a$, the hyperplane $\{x : p \cdot x = c\}$ separates A from B.

> "To separate two convex sets: subtract one from the other, support the difference at the origin, read off the separating direction. Then shift it toward the set you want to separate from."

12.5.3 Why support matters for economics

Supporting hyperplanes formalize what happens at an optimum. If an agent maximizes a linear objective over a convex constraint set, the optimal point is on the boundary, and the objective's gradient defines a supporting hyperplane there.

This is the geometric content of the **first-order conditions**. When we write $\nabla u(x^*) = \lambda p$ at a consumer's optimum, we are saying: the indifference surface through x^* is tangent to the budget hyperplane. The price vector p is the normal of the supporting hyperplane, and λ is a scaling factor.

In welfare economics, the supporting hyperplane theorem is used in the **First Welfare Theorem direction**: at a Walrasian equilibrium, the price hyperplane supports each consumer's "at least as good" set at their chosen bundle.

12.6 Incentive constraints as geometry

The geometric perspective extends naturally to mechanism design, where constraints are often linear in transfers and allocations.

Consider a simple mechanism design problem: a principal offers a menu of contracts (q, t) — a quality level q and a transfer t — to an agent whose type θ is private information. The agent's utility from a contract is

$$u(\theta) = \theta q - t.$$

12.6.1 Incentive compatibility as half-spaces

The **incentive compatibility** (IC) constraint says: each type must prefer her own contract to any other type's contract. If there are two types $\theta_H > \theta_L$ with contracts (q_H, t_H) and (q_L, t_L), the IC constraints are:

$$\theta_H q_H - t_H \geq \theta_H q_L - t_L \quad \text{(high type doesn't mimic low type)},$$

$$\theta_L q_L - t_L \geq \theta_L q_H - t_H \quad \text{(low type doesn't mimic high type)}.$$

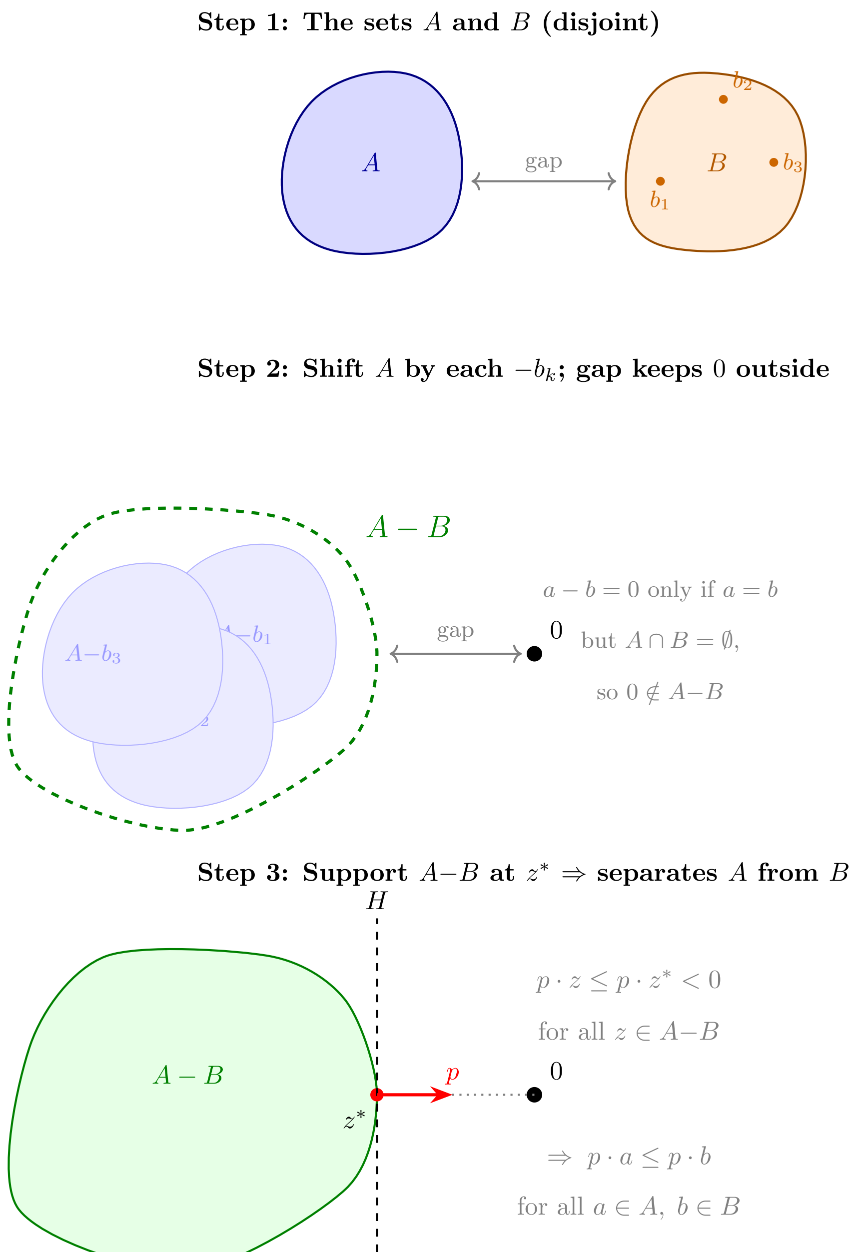


Figure 12.5: Separating from supporting

Each of these is a linear inequality in the contract variables (q_H, t_H, q_L, t_L). Each defines a half-space. The set of incentive-compatible contracts is the intersection of these half-spaces — a polyhedron.

12.6.2 Individual rationality as half-spaces

The **individual rationality** (IR) constraint says: each type must get at least her outside option. If the outside option is zero:

$$\theta_H q_H - t_H \geq 0, \qquad \theta_L q_L - t_L \geq 0.$$

Again, linear inequalities. Again, half-spaces.

12.6.3 The geometry of mechanism design

The full set of feasible mechanisms is the intersection of all IC and IR constraints — a polyhedron in the space of contracts. The principal's problem is to maximize revenue (or another linear objective) over this polyhedron.

This is a **linear programming** problem, and its geometry is clean:

- the feasible set is a polyhedron,
- the optimum is at a vertex (or a face) of this polyhedron,
- the binding constraints at the optimum tell us which incentive and participation conditions are "tight."

The result that "in the optimal mechanism, the low type's IR binds and the high type's downward IC binds" is a statement about which faces of the polyhedron the optimum lies on. The geometric perspective makes this visible rather than mysterious.

12.7 An economic example: the Second Welfare Theorem

Consider a two-consumer exchange economy. Consumer 1 has preferences over bundles $x_1 \in \mathbb{R}^2_+$ and consumer 2 over $x_2 \in \mathbb{R}^2_+$, with total endowment $\bar{\omega}$.

Let $(\hat{x}_1, \hat{x}_2)$ be a Pareto efficient allocation. Define:

- $A =$ the set of bundles that consumer 1 strictly prefers to $\hat{x}_1$,
- $B =$ the set of aggregate bundles that are feasible: $\{y : \bar{\omega} - y$ is in consumer 2's "at least as good as $\hat{x}_2$" set$\}$.

Because preferences are convex, both A and B are convex.[5] Because $(\hat{x}_1, \hat{x}_2)$ is

[5] Convexity of preferences ensures that the upper contour sets — the "at least as good as" sets — are convex. This is where the convexity assumptions from Topic 3 earn their place.

Pareto efficient, A and B are disjoint (there is no bundle that is strictly better for 1 and still feasible for 2).

By the separating hyperplane theorem, there exists a price vector p that separates A from B. This p is the vector of Walrasian equilibrium prices.

> "The Second Welfare Theorem says: efficient allocations can be decentralized by prices. The proof is: use the separating hyperplane theorem. The prices are the normal vector of the separating hyperplane."

This is perhaps the cleanest example of how geometry does real economic work.

12.8 Where this appears in Mas-Colell, Whinston, and Green (MWG)

- **MWG, Chapter 3** (budget sets and consumer choice — budget sets as half-spaces),
- **MWG, Chapter 16** (the Second Welfare Theorem — separating hyperplane argument),
- **MWG, Mathematical Appendix A** (convexity, separation theorems).

The geometric approach to mechanism design is not developed extensively in MWG but is standard in the mechanism design literature.

12.9 Why this matters for microeconomics

Linear constraints and separation theorems are the geometric foundation of:

- consumer theory (budget sets, demand, Slutsky),
- welfare economics (the First and Second Welfare Theorems),
- general equilibrium (existence of prices, Walrasian equilibrium),
- mechanism design (incentive and participation constraints as polyhedra),
- duality (shadow prices, Lagrange multipliers as supporting hyperplane normals).

The key insight running through all of this is: **convexity makes separation possible, and separation gives us prices**. This is why convexity assumptions (Topic 3) are not just technical convenience — they are what allows the price system to work.

12.10 What you should take away

After this topic, you should be comfortable with:

- reading hyperplanes, half-spaces, and polyhedra as geometric objects,
- seeing budget sets as half-spaces with the price vector as the normal,
- understanding the separating and supporting hyperplane theorems and their economic content,
- recognizing that incentive and participation constraints in mechanism design define polyhedra,
- and seeing prices as normal vectors of separating or supporting hyperplanes.

The geometry in this topic is not decoration. It is the language in which the deepest results of welfare economics and mechanism design are stated and proved.

Chapter 13

Topic 13: Topology in Practice

(How metrics structure change and space)

13.1 Picking up the threads

In Topic 4, when we introduced continuity, we also introduced the notion of a **metric**: a function $d : X \times X \to \mathbb{R}$ satisfying nonnegativity, the identity of indiscernibles, symmetry, and the triangle inequality. We noted that every topological concept from that topic—open sets, closed sets, convergent sequences, continuity, compactness—can be defined word-for-word in any metric space, with $d(x, x')$ replacing the Euclidean norm $\|x - x'\|$. And we deferred the question of why economists might need metrics on objects other than vectors.

In Topic 11 those "other" objects showed up, but we did not stop to explain it. The contraction mapping theorem requires a **complete** metric space. The space we used was the space of bounded functions $f : \Theta \to \mathbb{R}$, equipped with the **sup metric**

$$d(f, g) = \sup_{\theta \in \Theta} |f(\theta) - g(\theta)|.$$

The points in that metric space are functions — not vectors of numbers — and the metric measures the largest pointwise difference between two functions. We invoked completeness of that space as a fact. We did not ask: why the sup metric? What makes it the right one? What would go wrong with a different choice?

This topic aims to tie up those loose ends and asks one further question:

- When a correspondence maps parameters to *sets* of choices, how do we measure how much the set changes with the parameter? (The Hausdorff metric.)

A third natural question — when agents hold probability distributions over types or states, what is the right notion of "two distributions being close"? — requires the integral to state properly. We defer it to Topic 14, where we will have that machinery.

With three spaces already on the table — Euclidean space, function spaces, and spaces of sets — we cover most of what formal micro requires. The space of probability distributions joins them in Topic 14.

13.2 Why the metric matters: two spaces in economics

For most of this course, the objects we compare are vectors of numbers, and the Euclidean metric is the natural choice. But to handle more complex problems, we need to think about other spaces.

Value functions. A value function $V : \Theta \to \mathbb{R}$ assigns a number to each state. Two value functions are "close" if they assign similar numbers to every state. The sup metric captures this by measuring the maximum difference between the functions across all states. The Euclidean metric makes no sense here: V is not a finite-dimensional vector (because the set of states is often infinite and not countable). But then how should we measure distance?

Correspondences and their values. A budget correspondence $B(p, w)$ maps price-wealth pairs to subsets of the commodity space. As (p, w) changes, the budget set changes. Upper and lower hemicontinuity (Topic 5) were our way of saying the set changes "without jumps." But there is a cleaner formulation: there is a metric on compact subsets — the Hausdorff metric — such that hemicontinuity is just ordinary continuity with respect to that metric. The two-condition definition of continuity for correspondences was not an ad hoc invention; it is the natural decomposition of the Hausdorff distance into its two directed parts. Thus, the Hausdorff metric tells us why UHC and LHC are the correct conditions for continuity of correspondences.

Each of these spaces requires its own metric, and the choice of metric is not cosmetic: it determines what "convergent sequence" means, what "continuous" means, and what "compact" means.

13.3 The sup metric on function spaces

Let Θ be a nonempty set. Denote by $B(\Theta)$ the set of all **bounded** functions $f : \Theta \to \mathbb{R}$ — functions for which there exists $M < \infty$ with $|f(\theta)| \le M$ for all $\theta \in \Theta$.[1]

The **sup metric** on $B(\Theta)$ is

$$d_\infty(f, g) := \sup_{\theta \in \Theta} |f(\theta) - g(\theta)|.$$

Let's check that this is a metric.

- *Nonnegativity.* $|f(\theta) - g(\theta)| \ge 0$ for each θ, so $d_\infty(f, g) \ge 0$. [ok]
- *Identity.* $d_\infty(f, g) = 0$ if and only if $|f(\theta) - g(\theta)| = 0$ for all $\theta \in \Theta$, i.e., $f = g$. [ok]
- *Symmetry.* $|f(\theta) - g(\theta)| = |g(\theta) - f(\theta)|$, so $d_\infty(f, g) = d_\infty(g, f)$. [ok]
- *Triangle inequality.* For any θ and any three functions f, g, h:

  $$|f(\theta) - h(\theta)| \le |f(\theta) - g(\theta)| + |g(\theta) - h(\theta)| \le d_\infty(f, g) + d_\infty(g, h).$$

  Taking the supremum over θ: $d_\infty(f, h) \le d_\infty(f, g) + d_\infty(g, h)$. [ok]

So $(B(\Theta), d_\infty)$ is a metric space. Now: why this metric rather than, say, the integral metric $d_1(f, g) = \int_\Theta |f(\theta) - g(\theta)|\, d\theta$? (The integral will be developed properly in Topic 14; for now, think of d_1 as summing pointwise differences across all states.) Both are valid metrics for suitable function spaces. But they have different completeness properties, and completeness is what the Bellman operator needs.

13.3.1 Completeness and why it matters

$(B(\Theta), d_\infty)$ is **complete**: every Cauchy sequence in d_∞ converges to a limit in $B(\Theta)$.

Here is why. Suppose $\{f_n\}$ is a Cauchy sequence under d_∞. That means for every $\varepsilon > 0$, there exists N such that $d_\infty(f_m, f_n) < \varepsilon$ for all $m, n \ge N$.

Now under the sup norm we know that for every $\theta \in \Theta$ and every $m, n \ge N$, $|f_m(\theta) - f_n(\theta)| < \varepsilon$. So at each fixed θ, the real sequence $\{f_n(\theta)\}$ is Cauchy. Since $\mathbb{R}$ is complete, the pointwise limit $f(\theta) = \lim_n f_n(\theta)$ exists for every θ. One then checks that f is bounded and that $d_\infty(f_n, f) \to 0$.[2]

[1] In dynamic programming, we restrict to bounded functions because any candidate value function must be bounded if per-period payoffs are bounded and $\delta < 1$. The restriction is substantively harmless in those applications.

[2] The key step is that the Cauchy condition in d_∞ controls the error *uniformly* in θ: the same N works for every θ simultaneously. That is what "sup" over θ buys. This uniform control is what allows the pointwise limits to assemble into a limit in d_∞ rather than merely a pointwise limit.

The integral metric d_1 is not complete on $B(\Theta)$: a Cauchy sequence in d_1 can converge pointwise to a limit that is unbounded, or can fail to converge uniformly. For the Bellman operator, we need completeness to guarantee that the iterates $V_0, T(V_0), T(T(V_0)), \ldots$ converge to a fixed point *inside* the function space, not to a limit that doesn't belong to it. That is why the sup metric is the right metric for dynamic programming.

13.3.2 Convergence in d_∞ is uniform convergence

The statement "$f_n \to f$ in d_∞" is equivalent to the classical notion of **uniform convergence**: for every $\varepsilon > 0$, there exists N such that $|f_n(\theta) - f(\theta)| < \varepsilon$ for all $n \geq N$ and *all* $\theta \in \Theta$ simultaneously. The threshold N does not depend on θ.

This is stronger than another form of convergence, **pointwise convergence**. There N is allowed to depend on θ. If you only have pointwise convergence, you might have $f_n(\theta) \to f(\theta)$ at every individual state, but at some states the convergence is slower and slower, and there is no uniform bound on the error across all states.

An example would be $f_n(\theta) = \theta^n$ on $\Theta = [0, 1]$. For each fixed $\theta \in [0, 1)$, $f_n(\theta) \to 0$ as $n \to \infty$, so we have pointwise convergence to the zero function $f(\theta) = 0$. But for any n, $f_n(1) = 1$, so the sup $|f_n(\theta) - f(\theta)| = 1$ for all n, so we don't get uniform convergence.

For the Bellman operator, which processes the entire function at once, such non-uniform errors would invalidate the contraction argument. Uniform convergence—convergence under d_∞—ensures that indeed the functions are getting closer everywhere to the limit already before we reach said limit.

13.4 The Hausdorff metric on compact subsets

Now consider a correspondence $\Phi : T \to 2^X$ mapping a parameter t to a subset of X. We want a metric on sets so that continuity of Φ in that metric is equivalent to the hemicontinuity conditions from Topic 5. The answer is the Hausdorff metric.

13.4.1 Definition

Let (X, d) be a compact metric space and let $\mathcal{K}(X)$ denote the collection of all nonempty compact subsets of X.[3]

[3] Note that here we start out with a metric already defined on X for points. Now we are defining a new metric on sets of points building on that original metric. The Hausdorff metric is a way of measuring distance between sets based on the underlying metric on points.

For two sets $A, B \in \mathcal{K}(X)$, define the **one-sided distance** from A to B as

$$\vec{d}(A, B) = \sup_{a \in A} \inf_{b \in B} d(a, b).$$

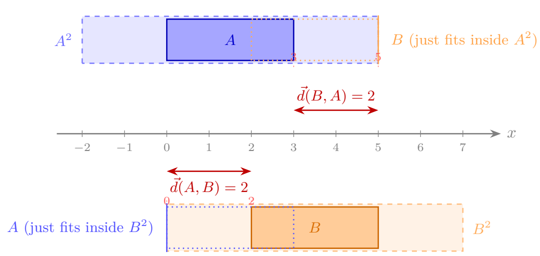
This asks: what is the farthest that any point of A can be from the set B? Equivalently, $\vec{d}(A, B)$ is the smallest $\varepsilon \geq 0$ such that every point of A lies within distance ε of some point of B.

The one-sided distance is not symmetric. If $A \subset B$ and $A \neq B$,[4] then every point of A is already in B, so $\vec{d}(A, B) = 0$; but some points of B are not in A, so $\vec{d}(B, A) > 0$.

Reading: $A = [0, 3]$, $B = [2, 5]$, $\varepsilon = d_H(A, B) = \max\{\vec{d}(A, B),\ \vec{d}(B, A)\} = \max\{2, 2\} = 2$.

Top strip: $A^2 = [-2, 5]$ expands A by $\varepsilon = 2$ in each direction. Its right end (5) coincides with the right end of B (5) — B just fits.

Bottom strip: $B^2 = [0, 7]$ expands B by $\varepsilon = 2$ in each direction. Its left end (0) coincides with the left end of A (0) — A just fits.



The Hausdorff distance is the smallest ε such that each set fits inside the ε-neighborhood of the other *simultaneously*.

Figure 13.1: Hausdorff metric

The **Hausdorff metric** symmetrizes by taking the larger of the two one-sided distances:

$$d_H(A, B) = \max\{\vec{d}(A, B),\ \vec{d}(B, A)\}.$$

In words: $d_H(A, B)$ is the smallest ε such that A fits inside the ε-neighborhood of B and B fits inside the ε-neighborhood of A — simultaneously.

13.4.2 It is a metric

- *Nonnegativity.* Clear. [ok]
- *Identity.* $d_H(A, B) = 0$ iff $\vec{d}(A, B) = 0$ and $\vec{d}(B, A) = 0$, i.e., every point of A is in the closure of B and vice versa. Since A and B are compact (hence closed), this means $A = B$. [ok]

[4]Normally writing $A \neq B$ is superfluous because $\subset$ is already strict. However, there are some conventions that treat $\subset$ as $\subseteq$, so I wanted to be clear.

- *Symmetry.* Immediate from the max. [ok]
- *Triangle inequality.* If every point of A is within ε of some point of B, and every point of B is within δ of some point of C, then every point of A is within $\varepsilon + \delta$ of some point of C. Taking $\varepsilon = d_H(A, B)$ and $\delta = d_H(B, C)$ gives $d_H(A, C) \leq d_H(A, B) + d_H(B, C)$. [ok]

When (X, d) is compact, $(\mathcal{K}(X), d_H)$ is itself compact—a fact known as Blaschke's selection theorem.[5]

13.4.3 Connection to hemicontinuity

Here is the payoff. A correspondence $\Phi : T \to \mathcal{K}(X)$ is:

- **upper hemicontinuous** at t (in the sense of Topic 5) if and only if $\vec{d}(\Phi(t_n), \Phi(t)) \to 0$ whenever $t_n \to t$,
- **lower hemicontinuous** at t if and only if $\vec{d}(\Phi(t), \Phi(t_n)) \to 0$ whenever $t_n \to t$.

It is **continuous in the Hausdorff metric** — meaning $d_H(\Phi(t_n), \Phi(t)) \to 0$ — if and only if it is both upper and lower hemicontinuous.

Upper hemicontinuity controls the one-sided distance from the image at the nearby parameter to the image at the limit: the set doesn't suddenly grow. Lower hemicontinuity controls the other direction: the set doesn't suddenly shrink. The Hausdorff metric demands both simultaneously. The two-condition definition of continuity for correspondences from Topic 5 was not an ad hoc invention; it is the natural decomposition of d_H into its two directed parts, $\vec{d}$ and $\vec{d}$ reversed.

In light of the Hausdorff metric, upper hemicontinuity then means that for any sequence of inputs $t_n \to t$ the distance of $\Phi(t_n)$ to $\Phi(t)$ goes to zero, but the distance of $\Phi(t)$ to $\Phi(t_n)$ does not necessarily go to zero.

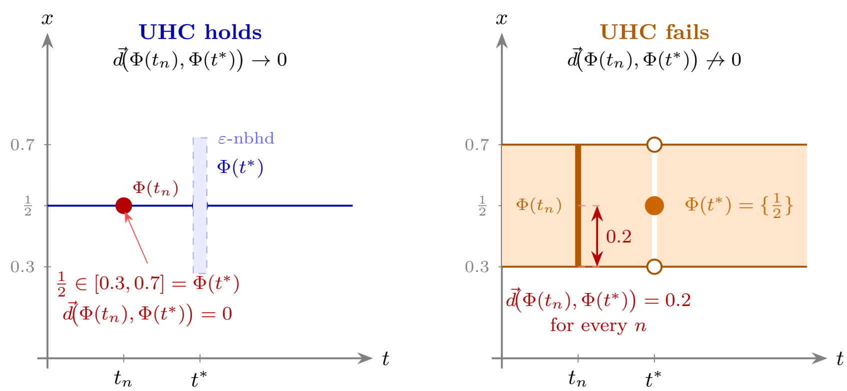
Lower hemicontinuity means that the distance of $\Phi(t)$ to $\Phi(t_n)$ goes to zero, but the distance of $\Phi(t_n)$ to $\Phi(t)$ does not necessarily go to zero.

Continuity in the Hausdorff metric means that both hold.

13.4.4 Economic example: the budget correspondence

The budget set $B(p, w) = \{x \in \mathbb{R}^L_+ : p \cdot x \leq w\}$ maps price-wealth pairs to compact subsets of the commodity space (when bounded). As we established in Topic 5, B is both upper and lower hemicontinuous in (p, w) at any (p, w) with $p \gg 0$ and $w > 0$. The Hausdorff translation: $d_H(B(p_n, w_n), B(p, w)) \to 0$ whenever $(p_n, w_n) \to (p, w)$. Small price changes produce small changes in the budget set, measured by the Hausdorff distance.

[5] Not in MWG. See Aliprantis and Border, *Infinite Dimensional Analysis*, 3rd ed., Springer, 2006, Theorem 3.85.



Left:Φ(t)=1/2 for $t \neq t^*$, $\Phi(t^*) = [0.3, 0.7]$. For $t_n \to t^*$, the limit image is *large*. That means as we walk along our sequence $t_n \to t^*$, $\Phi(t_n) \subset \Phi(t^*)$ always holds. therefore $\vec{d}(\Phi(t_n), \Phi(t^*)) = 0$ for all n. **UHC holds**.
Right: $\Phi(t^*) = \{\frac{1}{2}\}$, $\Phi(t_n) = [0.3, 0.7]$. Now, at the limit, the graph *collapses*. That means, as we walk along our sequence $t_n \to t^*$, we always have to increase $\Phi(t^*)$ to cover $\Phi(t_n)$, no matter the sequence. Therefore, $\vec{d}(\Phi(t_n), \Phi(t^*)) = 0.2 \not\to 0$. **UHC fails.**
Note: Because LHC requires the $\vec{d}$ in the other direction, in this picture the conclusion is flipped for LHC. Full Hausdorff continuity requires it to hold in *both* directions which is not true in either of the graphs.

Figure 13.2: Hausdorff metric and hemicontinuity

Contrast this with what happens when a price component approaches zero: the budget set expands discontinuously. That is, upper hemicontinuity fails at the non-negativity bound. The budget set is not continuous in the Hausdorff metric there.

13.5 Topological spaces: a brief orientation

Everything we have done so far has used a metric — a ruler that assigns a number to every pair of points. But there is a more primitive structure hiding underneath, and it is worth naming it, because it is what most of the vocabulary we actually use is built on.

13.5.1 From metrics to open sets

Recall from Topic 4: given a metric space (X, d), a set $U \subseteq X$ is **open** if every point in U has some breathing room inside U — formally, if for every $x \in U$ there exists $\varepsilon > 0$ such that the open ball $B(x, \varepsilon) = \{y \in X : d(x, y) < \varepsilon\}$ is entirely contained in U. A set is **closed** if its complement is open.

Every concept we have used in this topic and the preceding ones — convergent sequences, continuity, compactness, completeness — can be defined purely in terms of which sets are open, without ever mentioning the metric d directly. For instance:

- A sequence $x_n \to x$ if and only if every open set containing x eventually contains all x_n.
- A function $f : X \to Y$ is continuous if and only if the preimage of every open set in Y is open in X.
- A set K is compact if and only if every collection of open sets that covers K has a finite subcollection that still covers K.

So the metric does two things: it gives us a notion of distance, and it gives us a collection of open sets. The second thing is what all the topological concepts actually use.

13.5.2 The topology generated by a metric

Given a metric d on X, the collection of all open sets it generates — call it τ_d — satisfies three properties that can be verified directly:

1. $\emptyset \in \tau_d$ and $X \in \tau_d$,
2. any union (finite or infinite) of sets in τ_d is again in τ_d,
3. any *finite* intersection of sets in τ_d is again in τ_d.

These three properties are all you need. A **topological space** is a pair (X, τ) where τ is any collection of subsets of X satisfying these three properties —

whether or not there is a metric that generated τ. The sets in τ are called the **open sets of the topology**.

In this language: every metric space (X, d) is automatically a topological space, with topology τ_d. The metric is the extra structure; the topology is what you get when you strip it away and keep only the open sets. Conversely, not every topological space arises from a metric — there exist topologies on a set X that no metric could have generated.[6]

So the hierarchy is: metric space $\Rightarrow$ topological space, but not vice versa. A topological space is strictly more general.

13.5.3 Why go beyond metrics?

For most of the spaces in this course — $\mathbb{R}^n$, the space $B(\Theta)$ of bounded functions, the space $\mathcal{K}(X)$ of compact subsets — a metric is available and we have used it. The reason to work with topologies directly is that on some natural and important spaces, there is more than one useful metric, and the two metrics generate different open sets and hence different notions of convergence, continuity, and compactness.

The leading example in economics is the space of all utility functions, or all strategy profiles in a large game. There are two natural topologies available simultaneously:

- The **norm topology** (also called the strong topology): a sequence $f_n \to f$ if $d_\infty(f_n, f) \to 0$, i.e., uniform convergence. This is exactly the topology generated by the sup metric from §3.
- The **weak topology**: convergence is tested not by checking f_n and f uniformly, but by checking that every "measurement" applied to them yields similar values. The precise definition requires the integral, which we develop in Topic 14. What matters here is the structural point: weak convergence is genuinely weaker than norm convergence, and the two topologies are not equivalent in infinite dimensions.

These are two different topologies on the same underlying set X. The norm topology has more open sets (it is finer); the weak topology has fewer (it is coarser). They agree on finite-dimensional spaces like $\mathbb{R}^n$, but diverge in infinite dimensions.

The choice matters because of compactness. The unit ball in an infinite-dimensional space is compact under the weak topology (this is the Banach-Alaoglu theorem) but never compact under the norm topology. Since fixed-point arguments need compactness, the weak topology is usually the right one in infinite dimensions. We will see this concretely in Topic 14 when we study spaces of probability measures.

[6] Spaces whose topology is generated by some metric are called *metrizable*. Whether a given topological space is metrizable is a nontrivial question; the Urysohn metrization theorem gives sufficient conditions, but we will not need them here.

13.5.4 Product topology

One topology appears repeatedly and deserves a name. If $X = \prod_{i \in I} X_i$ is a Cartesian product, the **product topology** is the coarsest topology making all coordinate projections $\pi_i : X \to X_i$ continuous. A sequence $(x^{(k)})$ converges in the product topology if and only if $\pi_i(x^{(k)}) \to \pi_i(x)$ for each coordinate i separately — coordinatewise convergence.

For a *finite* product, the product topology is the same as the topology from any natural metric (e.g., the Euclidean metric in $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$). For an *infinite* product, the product topology is strictly coarser: it requires convergence in each coordinate, but the speed of convergence is not controlled uniformly across coordinates. This is exactly the right topology for the Mertens-Zamir universal type space, where a type is an infinite hierarchy of beliefs (first-order beliefs over the state, second-order beliefs over others' first-order beliefs, and so on): two types are close if they agree on the first several levels of the hierarchy, with the required agreement growing as the distance threshold shrinks.[7]

13.6 Compactness in general metric spaces

In Topic 4, we established that in $\mathbb{R}^n$, compactness is equivalent to being closed and bounded (Heine-Borel). In a general metric space, closed and bounded is not enough, and we need the correct generalization.

In any metric space, compactness is equivalent to sequential compactness: every sequence has a convergent subsequence.

The proof uses the metric structure essentially — it is not true in all topological spaces, but for metric spaces the equivalence is standard.[8]

13.6.1 Total boundedness

A metric space (X, d) is **totally bounded** if for every $\varepsilon > 0$, the set X can be covered by *finitely many* open balls of radius ε. In $\mathbb{R}^n$, total boundedness is equivalent to ordinary boundedness. But in an infinite-dimensional space, total boundedness is strictly stronger: a ball of radius 1 contains infinitely many

[7] See Mertens and Zamir (1985), "Formulation of Bayesian analysis for games with incomplete information," *International Journal of Game Theory*; and Brandenburger and Dekel (1993), "Hierarchies of beliefs and common knowledge," *Journal of Economic Theory*. These papers construct the universal type space as the completion of the space of all finite-level belief hierarchies under the product topology.

[8] Proof sketch: compactness implies sequential compactness by a diagonalization argument. Sequential compactness implies total boundedness (if the space were not totally bounded, one could find a sequence with no convergent subsequence) and completeness (any Cauchy sequence is a convergent subsequence of itself). Completeness plus total boundedness implies compactness by a covering argument.

points at mutual distance $\sqrt{2}$ from each other, so it is bounded but not totally bounded.

A metric space is compact if and only if it is complete and totally bounded. This generalizes Heine-Borel: in $\mathbb{R}^n$, closed means complete (as a subspace of $\mathbb{R}^n$) and bounded means totally bounded, so the two conditions reduce to the familiar pair.

13.6.2 Why compactness fails in infinite dimensions

In $B(\Theta)$ with the sup metric, the unit ball $\{f : d_\infty(f, 0) \leq 1\}$ is not compact. The sequence $f_n = \mathbf{1}_{\{\theta \geq n^{-1}\}}$ (indicator of a moving interval) is bounded by 1 everywhere, but any two of these functions are at sup distance 1 from each other, so no subsequence is Cauchy.[9]

For the applications in this course — compact type and action spaces, bounded payoffs — the relevant spaces are compact in the appropriate topology, and the tools from Topics 9 and 11 apply directly. The point of raising the failure of Heine-Borel in infinite dimensions is to make clear *why* those compactness assumptions were made: they are not automatic.

13.7 Two applications in the syllabus

13.7.1 Dynamic programming

The Bellman operator $T : B(\Theta) \to B(\Theta)$ is a contraction under d_∞: applying T brings any two value functions closer together by a factor of $\delta < 1$. Completeness of $(B(\Theta), d_\infty)$ ensures the iterates converge. A different metric on functions would not have guaranteed completeness, and the contraction argument would have been vacuous — the iterates might form a Cauchy sequence that converges "outside" the space. The sup metric is not a coincidence; it is the right metric for this problem.

13.7.2 Type spaces and equilibrium existence: a preview

Two further applications — the compactness of spaces of probability measures and its consequences for Bayesian equilibrium existence and mechanism design — require the integral to state precisely. Topic 14 closes this loop: once we know what it means to integrate against a probability measure, we can define weak convergence of distributions, state Prohorov's theorem (which gives compactness

[9] The correct compactness criterion for continuous functions on a compact set K is the Arzelà-Ascoli theorem: a closed and bounded family in $C(K)$ is compact if and only if it is *equicontinuous* — the modulus of continuity is uniform across the family. This is not needed in this course, but it is worth knowing that compactness in function spaces requires an extra condition beyond boundedness.

of $\Delta(\Theta)$ when Θ is compact), and explain why compactness of the type space is the standing assumption that purchases everything else in incomplete-information theory. The topological framework built here — metric spaces, completeness, the distinction between norm and weak topologies — is the scaffolding those results sit on.

13.8 Where this appears in MWG

The foundational material — metric spaces, open and closed sets, Cauchy sequences, continuity, compactness — is in **MWG, Mathematical Appendix A**. The contraction mapping theorem and the completeness of $B(\Theta)$ under d_∞ are stated in **MWG, Mathematical Appendix C**.

The Hausdorff metric, the product topology on infinite products, and weak convergence of probability measures are not developed in MWG. The standard economics reference for all of these is:

> Aliprantis, C. D. and K. C. Border, *Infinite Dimensional Analysis: A Hitchhiker's Guide*, 3rd ed., Springer, 2006.

Chapter 3 covers the Hausdorff metric and Blaschke's selection theorem. Chapters 15 and 18, covering weak convergence and Prohorov's theorem, will be referenced again in Topic 14. The treatment is self-contained and written for economists.

The type-space construction using the product topology is in Mertens and Zamir (1985) and Brandenburger and Dekel (1993), cited in the footnote above.

Chapter 14

Topic 14: Measurability and Integration

(Giving $\int f\,d\mu$ a rigorous home)

14.1 Why measure theory enters now

In Topic 8, we introduced probability informally. A random variable was a function $X : \Omega \to \mathbb{R}$, a distribution assigned probabilities to values of X, and expectation was a weighted average:

$$\mathbb{E}[X] = \sum_i p_i x_i \quad \text{(discrete)}, \qquad \mathbb{E}[X] = \int x\, f(x)\, dx \quad \text{(continuous)}.$$

That was enough for auctions, Bayesian updating, and mixed strategies in finite games.

But two things have since changed. In Topic 13, the integral $\int f\,d\mu$ appeared—in the definition of the d_1 metric on function spaces and in the preview of weak convergence—and was explicitly deferred: "we develop this in Topic 14." And the applications that Topic 13 could not finish—weak convergence of probability measures, Prokhorov's theorem, compactness of the space of distributions $\Delta(\Theta)$, and Bayesian Nash equilibrium existence for infinite type spaces—all require that we know what it means to integrate a function against a measure that is neither a finite sum nor something with a density.

The goal of this topic is narrow. Here is what we want to do:

1. give $\int f\,d\mu$ a precise meaning,

2. state the convergence theorems (dominated convergence, Fatou's lemma) that let us pass limits through integrals,
3. connect conditional expectation back to Topic 8 in a way that generalizes beyond finite state spaces,
4. define weak convergence and state Prokhorov's theorem,
5. close the two applications deferred from Topic 13.

We do *not* need the full machinery of a real analysis course. We will not construct Lebesgue measure on $\mathbb{R}$ from scratch, and we will not prove dominated convergence in full detail. The point is to make the key objects precise and the key results usable, so that the microeconomic applications rest on solid ground—for more, you'll have to hope for the best in the probability theory part of the PhD sequence.

14.2 Sigma-algebras: which subsets can we measure?

In Topic 8, we said that a probability assigns numbers to events, where an event is a subset of the state space Ω. But we were vague about *which* subsets. For finite Ω, this is harmless: every subset is an event. For uncountable Ω—a type space $[0, 1]$, or $\mathbb{R}^n$—not every subset can be assigned a probability in a coherent way.[1]

To solve this, we need to specify in advance which subsets we will measure. This collection is called a **sigma-algebra** (or σ-algebra).

Definition. A σ-algebra on a set Ω is a collection $\mathcal{F}$ of subsets of Ω satisfying:

1. $\Omega \in \mathcal{F}$ (the whole space is measurable),
2. if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ (complements of measurable sets are measurable),
3. if $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$ (countable unions of measurable sets are measurable).

> "A σ-algebra tells us what classifies as an event within our set Ω."

The pair $(\Omega, \mathcal{F})$ is called a **measurable space**.

[1]The existence of non-measurable sets is a consequence of the axiom of choice. Here is the intuition for why the problem arises. On $[0, 1]$, we want "length" to satisfy three properties: (i) the length of an interval $[a, b]$ is $b - a$, (ii) shifting a set does not change its length, and (iii) countable additivity holds. Now imagine partitioning $[0, 1]$ into uncountably many disjoint pieces (the axiom of choice lets us do this in a very irregular way — the standard construction is due to Vitali). If each piece had length zero, their union would have length zero by countable additivity, contradicting the fact that $[0, 1]$ has length 1. If each piece had positive length, there would be uncountably many pieces of positive length, and summing even countably many of them would give infinity, again a contradiction. The only escape is to refuse to assign a length to these pathological pieces at all. That is what the σ-algebra does: it restricts attention to the "well-behaved" subsets that can receive a consistent size.

14.2.1 Why these three properties?

Property 1 says we can always ask "did *something* happen?" Property 2 says that if we can ask "did A happen?" then we can also ask "did A *not* happen?" Property 3 says we can ask "did at least one of $A_1, A_2, \ldots$ happen?" Combined with property 2, this also gives us countable intersections (by De Morgan's laws).[2] So the collection is closed under all the set operations we would ever need.[3]

14.2.2 The leading example: the Borel σ-algebra

On $\mathbb{R}$ (or $\mathbb{R}^n$), the most important σ-algebra is the **Borel σ-algebra** $\mathcal{B}(\mathbb{R})$. It is the smallest σ-algebra *containing all* open sets. It is easy to generate: start with taking all open intervals (a, b) and include $\emptyset$ (which is open). Then include complements and countable unions. The result contains all open sets, all closed sets, all countable unions of closed sets, and so on.

For a compact metric space Θ (the typical type space in mechanism design), the Borel σ-algebra $\mathcal{B}(\Theta)$ is generated by the open sets of the metric topology. This is the σ-algebra we will always use.

14.2.3 Two trivial σ-algebras

The smallest σ-algebra on Ω is $\{\emptyset, \Omega\}$—we can only ask whether "something" or "nothing" happened. The largest is 2^Ω, the power set — every subset is measurable. For finite Ω, the power set is the natural choice and recovers the informal setting of Topic 8. For uncountable Ω, we typically work with the Borel σ-algebra, which sits strictly between these two extremes.

14.2.4 Connection to topologies

In Topic 13 we introduced a topology τ on Ω: a collection of subsets (called "open") satisfying three axioms ($\emptyset$ and Ω belong, arbitrary unions stay in, finite intersections stay in). A σ-algebra $\mathcal{F}$ is also a collection of subsets satisfying three axioms. The two structures look similar, and it is worth being precise about how they differ.

	Topology τ	σ-algebra $\mathcal{F}$
Contains Ω	yes	yes
Closed under complements	**no**	yes

[2]If $A_1, A_2 \in \mathcal{F}$ so is A_1^c and A_2^c (property 2). But, then so is $A_1^c \cup A_2^c$ (property 3). Now $A_1 \cap A_2 \in \mathcal{F}$ because $A_1 \cap A_2 = (A_1^c \cup A_2^c)^c$ which (applying property 2 once more) is also in $\mathcal{F}$.

[3]By *closed under* we mean you can apply all such operations and whatever set you generate, it will be part of $\mathcal{F}$.

	Topology τ	σ-algebra $\mathcal{F}$
Closed under unions	arbitrary (even uncountable)	countable
Closed under intersections	finite only	countable (via complements)

The key differences are two. First, a σ-algebra is closed under complements; a topology is not—the complement of an open set is closed, not open. Second, a topology allows arbitrary (uncountable) unions but restricts to finite intersections, while a σ-algebra restricts both to countable operations.

The reason for the difference is the job each structure does. A topology is designed for continuity and convergence: it defines "nearness." A σ-algebra is designed for measurement: it defines which sets can be assigned a size. The two tasks impose different closure requirements.

They are connected by the Borel construction. Given a topology τ on Ω, the Borel σ-algebra $\mathcal{B}(\Omega)$ is the smallest σ-algebra containing every set in τ. It starts from the open sets and then closes under complements and countable unions, adding closed sets, countable intersections of open sets, and so on. In symbols: $\tau \subseteq \mathcal{B}(\Omega)$, but typically $\mathcal{B}(\Omega)$ is strictly larger than τ. So the topology generates the σ-algebra, but the two are not the same collection of sets.

For the compact metric spaces in this course, the Borel σ-algebra is the right choice: it is rich enough to measure all the sets we care about (open, closed, their countable combinations) and compatible with the topology we use for continuity arguments.

14.2.5 Why not just use the power set?

Since the power set 2^Ω is the largest σ-algebra, one might ask: why not declare every subset measurable and avoid the whole issue?

For finite Ω, that works — and it is what we did implicitly in Topic 8. But for uncountable Ω like $[0, 1]$, the power set is too large. As the footnote in §2 explained, no measure on $2^{[0,1]}$ can simultaneously assign intervals their natural length, respect countable additivity, and be translation-invariant. The Vitali-type sets that cause the contradiction are deeply pathological — they require the axiom of choice to construct and have no economic interpretation.

The Borel σ-algebra avoids the problem by starting from the topology (the open sets we already know are well-behaved) and closing under the operations we need (complements, countable unions). The resulting collection $\mathcal{B}(\Omega)$ sits between the two extremes: $\tau \subseteq \mathcal{B}(\Omega) \subsetneq 2^\Omega$. It is large enough to contain every set an economist would write down — intervals, level sets of continuous functions, countable unions and intersections of these — but small enough that measures on it are well-defined and well-behaved.

14.3 Measures and probability measures

A σ-algebra says which sets are measurable. A **measure** assigns sizes to those sets.

Definition. A measure on a measurable space $(\Omega, \mathcal{F})$ is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying:

1. $\mu(\emptyset) = 0$ (the empty set has size zero),
2. if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

(**countable additivity**).

The triple $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**.

The measure is pretty intuitive; it only says the following:

> "A measure is a way of assigning sizes to sets that respects basic algebra: the size of a disjoint union is the sum of the sizes."

14.3.1 Probability measures

There are several types of measures. But the most relevant for us is the probability measure. Here is the definition:

> A measure μ is a **probability measure** if $\mu(\Omega) = 1$.

In that case, $\mu(A)$ is the probability of the event A, and we recover the Kolmogorov axioms from Topic 8—except now we have countable additivity rather than just finite additivity, and the events are required to come from a σ-algebra.

The notation Pr or P from Topic 8 is just a probability measure in the present language. The upgrade is that we now know precisely which sets Pr is defined on (namely, the sets in $\mathcal{F}$), and the additivity extends to countable collections.

14.3.2 Why countable additivity?

Finite additivity—$\mu(A \cup B) = \mu(A) + \mu(B)$ when $A \cap B = \emptyset$—was all we needed in Topic 8's discrete setting. But limits require countable additivity. If $A_n \uparrow A$ (i.e., $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_n A_n$), then countable additivity implies $\mu(A_n) \to \mu(A)$. Without this, we could not take limits of probabilities, and convergence theorems for integrals would fail.[4]

[4] The key here is we want to be able to say something about what happens if we chop (say) a set in half (in size) infinitely often. This operation remains countable (by construction), and

14.3.3 Three examples that cover the course

1. **Counting measure on a finite set.** Let $\Omega = \{1, \ldots, n\}$, $\mathcal{F} = 2^\Omega$, and $\mu(\{i\}) = p_i$ with $\sum p_i = 1$. This is the discrete distribution from Topic 8. Integration against μ is summation: $\int f \, d\mu = \sum_i p_i f(i)$.
2. **Lebesgue measure on** $[0, 1]$**.** The unique measure on $\mathcal{B}([0, 1])$ with $\mu([a, b]) = b - a$ for every subinterval. This is "length." A continuous distribution with density g on $[0, 1]$ has $\mu(A) = \int_A g(x) \, dx$—a probability measure that is absolutely continuous with respect to Lebesgue measure.
3. **Point mass (Dirac measure).** For a fixed $\theta_0 \in \Theta$, define $\delta_{\theta_0}(A) = 1$ if $\theta_0 \in A$ and 0 otherwise. This is a probability measure concentrated at a single point. It will appear repeatedly in the weak-convergence section.

14.4 Measurable functions: which functions can we integrate?

In Topic 8, a random variable was any function $X : \Omega \to \mathbb{R}$. When Ω is finite, every such function is fine. For general Ω, we need a compatibility condition between the function and the σ-algebra.

Definition. A function $f : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is **measurable** (or $\mathcal{F}$-measurable) if for every Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$f^{-1}(B) = \{\omega \in \Omega : f(\omega) \in B\} \in \mathcal{F}.$$

> "A function is measurable if pulling back any 'reasonable' subset of $\mathbb{R}$ lands in the σ-algebra. Equivalently: for every real number c, the set $\{\omega : f(\omega) \leq c\}$ is in $\mathcal{F}$."

The second phrasing — checking only the sets $\{f \leq c\}$ — is equivalent because these sets generate the Borel σ-algebra.[5]

14.4.1 A concrete example

Think of a bidder whose valuation depends on an underlying state $\omega \in [0, 1]$, with $v(\omega) = \omega^2$. The question "what is the probability that the valuation is at most 1/2?" amounts to computing $\mu(\{v \leq 1/2\})$. Now, $\{v \leq 1/2\} = \{\omega : \omega^2 \leq$

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the size of each set remains positive (also by construction). But then we can take all left halves we generated in this process (a countably infinite number) and take their union with the right half of the initial chop to recover the initial set (once more by construction). This is exactly the kind of operation we do when we take limits, so we need to make sure our math object somehow speaks to this.

[5]This is a standard result: a function is Borel measurable if and only if $\{f \leq c\} \in \mathcal{F}$ for every $c \in \mathbb{R}$. The proof uses the fact that the half-lines $(-\infty, c]$ generate $\mathcal{B}(\mathbb{R})$, and then the preimage of a generator is in $\mathcal{F}$, so the preimage of any Borel set is in $\mathcal{F}$.

$1/2\} = [0, 1/\sqrt{2}]$. This is a closed interval, hence a Borel set, hence in $\mathcal{B}([0,1])$, so the measure can evaluate it. That is measurability at work for the threshold $c = 1/2$.

Measurability says this works not just for $c = 1/2$ but for *every* threshold c. For the function $v(\omega) = \omega^2$, the set $\{v \leq c\}$ is always an interval $[0, \sqrt{c}]$ (for $c \in [0,1]$), and intervals are always Borel sets. So v is measurable. Any continuous function has this property: the preimage of a "nice" set in $\mathbb{R}$ (an interval, or anything built from intervals by countable operations) is a "nice" set in Ω — and "nice" here means "belongs to the Borel σ-algebra."

14.4.2 Why measurability matters

The point is simple: to compute $\mu(\{f \leq c\})$ — the probability that f is no larger than c — the set $\{f \leq c\}$ must be in $\mathcal{F}$, or the measure μ has nothing to say about it. Measurability is the condition that makes this work for every threshold c.

In the finite-Ω setting of Topic 8, every function is measurable with respect to the power set 2^Ω, which is why we never needed the concept. For uncountable spaces, measurability is automatic for every function you will encounter in (most of) economics—continuous functions are measurable, monotone functions are measurable, limits of measurable functions are measurable—so the concept is a safeguard, not a restriction.[6]

14.4.3 A random variable, upgraded

A **random variable** in the measure-theoretic sense is a measurable function $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. This definition is interesting, because the condition does not rely on the definition of a probability measure—we could define RVs also in other types of measures (those that do not sum to 1).

Of course, the CDF $F(c) = \mu(\{X \leq c\})$ is well-defined precisely because measurability guarantees $\{X \leq c\} \in \mathcal{F}$, so μ can evaluate it. This is the same CDF from Topic 8, now on rigorous footing.

14.5 The Lebesgue integral: motivation and construction

We now have a measure space $(\Omega, \mathcal{F}, \mu)$ and a measurable function $f : \Omega \to \mathbb{R}$. The question is: what does $\int f \, d\mu$ mean?

[6] The statement here says already something about economic modeling. If a function were not measurable, typically, the model makes little sense simply because we cannot say what exactly this function does and, ultimately, we try to model behavior so we should in fact model measurable things.

14.5.1 What is wrong with the Riemann integral?

The integral from calculus (the *Riemann integral*)—partition the x-axis into small intervals, form upper and lower sums, check that they converge—works well for continuous functions on $[a, b]$. But it has limitations that matter for probability:

1. **The domain must look like an interval.** The Riemann integral partitions the x-axis into small subintervals. If Ω is an abstract set (a type space, a set of strategies), there are no intervals to partition, and the construction does not get off the ground.

2. **Limits behave badly.** Take a sequence of functions that the Riemann integral can handle, and let them converge pointwise. The limit need not be Riemann-integrable.[7] This makes limit theorems — precisely the tools we need for passing limits through expected payoffs — unreliable in the Riemann framework.

3. **It only works with densities.** The Riemann integral $\int f(x)\, g(x)\, dx$ requires a density function g. But many probability measures do not have densities: a point mass δ_{θ_0} concentrates all its weight at a single point, and a mixture of a continuous and a discrete distribution (say, "with probability $1/2$ the type is drawn uniformly from $[0, 1]$, and with probability $1/2$ the type is exactly θ_0") has no density at all. The Riemann framework has no natural way to write $\int f\, d\mu$ for such measures.

The Lebesgue integral solves all three problems by building the integral from the measure directly.

14.5.2 The idea: partition the range, not the domain

The Riemann integral chops up the domain into small intervals and asks how the function behaves on each interval. The Lebesgue integral does the opposite: it chops up the *range* of f into small intervals $[y_k, y_{k+1})$ and asks how much of the domain lands in each range-slice.

More precisely, the "amount of domain" where f takes values in $[y_k, y_{k+1})$ is $\mu(\{y_k \leq f < y_{k+1}\})$. This is where the measure enters. The approximate integral is

$$\sum_k y_k \cdot \mu(\{y_k \leq f < y_{k+1}\}).$$

As the partition of the range gets finer, this sum converges to $\int f\, d\mu$.

[7] A classical example: enumerate the rationals in $[0, 1]$ as $r_1, r_2, \ldots$ and define $f_n = \mathbf{1}_{\{r_1,\ldots,r_n\}}$. Each f_n is zero except at finitely many points, so $\int f_n\, dx = 0$ in the Riemann sense. But the pointwise limit is $\mathbf{1}_{\mathbb{Q}}$, which equals 1 on the rationals and 0 on the irrationals. This function oscillates too wildly for the Riemann upper and lower sums to agree, so it is not Riemann-integrable.

“The Riemann integral asks: on each small piece of the domain, what value does f take? The Lebesgue integral asks: for each small range of values, how much of the domain does f occupy? The second question makes sense on any measure space, not just on intervals of $\mathbb{R}$.”

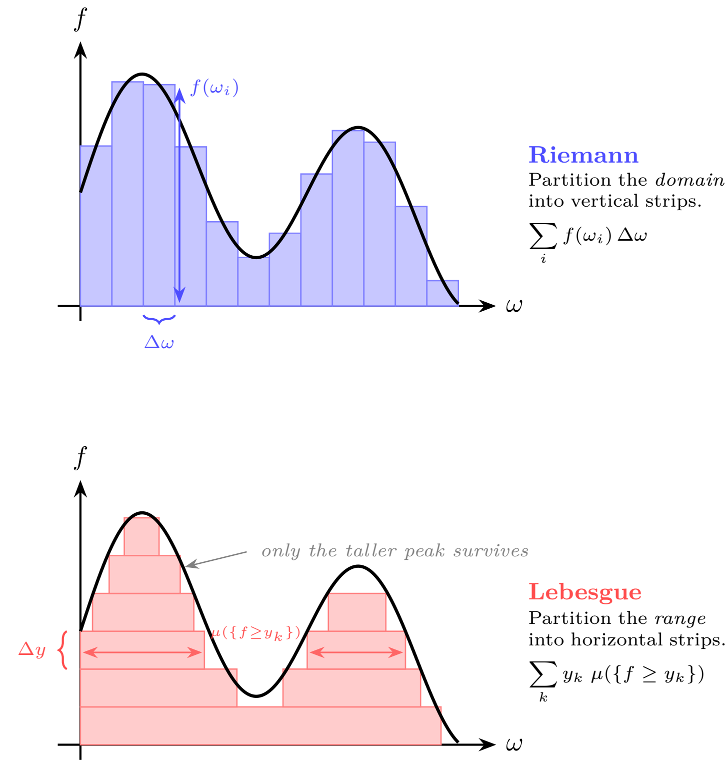

Figure 14.1: Riemann vs Lebesgue partitioning

14.5.3 The construction: from the simplest case to the general one

We want to define $\int f\,d\mu$ for every measurable function f. But we cannot do this in one shot: the whole point of the Lebesgue integral is that it works on abstract spaces where there is no “partition the x-axis into intervals” trick available. So we need a different strategy.

The idea is to start from the *one* case where the answer is completely obvious, and then extend, step by step, to larger and larger classes of functions. Each step defines the integral for a new class using only what the previous step already established. By the end, we reach all measurable functions.

Step 1: Indicator functions — the seed. An *indicator function* $\mathbf{1}_A(\omega)$ equals 1 if $\omega \in A$ and 0 elsewhere.[8] There is only one reasonable answer for its integral: the "area under the curve" of a function that is 1 on A and 0 everywhere else should be the size of A. So we *define*

$$\int \mathbf{1}_A \, d\mu = \mu(A).$$

This is where the measure enters the integral. Everything that follows is built on this single equation.

Step 2: Simple functions — finite sums of indicators. A **simple function** takes finitely many values:

$$s = \sum_{k=1}^{n} c_k \, \mathbf{1}_{A_k},$$

where $c_k \geq 0$ and $A_k \in \mathcal{F}$. Think of it as a staircase: the function is constant on each piece A_k, and it jumps between the levels $c_1, \ldots, c_n$. The integral should be the weighted sum of these levels, weighted by the measure of each piece. So we *define*

$$\int s \, d\mu = \sum_{k=1}^{n} c_k \, \mu(A_k).$$

This is just linearity applied to Step 1. Nothing new is assumed; we are using the indicator integral n times and adding up.

Step 3: Nonnegative measurable functions — approximation from below. Now the key step. A general nonnegative measurable function $f \geq 0$ is not a staircase — it takes infinitely many values, possibly a continuum. But we can *approximate* it from below by simple functions: chop the range into finer and finer levels, and at each resolution, round f down to the nearest level. This produces a sequence of staircases $s_1 \leq s_2 \leq \cdots$ that climb toward f from below.[9] Since each s_n is a simple function, we already know $\int s_n \, d\mu$ from Step 2. As the staircase gets finer, these integrals increase (more of f is captured). The integral of f is the limit:

$$\int f \, d\mu = \sup \left\{ \int s \, d\mu : s \text{ simple}, \ 0 \leq s \leq f \right\}.$$

This is where the range-partitioning picture from above connects to the formal construction: each staircase approximation corresponds to a particular partition of the range.

[8] The concept of the indicator function is good to remember. It will show up often in economic problems. Also note that this is very closely related to Dirac measures.

[9] Concretely: define $s_n(\omega) = k/2^n$ if $k/2^n \leq f(\omega) < (k+1)/2^n$, for $k = 0, 1, \ldots, n2^n - 1$, and $s_n(\omega) = n$ if $f(\omega) \geq n$. Each s_n is a simple function, $s_n \leq f$, and $s_n(\omega) \uparrow f(\omega)$ for every ω.

Step 4: General measurable functions — handling signs. The previous step required $f \geq 0$. For a function that can take negative values, we split it into its positive and negative parts: $f = f^+ - f^-$, where $f^+(\omega) = \max(f(\omega), 0)$ and $f^-(\omega) = \max(-f(\omega), 0)$. Both are nonnegative, so Step 3 handles each one. Define

$$\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu,$$

provided at least one of these is finite. If both are finite, we say f is **integrable** (or μ-integrable).

The chain is now complete: indicator $\to$ simple $\to$ nonneg $\to$ general. Each step used only the previous one. And the only external input was the measure μ, which entered once, in Step 1.

14.5.4 Recovering the familiar cases

When $\Omega = \{1, \ldots, n\}$ with $\mu(\{i\}) = p_i$,

$$\int f \, d\mu = \sum_{i=1}^{n} p_i \, f(i) = \mathbb{E}[f(X)]$$

where X is the identity random variable. This is the discrete expectation from Topic 8.

When $\Omega = \mathbb{R}$ with $d\mu = g(x) \, dx$ for a density g,

$$\int f \, d\mu = \int_{-\infty}^{\infty} f(x) \, g(x) \, dx,$$

which is the continuous expectation from Topic 8 (and coincides with the Riemann integral when f and g are well-behaved).

When $\mu = \delta_{\theta_0}$ is a point mass, we can trace the construction to see what happens. Start with an indicator: $\int \mathbf{1}_A \, d\delta_{\theta_0} = \delta_{\theta_0}(A)$, which is 1 if $\theta_0 \in A$ and 0 if not. In other words, $\int \mathbf{1}_A \, d\delta_{\theta_0} = \mathbf{1}_A(\theta_0)$. Now extend to simple functions: if $s = \sum_k c_k \mathbf{1}_{A_k}$, then $\int s \, d\delta_{\theta_0} = \sum_k c_k \, \delta_{\theta_0}(A_k) = \sum_k c_k \, \mathbf{1}_{A_k}(\theta_0) = s(\theta_0)$. Finally, since every measurable f is a limit of simple functions and the integral passes through the limit, we get

$$\int f \, d\delta_{\theta_0} = f(\theta_0).$$

> "Integrating any function against a point mass simply evaluates the function at that point."

This will be important in §10 when we verify that weak convergence works correctly for sequences of point masses.

So the Lebesgue integral unifies summation, Riemann integration, and evaluation at a point under a single notation: $\int f \, d\mu$. The measure μ tells you which one you are doing.

14.6 Key properties of the integral

Operationally, the Lebesgue integral can be treated like the Riemann integral. All operations carry over. In particular:

Linearity. For measurable f, g and constants a, b,

$$\int (af + bg)\, d\mu = a \int f\, d\mu + b \int g\, d\mu.$$

This is linearity of expectation (Topic 8) in its general form.

Monotonicity. If $f \leq g$ μ-almost everywhere, then $\int f\, d\mu \leq \int g\, d\mu$.

Absolute integrability. f is integrable if and only if $\int |f|\, d\mu < \infty$.

These properties follow directly from the construction: they hold for simple functions by finite arithmetic, and they extend to general functions by taking limits.

A difference, if you like, is:

$$\int_{\mathbb{Q}\cap[0,1]} 1\, \mathrm{d}\mu = 0$$

under Lebesgue integration (using the usual Borel σ-algebra), whereas a Riemann integral to express this notion does not exist.

So Lebesgue integration is conceptually new (and appropriate here); but once there is an integral, we can treat it just like the familiar Riemann integral from high school. We now turn to why and when Lebesgue integration is useful for us.

14.7 Convergence theorems: passing limits through integrals

The real payoff of the Lebesgue integral is not the integral itself—it is the convergence theorems it allows us to derive. Limits are important in economics. They matter to cleanly understand

- thinking on the margin,
- (approximate) optimality,
- benchmark cases,
- existence conditions for equilibria.

In short, we constantly need to evaluate limits of expected payoffs, exchange the order of limits and expectations, or argue that a sequence of value functions converges to the right limit. The convergence theorems tell us when $\lim_n \int f_n\, d\mu = \int \lim_n f_n\, d\mu$.

This exchange is *not* automatic. Consider the set of functions $f_n = n \cdot \mathbf{1}_{(0,1/n]}$ on $[0,1]$ for $n \in \mathbb{N}$. Each f_n is nonnegative by construction and $\int f_n \, d\mu = n(1/n) = 1$. But for any given $\omega \in (0,1]$, $f_n(\omega) \to 0$. Now that means that $\int \lim_{n\to\infty} f_n \, d\mu = 0 \neq 1 = \lim_{n\to\infty} \int f_n \, d\mu$. The functions spike higher and higher on a shrinking region; the mass "escapes to infinity." So the measure of the limit does not converge to the limit of the measures.

14.7.1 Monotone Convergence Theorem

We say $f_n \uparrow f$ *pointwise* (read converges pointwise monotone) if there is pointwise convergence (that is, for all ω, $f_n(\omega) \to f(\omega)$) *and* in addition $f_n(\omega) \le f_{n+1}(\omega)$ for all ω and n.

Theorem (Monotone Convergence, Beppo Levi). Let $f_n \ge 0$ be measurable and $f_n \uparrow f$ pointwise. Then

$$\int f_n \, d\mu \uparrow \int f \, d\mu.$$

> "If a sequence of nonnegative functions increases pointwise to a limit, the integrals increase to the integral of the limit. Monotonicity prevents mass from escaping."

Proof sketch. Since $f_n \le f$, monotonicity of the integral gives $\int f_n \, d\mu \le \int f \, d\mu$, so $\lim \int f_n \, d\mu \le \int f \, d\mu$. For the reverse inequality: for any simple $s \le f$ and any $c < 1$, the sets $A_n = \{f_n \ge cs\}$ increase to Ω (because $f_n \uparrow f \ge s > cs$ wherever $s > 0$). So $\int f_n \, d\mu \ge c \int_{A_n} s \, d\mu$, and as $n \to \infty$, $\int_{A_n} s \, d\mu \to \int s \, d\mu$. Taking the supremum over simple $s \le f$ gives $\lim \int f_n \, d\mu \ge c \int f \, d\mu$. Since this holds for all $c < 1$, we get equality.

14.7.2 Fatou's Lemma

Theorem (Fatou's Lemma). Let $f_n \ge 0$ be measurable. Then

$$\int \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int f_n \, d\mu.$$

> "Even without monotonicity, the integral of the pointwise liminf is no larger than the liminf of the integrals. Mass can escape to infinity but cannot appear from nowhere."

Proof sketch. Define $g_k = \inf_{n \ge k} f_n$. Then $g_k \le f_n$ for all $n \ge k$, so $\int g_k \, d\mu \le \int f_n \, d\mu$ for all $n \ge k$, hence $\int g_k \, d\mu \le \inf_{n \ge k} \int f_n \, d\mu$. Now $g_k \uparrow \liminf f_n$, so by monotone convergence $\int g_k \, d\mu \to \int \liminf f_n \, d\mu$. Taking $k \to \infty$ on the right side gives $\liminf \int f_n \, d\mu$.

Fatou's lemma is used whenever a bound suffices and exact convergence is not available. It is the safety net behind many limit arguments in optimization and equilibrium theory.

14.7.3 Dominated Convergence Theorem

Theorem (Dominated Convergence, Lebesgue). Let f_n be measurable with $f_n \to f$ pointwise. If there exists an integrable function g with $|f_n| \leq g$ for all n, then

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.$$

> "If a sequence of functions converges pointwise and is uniformly bounded by an integrable function, then the limit passes through the integral."

Proof sketch. The functions $g + f_n \geq 0$ and $g - f_n \geq 0$. Apply Fatou's lemma to each:

$$\int (g+f)\, d\mu \leq \liminf \int (g+f_n)\, d\mu, \qquad \int (g-f)\, d\mu \leq \liminf \int (g-f_n)\, d\mu.$$

Since $\int g \, d\mu < \infty$, cancel it. The first gives $\int f \, d\mu \leq \liminf \int f_n \, d\mu$. The second gives $-\int f \, d\mu \leq \liminf(-\int f_n \, d\mu) = -\limsup \int f_n \, d\mu$, i.e., $\limsup \int f_n \, d\mu \leq \int f \, d\mu$. Together: $\lim \int f_n \, d\mu$ exists and equals $\int f \, d\mu$.

Where each assumption is used:

- *Pointwise convergence* defines the limit f.
- *The dominating function* g prevents the "escaping mass" problem. In the counterexample above ($f_n = n \cdot \mathbf{1}_{(0,1/n]}$), no integrable g dominates all f_n because their heights grow without bound.

The result is geometrically very intuitive. If there is a cap on how high f_n can go (given by g) and g is integrable (i.e. not disappearing into infinity), then of course f_n cannot disappear to infinity either. That means laying bricks to approximate the area while not requiring parts of the wall to be infinitely high.

14.7.4 Why economists need dominated convergence

In mechanism design, we often need to show that a functional like expected revenue $\int_0^1 R(\theta, \mu_n)\, d\theta$ converges to $\int_0^1 R(\theta, \mu)\, d\theta$ when a sequence of mechanisms (which correspond to distributions over revenues, i.e. measures) converges. If payoffs are bounded — say $|R| \leq M$ because types and payments live in compact sets — then the constant function $g = M$ dominates everything, and dominated convergence applies immediately. This is why "bounded payoffs on a compact type space" is the standing assumption throughout mechanism design: it is precisely the condition that makes dominated convergence available.[10]

Similarly, in continuity arguments for expected utility (Topic 5's Berge theorem applied to expected payoffs), we need the expected payoff $\int u(a, \theta)\, d\mu(\theta)$ to be

[10] Recall here again, this assumption is a feature not a bug. Sure, mathematically, we lose generality, but economically, a model in which revenue can get infinite is (at best) valuable as a benchmark, but for sure not realistic.

continuous in a when u is continuous and bounded. Dominated convergence (with the bound M) is exactly what delivers this.[11]

14.8 Conditional expectation, revisited

In Topic 8, conditional expectation was defined for discrete random variables:

$$\mathbb{E}[X \mid Y = y] = \sum_i x_i \Pr(X = x_i \mid Y = y).$$

This worked because each event $\{Y = y\}$ had positive probability. In the continuous case, it has to be that $\Pr(Y = y) = 0$ for every single y.[12] But then, for whatever A I am looking for, the ratio $\Pr(A \cap \{Y = y\}) / \Pr(Y = y)$ is $0/0$. So the measure-theoretic definition of conditional expectation needs to make sure to handle this.

14.8.1 The idea: averaging over information

A σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ represents the information available to the agent. If $\mathcal{G}$ is coarser than $\mathcal{F}$ (fewer sets), the agent cannot distinguish states that $\mathcal{F}$ can distinguish—she has less information.[13]

The conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is defined as the unique $\mathcal{G}$-measurable function[14] that satisfies

$$\int_G \mathbb{E}[X \mid \mathcal{G}]\, d\mu = \int_G X\, d\mu \qquad \text{for every } G \in \mathcal{G}.$$

> "The conditional expectation of X given $\mathcal{G}$ is the best $\mathcal{G}$-measurable approximation to X: it preserves the integral over every set the agent can observe."

14.8.2 Why this generalizes Topic 8

When $\mathcal{G} = \sigma(Y)$ (the σ-algebra generated by Y, i.e., the information revealed by observing Y), the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is a function of Y alone. In

[11] Recall that in Topic 11, the contraction mapping argument for the Bellman operator required bounded per-period payoffs. That same boundedness is what makes dominated convergence available here.

[12] Think about it this way. If I remove one number from the interval $[0, 1]$, the size of the interval has to be the same, because otherwise I could remove infinitely many such numbers and that would violate that the probability that any number appears becomes larger than 1.

[13] Recall, the σ-algebra essentially measures what we (the agent) consider as an event within a finer underlying set.

[14] This excludes measure 0 sets. So the function is unique up to changes on those, of course. Funnily enough there may be (uncountably) many such functions, but the σ-algebra makes sure that those are irrelevant for the agent a priori.

the discrete case, this function evaluated at $Y = y$ recovers $\mathbb{E}[X \mid Y = y]$ from Topic 8. The law of iterated expectations,

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid \mathcal{G}]\big],$$

is now a consequence of the defining property: take $G = \Omega$ in the integral identity.

As in Topic 8, there is the simple interpretation of constraining the set of possibilities: Initially, there was a large set of possible events (the initial σ-algebra). Now, since we know Y, the set of possible events is smaller (this is the *information generated by* Y). Now, we make sure that mass actually still integrates to 1 over this restricted set.

14.8.3 The projection interpretation

There is another interpretation that is (at first glance) a bit more complicated to describe, but essentially does the same thing. It is useful in applied work, because it is the foundation of regression functions.

If X is square-integrable ($\mathbb{E}[X^2] < \infty$), the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is the **orthogonal projection** of X onto the space of $\mathcal{G}$-measurable square-integrable functions. "Orthogonal" means it minimizes the mean-squared error:

$$\mathbb{E}[X \mid \mathcal{G}] = \underset{Z \text{ is } \mathcal{G}\text{-measurable}}{\operatorname{argmin}} \mathbb{E}\big[(X - Z)^2\big].$$

This is exactly analogous to projection in $\mathbb{R}^n$: conditional expectation projects onto the subspace of functions that are "knowable" given the information $\mathcal{G}$, and the projection minimizes the distance.[15]

14.9 L^p spaces (briefly)

The projection interpretation of conditional expectation uses the idea of "distance" between random variables. This is formalized by so-called L^p spaces.

Definition. For $p \geq 1$, the space $L^p(\Omega, \mathcal{F}, \mu)$ is the set of measurable functions f with

$$\|f\|_p := \left(\int |f|^p \, d\mu \right)^{1/p} < \infty.$$

Two functions that differ only on a set of measure zero are identified.[16]

[15]This projection interpretation is the foundation of the regression function in econometrics. The conditional expectation $\mathbb{E}[Y \mid X]$ is the function of X that best predicts Y in the mean-squared sense. The "residual" $Y - \mathbb{E}[Y \mid X]$ is orthogonal to every function of X.

[16]Formally, elements of L^p are equivalence classes of functions. The difference between a function and its equivalence class is a technicality that never matters in applications: we never ask what a function does on a single point of a continuous type space, only what its integral does over measurable sets.

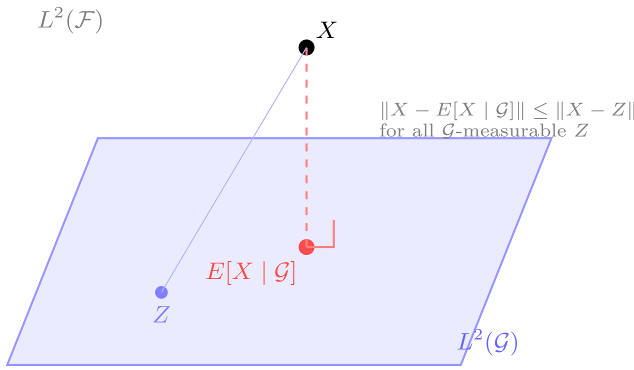

Figure 14.2: Conditional expectation as projection in L^2

The two most important cases for us:

- L^1: integrable functions. $\|f\|_1 = \int |f| \, d\mu < \infty$. This is the space of random variables with finite expectation.
- L^2: square-integrable functions. $\|f\|_2 = (\int |f|^2 \, d\mu)^{1/2} < \infty$. This is a Hilbert space[17] with inner product $\langle f, g \rangle = \int fg \, d\mu$, and conditional expectation is an orthogonal projection in this space.

L^p is a complete metric space under $\|f - g\|_p$ (this is the Riesz-Fischer theorem). Completeness is what makes the projection in L^2 well-defined: the minimizer exists because the space has no "holes."

For economics, L^2 matters because it gives a geometric language for conditional expectation, and L^1 matters because dominated convergence says: if $f_n \to f$ pointwise and $|f_n| \le g \in L^1$, then $f_n \to f$ in L^1 as well (i.e., $\|f_n - f\|_1 \to 0$).

14.10 Weak convergence of probability measures

We now have everything we need to close the loop from Topic 13. Recall the setup: Θ is a compact metric space (a type space, a state space) and $\Delta(\Theta)$ is the set of all Borel probability measures on Θ. Topic 13 asked: what is the right notion of "two distributions being close"?

[17] A Hilbert space is a generalization of the Euclidean space allowing for infinite dimensions.

14.10.1 Why the Kolmogorov metric fails

The most obvious idea—use the sup distance between CDFs—is the **Kolmogorov metric**:

$$d_\infty(\mu, \nu) = \sup_x |F_\mu(x) - F_\nu(x)|.$$

Topic 13 already exhibited the failure. Consider point masses $\mu_n = \delta_{1/n}$ on $[0, 1]$. As $n \to \infty$, the mass slides toward 0, and the "limit" should be δ_0. But the CDF of $\delta_{1/n}$ jumps at $1/n$ and the CDF of δ_0 jumps at 0, so $d_\infty(\mu_n, \delta_0) = 1$ for every n. The Kolmogorov metric says these distributions are maximally far apart, even though the point masses are converging to 0.

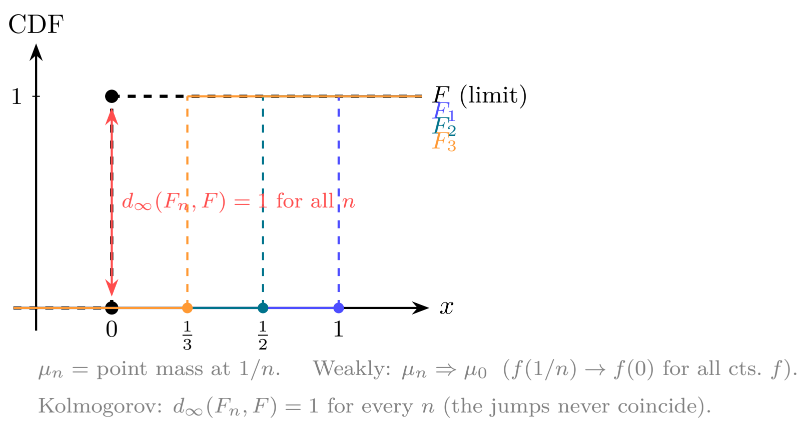

Figure 14.3: CDF failure of Kolmogorov metric

The problem is that the Kolmogorov metric demands that CDFs agree uniformly, including at their jump points. But for discrete distributions, the jump locations carry all the information, and a slight shift in the jump location registers as a maximal discrepancy.

14.10.2 Weak convergence: testing with continuous functions

The correct notion for economics is **weak convergence**. It uses the integral we have just built.

Definition. A sequence of probability measures μ_n on a metric space Θ **converges weakly** to μ (written $\mu_n \Rightarrow \mu$) if

$$\int f \, d\mu_n \to \int f \, d\mu \qquad \text{for every bounded continuous function } f : \Theta \to \mathbb{R}.$$

> "Two distributions are close if they assign similar expected values to every continuous test function."

This is the definition that Topic 13 deferred. The integral $\int f\,d\mu$ now has a precise meaning (§5), and the definition makes sense for any Borel probability measure on a metric space.

14.10.3 Why it works

Return to the point-mass example. For any continuous $f : [0,1] \to \mathbb{R}$,

$$\int f\,d\delta_{1/n} = f(1/n) \to f(0) = \int f\,d\delta_0$$

by continuity of f. So $\delta_{1/n} \Rightarrow \delta_0$ — exactly what we wanted. The Kolmogorov metric failed because it tested CDFs, which are discontinuous for point masses; weak convergence tests continuous functions, so it does not see the jumps.

14.10.4 The Portmanteau theorem

Weak convergence has several equivalent formulations, collected in the Portmanteau theorem.

The following are equivalent for probability measures on a metric space Θ:

1. $\int f\,d\mu_n \to \int f\,d\mu$ *for every bounded continuous f (the definition above),*
2. $\limsup_n \mu_n(C) \le \mu(C)$ *for every closed set C,*
3. $\liminf_n \mu_n(U) \ge \mu(U)$ *for every open set U,*
4. $\mu_n(A) \to \mu(A)$ *for every Borel set A with $\mu(\partial A) = 0$ (a "continuity set" — one whose boundary has zero measure under the limit).*

Condition 4 explains the CDF failure: the point 0 is a continuity point of F_{δ_0} only from the right; the set $(-\infty, 0]$ has boundary $\{0\}$ with $\delta_0(\{0\}) = 1 \neq 0$, so it is not a continuity set.

We will not prove the Portmanteau theorem. We leave that to your probability theory course.

14.10.5 The Prokhorov metric

Weak convergence can be metrized. On $\Delta(\Theta)$ when Θ is a separable metric space, the **Prokhorov metric** is defined by

$$d_P(\mu,\nu) = \inf\{\varepsilon > 0 : \mu(A) \le \nu(A^\varepsilon) + \varepsilon \text{ for all Borel } A\},$$

where $A^\varepsilon = \{x : d(x,A) < \varepsilon\}$ is the ε-fattening of A.

> "Two distributions are Prokhorov-close if neither assigns much more probability to any set than the other assigns to a slightly enlarged version of the same set."

Convergence in d_P is equivalent to weak convergence. The Prokhorov metric turns $\Delta(\Theta)$ into a metric space, so all the tools from Topics 4 and 13 — continuity, compactness, convergent sequences — apply.[18]

14.11 Prokhorov's theorem: compactness of $\Delta(\Theta)$

In §10 we defined weak convergence and the Prokhorov metric on the set $\Delta(\Theta)$—the set of all Borel probability measures on Θ. Let us be precise about what this set contains. If $\Theta = [0, 1]$, then $\Delta([0, 1])$ includes every way of spreading probability across $[0, 1]$: the uniform distribution, the point mass $\delta_{1/2}$, the Beta distributions, mixtures of continuous and discrete parts, and so on. Each element of $\Delta(\Theta)$ is a probability measure μ on $(\Theta, \mathcal{B}(\Theta))$; the Prokhorov metric d_P tells us when two such measures are close.

The question is: is $\Delta(\Theta)$ compact under this metric?

This matters because compactness is the engine behind existence arguments. Recall the pattern from earlier topics: to show that a fixed point, an optimum, or an equilibrium exists, we need a compact domain (so that sequences have convergent subsequences) and continuity (so that limits of near-solutions are actual solutions). In Topic 9, the domain was the simplex $\Delta(S_i)$ of mixed strategies over a finite action set — compact and convex, so Kakutani applied. But in games with continuous type spaces, strategies are probability-measure-valued functions, and the relevant domain is (a subset of) $\Delta(\Theta)$. If $\Delta(\Theta)$ is not compact, the Kakutani argument has no ground to stand on.

Topic 13 (§7) previewed exactly this issue: it stated that the compactness of $\Delta(\Theta)$ and its consequences for Bayesian equilibrium existence required the integral, and deferred the result to this topic. We can now deliver it.

Theorem (Prokhorov). If Θ is a compact metric space, then $\Delta(\Theta)$ is compact under the Prokhorov metric (equivalently, under weak convergence).

> "If the underlying space is compact, the space of all probability distributions over it is also compact."

This is a powerful result. It says that any sequence of distributions on a compact type space has a weakly convergent subsequence. No additional assumptions are needed.

Proof sketch. The argument has two parts.

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[18] The Prokhorov metric is defined and studied in Billingsley, *Convergence of Probability Measures*, 2nd ed., Wiley, 1999, Chapter 1. In the economics literature, the metric is also discussed in Aliprantis and Border, *Infinite Dimensional Analysis*, 3rd ed., Springer, 2006, Chapter 15.

Tightness. A family $\mathcal{M}$ of probability measures on Θ is **tight** if for every $\varepsilon > 0$ there exists a compact $K \subseteq \Theta$ with $\mu(K) \geq 1 - \varepsilon$ for all $\mu \in \mathcal{M}$. When Θ is itself compact, every probability measure satisfies $\mu(\Theta) = 1$, so every family is automatically tight.

Prokhorov's theorem (general form). On a separable complete metric space, a family of probability measures is relatively compact (every sequence has a weakly convergent subsequence) if and only if it is tight.[19]

When Θ is compact, tightness is free, so $\Delta(\Theta)$ is compact.[20]

14.11.1 Completeness

$(\Delta(\Theta), d_P)$ is also complete when Θ is complete and separable. So for compact Θ, we have: $\Delta(\Theta)$ is a compact metric space. This means all the tools that require a compact metric space — Berge's theorem (Topic 5), the contraction mapping theorem (Topic 11), Blaschke's selection theorem (Topic 13) — are available on the space of distributions over compact Θ.

14.12 Application 1: mechanism design and type spaces

We can now state precisely why "compact type space" is the standing assumption in mechanism design and information economics.

14.12.1 The setup

A mechanism designer faces agents whose types θ_i are drawn from a compact metric space Θ_i. A **prior** is a probability measure $\mu \in \Delta(\Theta_1 \times \cdots \times \Theta_n)$. The designer chooses an allocation rule $q : \Theta \to Q$ and a payment rule $t : \Theta \to \mathbb{R}^n$, and agents who have a utility $v_i(q(m), \theta_i) - t_i$ choose a message $m_i \in \Theta_i$ not knowing the realization of θ_{-i}.

Assuming everyone else selects $m_{-i} = \theta_{-i}$, the expected payoff for i is

$$U_i(\theta_i) = \int_{\Theta_{-i}} \big[v_i(q(\theta_i, \theta_{-i}), \theta_i) - t_i(\theta_i, \theta_{-i})\big]\, d\mu_{-i}(\theta_{-i} \mid \theta_i),$$

[19] The direction "tight $\Rightarrow$ relatively compact" is the hard part and requires a diagonalization argument over a countable dense set of test functions. The converse is easier. See Billingsley, *Convergence of Probability Measures*, 2nd ed., Theorem 5.1 and 5.2.

[20] The weak-$*$ compactness of $\Delta(\Theta)$ can also be seen as a consequence of the Banach-Alaoglu theorem, since probability measures on Θ sit inside the unit ball of $C(\Theta)^*$ (the dual of the space of continuous functions). Prokhorov's theorem gives the same conclusion via a probabilistic route, and the two are equivalent for compact metric spaces.

where $\mu_{-i}(\cdot \mid \theta_i)$ is the conditional distribution of others' types given θ_i. The integral is the Lebesgue integral from §5; the conditional distribution is the conditional expectation from §8 applied to indicator functions.

14.12.2 Why compactness of $\Delta(\Theta)$ matters

When proving existence of optimal mechanisms or characterizing incentive compatibility, we need to take limits: sequences of mechanisms, sequences of type distributions, sequences of allocation rules. Compactness of $\Delta(\Theta)$ guarantees that:

1. Any sequence of posteriors has a convergent subsequence (Prokhorov), so limit arguments do not "leave the space."
2. Expected payoffs are continuous in the distribution (dominated convergence, since payoffs are bounded on a compact space), so the limits we take are well-behaved.
3. The set of incentive-compatible mechanisms is closed (because the IC constraints are defined by integrals that are continuous in the relevant arguments).

Without compactness of the type space, none of these closures are automatic. The "compact type space" assumption is not a simplification for convenience — it is the condition that makes the mathematical infrastructure (compactness of $\Delta(\Theta)$, dominated convergence, continuity of expected payoffs) available.

14.12.3 Hierarchies of beliefs

In models of incomplete information, agents may have beliefs about others' types, beliefs about others' beliefs, and so on. The universal type space (Mertens-Zamir, referenced in Topic 13) is built from the product space $\Theta \times \Delta(\Theta) \times \Delta(\Theta \times \Delta(\Theta)) \times \cdots$. Compactness of $\Delta(\Theta)$ at each level, combined with Tychonoff's theorem (which says that the product of compact spaces is compact in the product topology), ensures that the universal type space is also compact. This is what makes the Harsanyi program—encoding any game of incomplete information as a game with a common prior on a type space—mathematically coherent.

14.13 Application 2: Bayesian Nash equilibrium existence

We can now also prove the existence result that Topic 9 (Nash equilibrium in finite games) did not cover: existence of Bayesian Nash equilibrium in games with continuous type spaces.

14.13.1 The setup

Consider a Bayesian game:

- n players,
- type spaces Θ_i (compact metric spaces),
- action spaces A_i (compact metric spaces),
- utility functions $u_i : A \times \Theta \to \mathbb{R}$ (continuous and bounded),
- a common prior μ on $\Theta = \Theta_1 \times \cdots \times \Theta_n$.

A **(behavioral) strategy** for player i is a measurable mapping $\sigma_i : \Theta_i \to \Delta(A_i)$—for each type, a distribution over actions. Equivalently, it is an element of $\Delta(A_i)^{\Theta_i}$, or after appropriate identification, an element of the space of all probability-measure-valued measurable functions.

14.13.2 The existence argument

The proof follows the same Kakutani logic as in Topic 9, but now on an infinite-dimensional space. There are several routes in the literature; the cleanest for our purposes uses the "distributional strategy" formulation due to Milgrom and Weber (1985).[21]

Step 1: The strategy space. Instead of working with functions $\sigma_i : \Theta_i \to \Delta(A_i)$, rewrite a strategy as a *joint distribution* $\tau_i \in \Delta(A_i \times \Theta_i)$ whose marginal on Θ_i equals the prior marginal μ_i. The set of such distributions is a closed convex subset of $\Delta(A_i \times \Theta_i)$.

Since $A_i \times \Theta_i$ is a compact metric space, Prokhorov's theorem gives compactness of $\Delta(A_i \times \Theta_i)$. The marginal constraint is preserved under weak limits (because the marginal is a continuous map in the weak topology). So the strategy space is compact and convex.[22]

Step 2: Expected payoffs are continuous. Player i's expected payoff given the strategy profile $(\tau_1, \ldots, \tau_n)$ is

$$\Pi_i(\tau) = \int_{A \times \Theta} u_i(a, \theta)\, d\tau(a, \theta),$$

where τ is the joint distribution induced by the profile. Since u_i is continuous and bounded and the integral is continuous in the measure under weak convergence (this is exactly the definition of weak convergence), Π_i is continuous in the strategy profile.

Step 3: Best responses have convex values. Player i's best-response set is the set of strategies maximizing Π_i given others' strategies. Since Π_i is linear

[21] Milgrom, P. and R. Weber, "Distributional strategies for games with incomplete information," *Mathematics of Operations Research*, 10(4), 1985, pp. 619–632. This paper formulates Bayesian games so that the strategy space is a subset of $\Delta(A_i \times \Theta_i)$, making it a compact convex subset of a locally convex space.

[22] Convexity here is in the linear sense: mixtures of distributions are distributions. This is the infinite-dimensional analogue of the simplex $\Delta(S_i)$ from Topic 9.

in τ_i (expectation is linear), any convex combination of best responses is also a best response. So the best-response correspondence has convex values.

Step 4: Apply Kakutani (or its infinite-dimensional generalization). The strategy space is compact and convex (in a locally convex topological vector space). The best-response correspondence is upper hemicontinuous with nonempty convex values. By the Kakutani-Fan-Glicksberg fixed-point theorem — the infinite-dimensional generalization of Kakutani's theorem[23] — a fixed point exists. That fixed point is a Bayesian Nash equilibrium.

> "The existence of Bayesian Nash equilibrium for continuous type and action spaces follows the same Kakutani logic as for finite games (Topic 9), but on the space of distributions rather than on simplices. Prokhorov's theorem provides the compactness, and dominated convergence provides the continuity."

Notice how the entire argument rests on the infrastructure built in this topic: the integral (to define expected payoffs), dominated convergence (to get continuity), compactness of $\Delta(\Theta)$ (Prokhorov), and the projection from behavioral strategies to distributional strategies (which uses measurability).

14.14 Where this appears in MWG

Measure theory and the Lebesgue integral are not developed in MWG's mathematical appendices. MWG uses integrals throughout — expected utility in Chapter 6, expected payoffs in Chapters 8 and 23, conditional expectations in Chapters 13 and 23 — but treats them as known.

The standard references for the material in this topic, in roughly the order we covered it:

- σ-algebras, measures, the Lebesgue integral, convergence theorems: any graduate real analysis text. The standard is Royden and Fitzpatrick, *Real Analysis*, 4th ed., Prentice Hall, 2010 (Chapters 2–5 and 17–19). For a treatment aimed at economists, see Ok, *Real Analysis with Economic Applications*, Princeton, 2007 (Part III).

- Conditional expectation and L^p spaces: Williams, *Probability with Martingales*, Cambridge, 1991 (Chapters 1–10), is compact and rigorous. Ok (2007, Chapter G) gives a treatment for economists.

[23] Fan, K., "Fixed-point and minimax theorems in locally convex topological linear spaces," *Proceedings of the National Academy of Sciences*, 38(2), 1952, pp. 121–126. Glicksberg, I. L., "A further generalization of the Kakutani fixed point theorem, with application to Nash equilibrium points," *Proceedings of the American Mathematical Society*, 3(1), 1952, pp. 170–174.

- Weak convergence and Prokhorov's theorem: Billingsley, *Convergence of Probability Measures*, 2nd ed., Wiley, 1999 (Chapters 1 and 5). Aliprantis and Border, *Infinite Dimensional Analysis*, 3rd ed., Springer, 2006 (Chapters 15 and 18), is the encyclopedic economics-oriented reference.
- Bayesian Nash equilibrium existence via distributional strategies: Milgrom and Weber (1985), as cited above.

14.15 Why this matters for microeconomics

This topic closes a circle that began in Topic 8. There, probability was a finite weighted sum. Here, it is a measure, and expectation is an integral against that measure. The notation $\mathbb{E}[f]$ now stands for $\int f\,d\mu$, and the rules we used informally — linearity, iterated expectations, the ability to pass limits through expectations — are now theorems with precise conditions.

The infrastructure serves three distinct purposes in the microeconomics curriculum:

1. **Defining payoffs.** Expected utility, expected revenue, expected continuation payoffs — all involve $\int f\,d\mu$. Without a rigorous integral, these are undefined whenever the type space is uncountable.
2. **Taking limits.** Comparative statics, mechanism convergence, value function iteration — all require exchanging limits and integrals. Dominated convergence (with the bounded-payoff assumption) is the workhorse.
3. **Compactness of distribution spaces.** Prokhorov's theorem makes $\Delta(\Theta)$ compact, which in turn makes Kakutani applicable, which in turn gives equilibrium existence. This is the route from "types are drawn from a compact space" to "a Bayesian Nash equilibrium exists."

The material is admittedly more abstract than earlier topics. But the abstraction buys a lot: once the integral and the convergence theorems are in place, the arguments in mechanism design, information economics, and game theory under incomplete information become clean applications of tools already in hand.

Chapter 15

Topic 15: Proof Techniques in Economic Theory

(Making conscious what you have been doing all along)

15.1 Why this topic exists

You have now read, and in places worked through, a substantial number of mathematical arguments. The IVT was proved by chasing the supremum of a sublevel set (Topic 4). Brouwer in one dimension was reduced to the IVT; in higher dimensions the argument proceeded by assuming no fixed point and deriving a geometric impossibility (Topic 9). Kakutani was proved by constructing a sequence of ordinary functions that approximate the correspondence, applying Brouwer to each, and passing to a limit (Topic 9). The supporting hyperplane was found by approaching a boundary point from outside, projecting at each step, and extracting a convergent subsequence (Topic 12). The Lebesgue integral was built by summing horizontal bricks of width ε and letting $\varepsilon \to 0$ (Topic 14).

Each of these arguments worked. But at no point were you told *why* a particular strategy was chosen, or how you would have known to try it if you were starting from scratch. The goal of this topic is to fill that gap.

We will give names to the techniques you have already seen, identify the situations in which each technique is the right one, and look across the whole course to find the recurring *patterns* that economic theorems follow. The goal is not to turn you into a logician. It is to make you a more deliberate reader and writer of proofs: to give you a small number of named strategies to try when you are stuck, and a sense for which economic structures call for which strategies.

15.2 What proofs actually do in economics

There is a standard line about proofs in mathematics: they establish truth rigorously. This is obviously important, and one of the reasons that separate "vibe" arguments from (at least) internally consistent ones. Moreover, given the degrees of freedom given to us in modeling economics, there is another reason for a proof, especially in theory.

A proof tells us which assumptions are doing what work. This is not the same as establishing a conclusion. A statement establishes a non-trivial causal relationship: *if* these conditions hold, *then* this conclusion follows. The proof is the chain that runs from the conditions to the conclusion, delivering (ideally) an elegant line of reasoning.

Consider the Weierstrass theorem (Topic 4): a continuous function on a compact set attains its maximum. The proof uses compactness to extract a convergent subsequence, then uses continuity to pass the limit through f. If you remove compactness, the argument fails at the first step (there is no convergent subsequence). If you remove continuity, it fails at the second step (you cannot pass the limit through f). The proof therefore tells you something economically meaningful: compactness is what prevents the optimum from escaping to infinity or to the boundary of the domain, and continuity is what ensures the objective value is preserved in the limit. Neither assumption is there without a reason.

The same logic runs through every theorem in the course. Kakutani needs convex values because the approximating functions are constructed by averaging, and averaging only stays inside the correspondence's image if those images are convex (Topic 9). The contraction mapping theorem needs $\beta < 1$ because that is precisely the condition that makes the geometric series $\sum \beta^n$ converge and drives the Cauchy estimate to zero (Topic 11). The separating hyperplane theorem needs the sets to be convex because the closest-point projection only behaves as required for convex sets (Topic 12).

This is why proofs matter. They are the only reliable answer to the question "what happens if I weaken this assumption?" If you have a proof, you can look at which steps use the assumption and ask whether they can be rescued. The statement alone (even if correct) does not tell you that.

15.3 The basic toolkit

Every proof in this course—and in most of microeconomic theory—uses one or more of four elementary techniques. We describe each one, identify where it appeared in the course, and explain why it was the right choice in each case.

15.3.1 Direct proof

A **direct proof** starts from the hypotheses and constructs the conclusion step by step, using definitions, previously established results, and algebraic manipulation. There is no auxiliary assumption, no case that is supposed to be impossible, just a chain of deductions.

Most of the algebraic work in the course is direct proof. When we checked that the Bellman operator is a contraction (Topic 11), the argument was: write out $\|TW - TW'\|_\infty$, substitute the definition of T, use the fact that $|W(\theta) - W'(\theta)| \leq \|W - W'\|_\infty$ everywhere, and conclude that $\|TW - TW'\|_\infty \leq \delta \|W - W'\|_\infty$. Each line follows from the previous one by direct substitution or inequality. No detour through an impossible case, no side assumption that is later discarded. You start at the hypotheses and walk straight to the conclusion.

When is direct proof the right strategy? Almost always, when you can see a path from the hypotheses to the conclusion. The challenge is that "seeing the path" often requires knowing what objects to introduce along the way. In the Bellman operator argument, the key object was the sup norm $\|\cdot\|_\infty$, and the key move was bounding the pointwise difference uniformly. Developing an instinct for these intermediate objects is most of what it means to be good at direct proofs.

The IVT proof in Topic 4 is a more subtle example of a direct proof. You want to show a particular value is attained. The intermediate object introduced is the set $X = \{x \in [a, b] : f(x) \leq c\}$, whose supremum $\hat{x}$ turns out to be the point you want. The argument then rules out $f(\hat{x}) < c$ and $f(\hat{x}) > c$ by continuity, and concludes $f(\hat{x}) = c$ by elimination. Every step is direct deduction; there is no assumption that is later contradicted.

Direct proofs are constructive but not necessarily pedagogical. They tell us step by step how to move from assumptions to results—that is the constructive part. They are also sometimes lengthy and cumbersome to read. Algebra delivers only so much intuition—that is the more difficult part.

15.3.2 Proof by contradiction

A **proof by contradiction** assumes that the conclusion is *false*, and then derives a consequence that contradicts the hypotheses or a known fact. The logical structure is: if the conclusion were false, then something impossible would follow; therefore the conclusion must be true. The ideas (like the proof by contrapositive below) is based on the logical identity that if the statement P implies Q is true, that means that if Q is not true, then also P cannot be true. What the proof by contradiction does, is to say, suppose Q is falls (it is $\not Q$), can we find something in P to still generate our element in $\not Q$. If so, the statement cannot be true. If any such attempts leads us outside the set P, we cannot get there.

The higher-dimensional proof of Brouwer's theorem is the clearest example in the course (Topic 9). We want to show: every continuous self-map of B^n has a fixed point. We assume the opposite—that f has *no* fixed point—and ask what that implies. It implies we can draw a ray from $f(x)$ through x for every x, which defines a retraction $r : B^n \to S^{n-1}$ (a continuous map that keeps the boundary sphere fixed). But Borsuk's theorem says no such retraction can exist. Contradiction. Therefore our assumption was wrong, and a fixed point must exist.

Notice the structure. The conclusion ("a fixed point exists") is an *existence claim*, and existence claims are often hard to prove directly: you would need to construct the fixed point explicitly. But the negation ("no fixed point exists") is a universal claim ("for all x, $f(x) \neq x$"), and universal claims are often easier to work with because they give you a property that holds everywhere. That universal property—the existence of the ray at every point—is exactly what builds the retraction, and it is the retraction that leads to the contradiction.

A second example is the uniqueness argument in the contraction mapping theorem (Topic 11). We want to show there is at most one fixed point. We assume there are two, x^* and y^*. Then $d(x^*, y^*) = d(T(x^*), T(y^*)) \leq \beta\, d(x^*, y^*)$. Since $\beta < 1$ and $d(x^*, y^*) \geq 0$, this forces $d(x^*, y^*) = 0$, hence $x^* = y^*$. Contradiction.

The separating hyperplane proof also uses contradiction in an embedded form (Topic 12, Step 0). To show the closest point z separates y from C, we suppose some $x \in C$ is on the wrong side ($p \cdot x > p \cdot z$) and show that the midpoint between z and x would then be closer to y than z is—contradicting the minimality of z.

Proof by contradiction is often short and elegant. You show that if the statement was not true, then any construction involves that you fall out of the set of assumptions. However, the proof by contradiction is by construction not constructive (pun intended!). You only learn that something has to be true, not why. There is no chain of arguments delivering the results. Instead there is a chain of arguments that excludes other options.

15.3.3 Proof by contrapositive

The **contrapositive** of a statement "if P then Q" is "if not Q then not P." The two are logically equivalent, so proving the contrapositive proves the original statement. This is in somesense the direct version of a proof by contradiction. But instead of just showing we always end up in a contradiction, but we show why the contrapositive must be true in a constructive manner, that is, there is more to learn as to why at least the contrapositive is as it is. Contrapositive is most useful when the forward direction $P \Rightarrow Q$ is hard to see directly, but the reverse direction "no Q implies no P" has an obvious witness.

Contrapositive proofs are less prominent in the course than contradiction, but the logic underlies several arguments. Consider: "if the Bellman operator has $\delta \geq 1$, it need not be a contraction." This is the contrapositive framing of the necessity of $\delta < 1$: the assumption $\delta < 1$ is not just sufficient for a contraction, it is (essentially) necessary. The counterexamples that appear at the end of the convergence theorem section in Topic 14—-sequences that fail to converge when one hypothesis is dropped—-are all instances of contrapositive reasoning: they exhibit "not Q" (convergence fails) when "not P" (one hypothesis is violated), thereby confirming that P is genuinely needed.

When reading a theorem statement, it is always worth asking: "what is the contrapositive?" If the theorem says "compactness implies existence of a maximum," the contrapositive says "if the maximum does not exist, the set was not compact (or the function was not continuous)." This reframing often gives a clean intuition for *why* the hypothesis is needed.

15.3.4 Mathematical induction

Mathematical induction proves a statement $P(n)$ for all natural numbers n by: (i) verifying one version (typically for the starting value $n = 0$ or $n = 1$), and (ii) showing that if $P(n)$ holds then $P(n + 1)$ holds (the inductive step). The logic is that if the first domino falls and each domino knocks over the next, all dominoes fall.

Induction appears most naturally in the course in the analysis of geometric series (Topic 10). To show that $\sum_{k=0}^{n} \delta^k = (1 - \delta^{n+1})/(1 - \delta)$, you check it for $n = 0$ (both sides equal 1), then assume it holds for n and add the $(n + 1)$-th term to both sides. More substantially, the contraction mapping proof (Topic 11) is an inductive argument in disguise: the estimate $d(x_n, x_{n+1}) \leq \beta^n \, d(x_0, x_1)$ is proved by applying the contraction property n times – one application per step of the iteration – which is exactly the inductive structure.

Induction is also the right technique for arguing about *finitely iterated* procedures: backward induction in dynamic games, iterated elimination of dominated strategies, finite-period dynamic programs. Any time a result "propagates backwards" from a known endpoint, the formal justification is induction.

15.3.5 A brief note on "proof by cases"

Sometimes an argument splits into cases because the conclusion holds for different reasons depending on which case you are in. The IVT proof handles the case $f(\hat{x}) < c$ and $f(\hat{x}) > c$ separately. The supporting hyperplane proof handles $\hat{x} = a$, $\hat{x} = b$, and $\hat{x} \in (a, b)$ as sub-cases. This is not a technique in the same sense as the above—it is more of a bookkeeping device. The key discipline is: cover every case, and do not let cases overlap in a way that lets you avoid a hard sub-argument.

That said, cases are often prevalent for general statements because sometimes things hold in different regions, but for different reasons. To see why, it makes sense to break up the statement and prove it case-by-case.

15.4 Construction and approximation

Two techniques that have a distinctive flavor in analysis and in economic theory deserve separate treatment.

15.4.1 Construction proofs

A **construction proof** establishes existence by explicitly building the object whose existence is claimed. It is the most direct of all existence arguments: rather than assuming non-existence and deriving a contradiction, you point to the object and verify it has the required properties.

Tarski's fixed-point theorem (Topic 9) is proved constructively. The claimed fixed point is $x^* = \sup\{x \in X : x \preceq f(x)\}$ – the supremum of the set of points "below their image." You then verify: $f(x^*)$ is an upper bound for that set (so $x^* \preceq f(x^*)$), and $f(x^*)$ is itself in the set (so $f(x^*) \preceq x^*$), giving $x^* = f(x^*)$. The fixed point is not found by ruling out alternatives; it is built from the lattice structure and handed to you.

The Lebesgue integral (Topic 14) is constructed similarly. You want to define $\int f\, d\mu$ for a nonnegative measurable function. Step one: define the integral for step functions (finite sums of indicator functions over measurable sets), where it is just a weighted sum of measure times height. Step two: define the integral for a general nonneg function as the supremum of the integral over all step functions that lie below it. The construction works because the lattice structure of the real line ensures the supremum is well-defined and the monotone convergence theorem guarantees it is stable under limits. You never "prove the integral exists by contradiction"—you build it, brick by brick, from simpler objects.

Why construction? Because it delivers a clear intuition what is happening.[1]

15.4.2 Approximation-then-limit

An **approximation-then-limit** proof proceeds in three steps:

1. Replace the object of interest (which is complicated or infinite-dimensional) with a sequence of simpler approximations.
2. Prove the desired result for each approximation.
3. Show that the limit of the approximations inherits the result.

The transition from step 2 to step 3 is always the hard step, and it always requires a *compactness* or *closedness* argument that ensures the limit is well-behaved.

The Kakutani proof (Topic 9) is the clearest example. The correspondence F is complicated – it is set-valued, and we cannot apply Brouwer directly.

[1] Brouwer, of our fixed-point theorem had a very distinct opinion that proofs need to be "constructible," for—in his mind—they do not carry meaning otherwise. Hilbert, another famous mathematician, argued that this is like not allowing the boxer to use their fists. The fight between the two led to Hilbert (who was the editor-in-chief) removing Brouwer from the editorial board of the *Mathematische Annalen* a leading journal at the time.

The approximating objects are piecewise linear functions f_n that "fill in" the correspondence on a grid of mesh $1/n$. Each f_n is an ordinary continuous function, and Brouwer gives a fixed point $x_n = f_n(x_n)$. Compactness of X ensures the sequence (x_n) has a convergent subsequence with limit x^*. The closed graph property of F (which follows from upper hemicontinuity and compact values – recall Topic 5) then ensures that the limit point satisfies $x^* \in F(x^*)$. The crucial phrase: *each* x_n is only an *approximate* fixed point of F, and it is the closed graph that converts "approximate" into "exact." Without that final ingredient, the approximation argument only gives you a sequence – not a conclusion.

The supporting hyperplane proof (Topic 12) follows the same template. The object of interest is the normal vector to a hyperplane tangent to C at a boundary point x^*. The approximations are the normal vectors $p_k = (y_k - z_k)/\|y_k - z_k\|$ obtained by projecting points $y_k \to x^*$ from outside C. Each p_k is a unit vector, so the sequence (p_k) lives in the compact unit sphere. Compactness gives a convergent subsequence with limit p. The inequality $p_k \cdot x \leq p_k \cdot z_k$ for all $x \in C$, satisfied by each approximation, survives the limit because the inner product is continuous. The supporting hyperplane is the limit of approximating separating hyperplanes.

The pattern is the same each time:

> Approximate $\to$ Apply the easy result $\to$ Extract a limit via compactness $\to$ Verify the limit inherits the property via closedness or continuity.

Recognizing this structure when you see it in a proof is half the battle.

15.5 The economist's pattern book

Economic theorems cluster into a small number of recurring proof patterns. Knowing the pattern tells you, before reading a single line of a proof, what strategy to expect and which assumptions will be critical.

15.5.1 Existence via continuity and compactness

Pattern: The object exists because a continuous function on a compact set attains its value.

The archetype is the Weierstrass theorem (Topic 4). In economics, this pattern underlies: the existence of a utility-maximizing bundle (continuous utility, compact budget set), the existence of a profit-maximizing output (continuous profit function, bounded feasible set), and the existence of a best response in a finite game (continuous payoff in mixed strategies, compact simplex).

Why it works: Compactness supplies a convergent maximizing sequence; continuity passes the objective value through the limit. The proof is direct.

What breaks without the assumptions: Without compactness, the maximum may not be attained (a firm with unbounded output set and strictly increasing profit). Without continuity, the function may skip over its supremum (a payoff function with a jump discontinuity).

15.5.2 Existence via fixed point

Pattern: The equilibrium is a fixed point of the best-response map; existence follows from Brouwer or Kakutani.

This is the Nash theorem (Topic 9). The best-response correspondence maps strategy profiles to strategy profiles. Upper hemicontinuity comes from Berge's theorem (Topic 5): the argmax of a continuous function varies upper hemicontinuously with the parameters when the constraint correspondence is continuous. Convex values come from the linearity of expected payoffs in one's own mixing probabilities. Kakutani then delivers a fixed point.

Why the strategy is correct: The conclusion is existence of an equilibrium, which is an existence claim. Direct construction is not available (there is no formula for the equilibrium of a general game). Contradiction would require assuming no equilibrium and deriving a contradiction – possible but roundabout. Kakutani is the direct tool because it is exactly designed for this situation: self-maps of compact convex sets with well-behaved set-valued structure.

What each assumption buys: Compactness of the strategy simplex keeps the domain manageable. Convexity of the simplex is needed for Kakutani. Upper hemicontinuity (via Berge) ensures the best-response map is well-behaved. Convex values (via linearity of payoffs) is the assumption that fails if we restrict to pure strategies – which is why pure-strategy Nash equilibria need not exist in finite games.

15.5.3 Existence via order

Pattern: The fixed point (hence the equilibrium) exists because the best-response mapping is monotone on a lattice, and Tarski's theorem applies.

This appeared in Topic 9 and connects to the monotone comparative statics of Topic 7. In games with strategic complements – where higher play by others makes higher play by oneself more attractive – the best-response mapping is monotone increasing on the appropriate lattice. Tarski then gives both existence of a fixed point and the result that the set of equilibria is itself a lattice (with a largest and a smallest equilibrium).

Why this is useful beyond Kakutani: Tarski requires no compactness, no convexity in the geometric sense, and no continuity. Its currency is order and monotonicity. For models where the natural structure is an ordering (strategies that are "higher" or "lower" effort levels) rather than a convex set, Tarski is the right tool. It also gives comparative statics for free: if the environment shifts in a way that makes the best-response mapping shift up, the largest equilibrium shifts up too.

15.5.4 Uniqueness via contradiction

Pattern: Assume two solutions exist. Derive that they must be equal. Conclude there is only one.

The cleanest instance is the uniqueness part of the contraction mapping theorem (Topic 11): $d(x^*, y^*) = d(Tx^*, Ty^*) \leq \beta\, d(x^*, y^*)$, which forces $d = 0$. The structure is: the two alleged solutions satisfy the same defining equation, and the equation's properties (here, the contraction property) are strong enough to force the distance between them to zero.

In consumer theory, the uniqueness of the optimal bundle under strict convexity of preferences follows the same structure: if two bundles were both optimal, the midpoint would be strictly preferred (by strict convexity), contradicting optimality of the original two.

Why uniqueness proofs are almost always by contradiction: Uniqueness is a *universal* statement ("there is no *other* solution"), and universal statements are most easily handled by assuming a counterexample (the second solution) and showing it cannot exist. A direct proof of uniqueness would need to show the solution is unique *without* referencing what a second solution would look like – which is awkward.

15.5.5 Characterization via necessary and sufficient conditions

Pattern: An object is characterized by a set of properties. The proof shows (a) any object with these properties satisfies the characterization and (b) any object satisfying the characterization has these properties.

This is the structure of the conditional expectation in Topic 14. The conditional expectation $\mathbb{E}[X|\mathcal{G}]$ is characterized as the unique $\mathcal{G}$-measurable random variable Z satisfying $\mathbb{E}[Z \cdot \mathbf{1}_A] = \mathbb{E}[X \cdot \mathbf{1}_A]$ for every $\mathcal{G}$-measurable event A. This is both a definition and a characterization: it says *exactly which property* pins down $\mathbb{E}[X|\mathcal{G}]$ uniquely, with no reference to a formula or an integral. The geometric interpretation – the conditional expectation is the orthogonal projection of X onto the subspace of $\mathcal{G}$-measurable functions – is a second characterization, and the proof that the two coincide is a proof that two sets of necessary and sufficient conditions describe the same object.

In mechanism design, the envelope theorem characterizes how the principal's value changes with a small change in the type distribution. In welfare economics, the first welfare theorem characterizes Pareto optimality: an allocation is Pareto optimal if and only if no feasible trade improves everyone. In each case, the value of the characterization is that it replaces a difficult search ("find all optimal allocations") with a testable condition ("check whether this condition holds at the candidate allocation").

First-order conditions are characterizations. The KKT conditions in constrained optimization (Topic 6) characterize the optimal solution: under constraint qualifications, x^* solves the program if and only if the KKT conditions hold at x^*. This is a characterization theorem. The proof has two halves: necessity (the optimum satisfies KKT) and sufficiency (any KKT point is optimal, given convexity). Both halves matter, and they use different tools – necessity uses the structure of the constraint set, sufficiency uses convexity of the objective.

15.5.6 Impossibility via contradiction

Pattern: Assume that the desired property holds. Derive a contradiction with an axiom or a structure theorem. Conclude the property is impossible.

Arrow's impossibility theorem is the canonical example in microeconomics (not developed in this pre-course but directly relevant). Assume a social welfare function satisfies unrestricted domain, Pareto efficiency, independence of irrelevant alternatives, and non-dictatorship simultaneously. Then derive a contradiction. Each step of the proof involves showing that the axioms, jointly, force the social welfare function to resemble a dictatorship in some sub-domain, and then extending this until full dictatorship is established. The conclusion ("there is no such function") is existential negation, and proof by contradiction is the only available strategy – you cannot prove non-existence by construction.

The same structure appears in the proof that $\delta \geq 1$ breaks the contraction argument (Topic 11): with $\delta = 1$, the geometric series diverges and the Cauchy estimate fails. This is not an impossibility theorem in the Arrow sense, but it is the same logical structure: given an assumption (here, $\delta \geq 1$), the desired conclusion (convergence) cannot hold.

15.6 Reading a proof in MWG

MWG's mathematical appendix and its main-text proofs are written in a compressed, symbol-dense style. Here is a protocol for reading one.

Step one: identify the claim. Before reading the proof, read the theorem statement carefully. Write down (or mark) what is *assumed* and what is *claimed*. These are the two endpoints of the chain you are about to read.

Step two: classify the strategy. Scan the first paragraph of the proof. Does it start "Suppose for contradiction..."? Then it is a contradiction proof, and you should look for where the contradiction appears. Does it start by constructing an object (a sequence, a set, a function)? Then it is probably a construction or approximation argument, and you should track what this object is and why it has the required properties. Does it start by writing out definitions and manipulating inequalities? Then it is a direct proof.

Step three: track the assumptions. Each hypothesis in the theorem statement should be used at least once in the proof. When you find a step that is not obvious, ask: which hypothesis justifies this? MWG proofs are tight enough that the answer is usually one specific assumption. If you cannot find it, that is a signal you have missed something – not that MWG made an error.

Step four: find the key step. Most proofs have one hard step and several easy ones. The hard step is usually the one where the most important assumption is used, or where an approximation is turned into an exact result, or where the contradiction is derived. Finding the key step is the same as finding the heart of the theorem. In the Kakutani proof, the key step is the closed graph argument that converts approximate fixed points into an exact one. In the IVT proof, it is the two-sided elimination of $f(\hat{x}) \neq c$. In the supporting hyperplane proof, it is the compactness of the unit sphere that gives the convergent subsequence of normals.

Step five: construct a counterexample for each hypothesis. For every hypothesis in the theorem, ask: if I drop this assumption and keep all others, does the conclusion still hold? If you can find a counterexample, you understand the theorem. If you cannot, you do not yet understand which direction of the argument needs the assumption. MWG occasionally provides counterexamples inline; the convergence theorems in Chapter 11 (and our Topic 14) explicitly include counterexamples for this reason.

15.7 Writing a proof: how to start and how to get unstuck

Starting a proof from scratch is harder than reading one, because you do not know the structure in advance. Here is a small toolkit of starting moves.

Write down what you have and what you want. At the top of a scratch page: "Given: [hypotheses]. Want: [conclusion]." Spell out the definitions of each term in the statement. Often this step alone reveals the structure of the argument, because the definition of the conclusion is already partway to what you need.

Match the conclusion type to a technique.

- Conclusion is "there exists x with property P": try direct construction first. If you cannot construct x, try contradiction (assume no such x exists and derive something impossible) or approximation (build a sequence of approximate solutions and pass to a limit).

- Conclusion is "for all x, $P(x)$ holds": try direct proof (pick an arbitrary x and verify $P(x)$). If $P(x)$ is an inequality, try bounding one side by the other.
- Conclusion is "P implies Q": assume P and try to derive Q directly. If direct proof is stuck, try contrapositive (assume not-Q and derive not-P) or contradiction (assume P and not-Q and derive a contradiction).
- Conclusion is "P and Q are equivalent": prove both $P \Rightarrow Q$ and $Q \Rightarrow P$ separately, by any technique.
- Conclusion is "there is exactly one x with property P": prove existence (see above) and uniqueness (assume two, derive they are equal, usually by contradiction).

If stuck on a direct proof, introduce the right intermediate object. Most proofs require something that is not in the hypotheses: a cleverly defined set (as in the IVT proof), a norm (as in the contraction proof), a sequence of approximations (as in Kakutani), or an auxiliary function (as in the Brouwer 1D proof, where $g(x) = f(x) - x$ converted a fixed-point question into a zero-finding question). There is no algorithm for finding this intermediate object. But the object is almost always motivated by what the conclusion requires: ask "what would I need in order to apply [the relevant tool]?" and define the intermediate object to supply it.

Use the hypotheses. If a proof is going nowhere, check whether you have used every hypothesis. Theorems are stated with exactly the assumptions needed. If you have an unused assumption, it almost certainly points to the missing step.

Work backwards. Start from the conclusion and ask "what would I need in order to conclude this?" Then ask "what would I need in order to have *that*?" Keep working backwards until you reach the hypotheses. If the forward and backward chains overlap, you have the proof.

15.8 A worked example

We close with an example that is small enough to be followed completely but rich enough to illustrate several techniques at once. It also has an immediate economic interpretation.

Claim. Let $f : [0, 1] \to [0, 1]$ be continuous and strictly decreasing. Then f has exactly one fixed point.

Economic interpretation. In a Cournot duopoly, suppose each firm's best response to the other's quantity is strictly decreasing (strategic substitutes: when my rival produces more, my optimal output falls). If the best-response functions map $[0, \bar{q}]$ to $[0, \bar{q}]$ and are continuous, this claim guarantees a unique Nash equilibrium.

15.8.1 Step 1: Existence (direct proof using the IVT)

Define $g : [0,1] \to \mathbb{R}$ by $g(x) = f(x) - x$.

Then $g(0) = f(0) - 0 = f(0) \geq 0$ (since $f(0) \in [0,1]$) and $g(1) = f(1) - 1 \leq 0$ (since $f(1) \in [0,1]$).

Since g is continuous (as the difference of two continuous functions – recall Topic 4) and $g(0) \geq 0 \geq g(1)$, the Intermediate Value Theorem (Topic 4, §7) gives a point $x^* \in [0,1]$ with $g(x^*) = 0$, i.e., $f(x^*) = x^*$.

Why this strategy? A fixed point is a zero of $g(x) = f(x) - x$. The IVT is the canonical tool for finding zeros of continuous functions on intervals. The reduction $f(x) = x \Leftrightarrow g(x) = 0$ is the key intermediate step, and the boundary conditions $g(0) \geq 0 \geq g(1)$ follow immediately from the assumption that f maps $[0,1]$ into itself.

15.8.2 Step 2: Uniqueness (proof by contradiction)

Suppose x^* and y^* are both fixed points with $x^* < y^*$. Then $f(x^*) = x^*$ and $f(y^*) = y^*$.

Since f is strictly decreasing and $x^* < y^*$, we have $f(x^*) > f(y^*)$. But $f(x^*) = x^*$ and $f(y^*) = y^*$, so $x^* > y^*$. This contradicts $x^* < y^*$.

Therefore no two distinct fixed points can exist.

Why contradiction? Uniqueness claims are universal statements ("there is no second fixed point"), and the natural move is to assume a second fixed point and show it leads to something impossible. Here the contradiction is immediate: the strict decrease of f forces $f(x^*) > f(y^*)$, but the fixed-point property forces the reverse inequality.

Which assumption did the work? Existence used continuity (to apply the IVT) and the self-mapping property (to get the right signs at the boundary). Uniqueness used strict decrease. If f were only weakly decreasing, there could be an interval of fixed points. If f were constant, every point in the interval would be a fixed point. Strict decrease is exactly what prevents this.

Summary of techniques used:

Step	Technique	Key assumption used
Reduction to g's zero	direct manipulation	definition of fixed point
Boundary signs	direct verification	$f : [0,1] \to [0,1]$
Existence	IVT (itself a direct proof)	continuity of f
Uniqueness	contradiction	strict decrease of f

15.9 Where this appears in MWG and beyond

MWG does not have a chapter on proof techniques as such. The relevant background is spread across:

- **MWG, Mathematical Appendix A** (sets, functions, relations – the language in which proofs are written),
- **MWG, Mathematical Appendix C** (fixed-point theorems stated, without proof),
- **MWG, Chapter 16 and 17** (welfare theorems and general equilibrium – where the separating hyperplane and Kakutani appear as working tools, not as pure mathematics).

For a longer treatment of proof technique oriented toward economists, Efe Ok's *Real Analysis with Economic Applications* (2007) is the standard reference. It covers all the theorems in this pre-course with full proofs and extensive economic motivation. Chapter A (logic and sets) is a good place to go if you want to sharpen the formal side of what this topic covers informally.

15.10 What you should take away

After this topic, you should be able to:

- identify the proof technique (direct, contradiction, contrapositive, induction, construction, approximation-then-limit) being used in a proof you are reading,
- explain, for each technique, which features of the conclusion or the hypotheses make it the appropriate choice,
- locate the key step in a proof and identify which hypothesis it uses,
- recognize the five recurring patterns in economic proofs (existence via fixed point, existence via order, uniqueness via contradiction or contraction, characterization via necessary and sufficient conditions, impossibility via contradiction),
- and start a proof of your own by matching the conclusion type to a technique and working forward from the hypotheses or backward from the conclusion.

You will not write novel existence proofs on a problem set. But you will write uniqueness arguments, verify first-order characterizations, check that a given mapping satisfies a fixed-point theorem's hypotheses, and follow multi-step proofs in MWG without getting lost. The techniques in this topic are the tools for all of that.